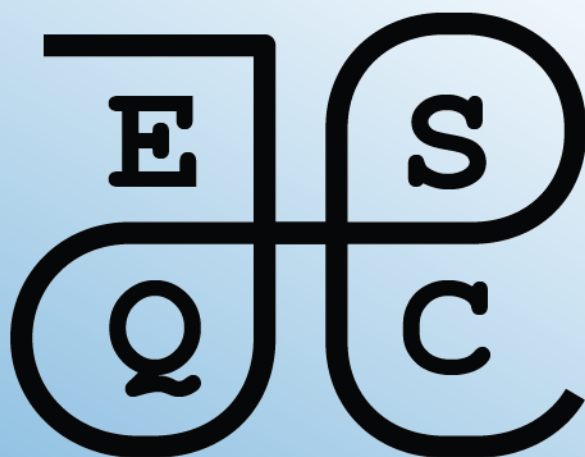




Mathematics Exercises

2026

ESQC mathematics exercises

2026 edition

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2026-09-07

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Preface

This is a collection of mathematics exercises for the European Summerschool in Quantum Chemistry (ESQC).

Many exercises have been written by Simen Kvaal, while many other (excellent) exercises and solutions are generated by GPT-5.6 Sol from OpenAI, and quality controlled by Simen Kvaal.

Part I

Exercise recommendations

Exercises for the mathematics tutorial

This chapter contains recommended exercises for the mathematics tutorial on the first day of the school. The focus is on *linear algebra*, with some *set theory* exercises as well, and then some exercises on *continuity and differentiation*. **Start with linear algebra — these are the most important exercises.** The book contains many more exercises as well, if you find time and inspiration!

The **beginner** list contains exercises that we consider to be especially useful for the students who feel that they need some refreshing to do before embarking on the remaining exercises.

The **intermediate** list contains more advanced exercises, accessible for the majority of the students at the school.

The **for the curious** list contains exercises that go deeper into either mathematics or applications, and can be useful for students that already master the mathematics material at an advanced level. That being said, they should be interesting for most students.

Beginner

- Exercise 3.1 — Very basic sums
- Exercise 3.2 — Simplifying as a sum
- Exercise 6.1 — Basic vector operations
- Exercise 6.2 — Basic matrix-vector products
- Exercise 6.3 — Matrix-matrix products
- Exercise 9.1 — Checking candidate eigenvectors
- Exercise 9.2 — Eigenvalues in chemical kinetics
- Exercise 10.1 — Cooking with vectors 1
- Exercise 1.1 — Days of the week as a set
- Exercise 1.9 — Classically allowed region for Morse potential
- Exercise 6.4 — Linear independence and basis
- Exercise 21.1 — A first discontinuity

- Exercise 21.3 — Continuity by direct substitution
- Exercise 21.11 — Continuity of a model potential energy
- Exercise 21.12 — First partial derivatives
- Exercise 21.13 — Partial derivatives as one-dimensional derivatives
- Exercise 21.15 — A first Jacobian

Intermediate

- Exercise 6.6 — Office coordinates
- Exercise 6.7 — Boat sailing on the ocean
- Exercise 6.8 — An exercise by Robert Beezer
- Exercise 6.9 — Compute the action of a matrix on vectors
- Exercise 6.10 — Matrix-matrix products, rows and columns
- Exercise 9.3 — Eigenvalues and diagonalization of a Hermitian matrix
- Exercise 9.4 — A matrix that cannot be diagonalized
- Exercise 10.2 — Cooking with vectors 2
- Exercise 1.7 — De Morgan's laws
- Exercise 1.10 — Molecules as sets
- Exercise 27.3 — The exponential, sine, and cosine series

For the curious

- Exercise 8.1 — Hybrid orbitals from vectors and matrices
- Exercise 9.6 — Eigenvalues of a reaction network
- Exercise 4.4 — Stirling's approximation
- Exercise 5.12 — A finite-dimensional quantum-mechanical impossibility
- Exercise 11.1 — Complex numbers as a real vector space
- Exercise 21.21 — Checking multivariable limits along different paths
- Exercise 21.23 — Explore a two-dimensional energy landscape

Recommended exercises by school topic

This chapter contains lists of mathematics exercises grouped by each quantum chemistry topic taught at the school. These exercises can be useful as a warm-up to the proper exercises given each day.

The list is not complete, and is continually updated!

Coupled-cluster theory

Beginner

- Exercise [29.1](#) — Approximating the exponential function
- Exercise [30.1](#) — Exponential of a diagonal matrix
- Exercise [30.2](#) — An exponential whose series terminates

Intermediate

- Exercise [29.6](#) — From the exponential to rotations
- Exercise [30.4](#) — A matrix exponential and rotations
- Exercise [30.6](#) — A terminating similarity transformation

Second quantization

Beginner

- Exercise [3.1](#) — Very basic sums
- Exercise [3.2](#) — Simplifying as a sum
- Exercise [9.1](#) — Checking candidate eigenvectors
- Exercise [3.4](#) — Reconstruct a vector

- Exercise 3.6 — The summation index is a dummy index
- Exercise 3.7 — Linearity of sums
- Exercise 3.8 — When something does not depend on the summation index
- Exercise 3.9 — Free and summed indices
- Exercise 3.10 — Shifting the summation index
- Exercise 3.11 — The Kronecker delta: basic exercise
- Exercise 4.5 — Basic binomial coefficients
- Exercise 4.6 — Pascal's triangle
- Exercise 4.9 — The binomial theorem
- Exercise 5.6 — Your first induction proof
- Exercise 22.1 — From components to a gradient
- Exercise 22.2 — Differentiating a quadratic form

Intermediate

- Exercise 9.3 — Eigenvalues and diagonalization of a Hermitian matrix
- Exercise 4.10 — Coefficients in binomial expansions
- Exercise 5.7 — Direct proof with a matrix
- Exercise 27.3 — The exponential, sine, and cosine series
- Exercise 27.7 — The Cauchy product of power series

For the curious

- Exercise 4.7 — Pascal's identity

Self-consistent field theory

No exercises selected.

Density functional theory

No exercises selected.

Response theory

Beginner

- Exercise [22.4](#) — Treating c and c^* as independent variables
- Exercise [30.1](#) — Exponential of a diagonal matrix
- Exercise [30.2](#) — An exponential whose series terminates

Intermediate

- Exercise [30.4](#) — A matrix exponential and rotations
- Exercise [30.5](#) — Anti-Hermitian matrices and unitary matrices
- Exercise [30.6](#) — A terminating similarity transformation

Relativistic quantum chemistry

Beginner

- Exercise [3.12](#) — The Levi-Civita symbol: basic exercise
- Exercise [17.2](#) — Roots of a cubic polynomial by guessing and polynomial long division

Molecular properties

Intermediate

- Exercise [7.1](#) — The vector triple product and the Levi-Civita symbol

Multiconfigurational methods

No exercises selected.

Part II

Mathematical foundations

1 Sets and functions

Sets and set notation appear many places in quantum chemistry. The goal of this chapter is to train you in the basic usage of sets and set notation.

Exercises: abstract sets and functions

Exercise 1.1 (Days of the week as a set). The days of the week form a simple example of a finite set. Recall that we write the elements of a set inside braces, such as

$$A = \{a, b, c\}.$$

- Write the days of the week as a set D using braces $\{ \}$.
- How many elements does D contain? Write your answer using the notation $|D|$ for the number of elements in a set.
- Is Monday an element of D ? Write your answer using the symbol $\in$.
- Is January an element of D ? Write your answer using the symbol $\notin$.
- Let W be the set consisting of Saturday and Sunday. Write W using braces. Is W a subset of D ? Express your answer using the symbol $\subseteq$.
- Let

$$M = \{\text{Monday, Tuesday, Wednesday, Thursday, Friday}\}.$$

What is $M \cup W$? What is $M \cap W$?

g. Suppose another student writes the days in a different order,

$$D' = \{\text{Sunday, Saturday, Friday, Thursday, Wednesday, Tuesday, Monday}\}.$$

Is $D' = D$?

Exercise 1.2 (Ordered pair). An *ordered pair* (x, y) is a two-element “set” where the order matters. Two ordered pairs (x, y) and (u, v) are equal if and only if $x = u$ and $y = v$. From this it follows that $(x, y) = (y, x)$ if and only if $x = y$. A definition of an ordered pair as a *set* is

$$(x, y) = \{\{x\}, \{x, y\}\}.$$

Show that $(x, y) = (y, x)$ if and only if $x = y$ using the set theoretic definition.

Exercise 1.3 (Rational numbers as ordered pairs). A nonnegative rational number p/q , $p, q \in \mathbb{N}$, $q > 0$, can be written as an ordered pair (p, q) . Write down the rational numbers $1/4$, $2/3$, 0 , $1/3$ as ordered pairs using set theory, both for the pair and for each integer. (Strictly speaking we the number itself is an *equivalence class* of such pairs, i.e., $pn/qn = p/q$, so we must identify (p, q) and (np, nq) . Ignore equivalence classes in this exercise.)

Exercise 1.4 (Cartesian product of two sets). The *cartesian product* of two sets A and B is

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

Write down the cartesian product of $\{\heartsuit, \diamondsuit, \spadesuit, \clubsuit\}$ and $\{1, 2, 3\}$, using the set theoretic definition for the ordered pair. You can skip spelling out the set theoretic definition of the natural numbers.

Exercise 1.5 (Ordered pair as a two-element set). Similar to the previous exercise, write down $\{1, 2, 3\} \times \{2, 3, 4\}$.

Exercise 1.6 (Function from one set to another set). A *function* $f : A \rightarrow B$ from one set A to another set B is a rule that assigns to every $a \in A$ precisely one $b \in B$. In terms of set theory, a function $f : A \rightarrow B$ is a subset of $A \times B$, such that

- For all $a \in A$ there exists $b \in B$ such that $(a, b) \in f$.

- For all $a \in A$ and $b, b' \in B$, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.

Write down the function $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$, $f(1) = 2$, $f(2) = 3$, $f(3) = 1$, using the set theoretic definition of ordered pairs and the cartesian product.

De Morgan's laws are useful when discussing subsets A, B of a larger set X . Recall that the complement of A relative to X is $A^c = X \setminus A$. De Morgan's laws state that:

- $(A \cup B)^c = A^c \cap B^c$.
- $(A \cap B)^c = A^c \cup B^c$.

Exercise 1.7 (De Morgan's laws). Draw a picture, representing X as the whole sheet, A and B as overlapping shapes, e.g., circles. Label A , B , $A \cap B$, and $A \cup B$, making another drawing if necessary. Convince yourself that De Morgan's laws are correct.

Exercise 1.8 (De Morgan's laws 2). Prove De Morgan's laws mathematically. Note that the complement operation acts like negation of truth, i.e., $a \in A^c$ if and only if $a \in X$ and $a \notin A$.

Exercises: Sets in quantum chemistry

Exercise 1.9 (Classically allowed region for Morse potential). Figure Figure 1.1 shows the *Morse potential* - a common model potential for vibrational motion of a diatomic molecule:

$$V(r) = D_e (1 - e^{-a(r-r_e)})^2 - D_e,$$

where $D_e > 0$ is the dissociation energy, $r_e > 0$ is the equilibrium bond length, and $a > 0$ is a parameter that determines the width of the potential well. The total energy of the molecule is denoted by $E \in \mathbb{R}$. The classically allowed region for the total energy E is the set S_E of all bond lengths r such that $V(r) \leq E$:

$$S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$$

- Explain the notation $S_E = \{r \in \mathbb{R} \mid V(r) \leq E\}$ in words. In particular, what does the vertical bar $\mid$ mean?

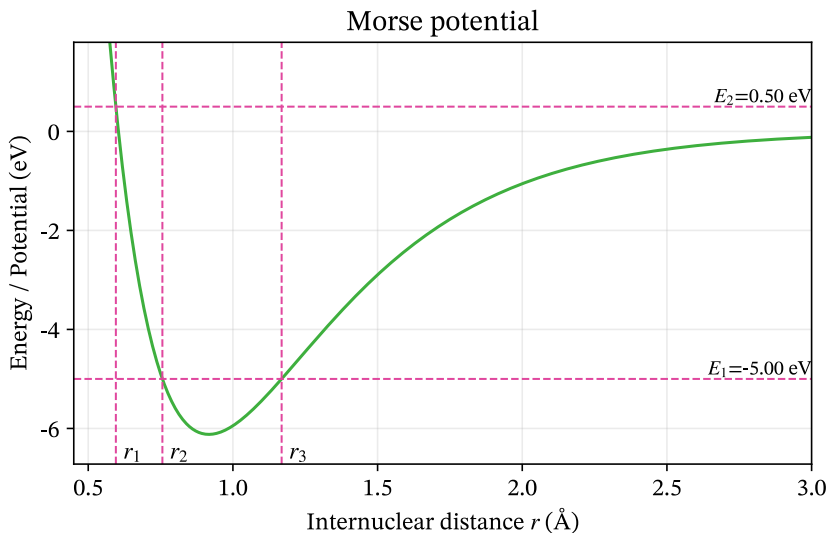


Figure 1.1: Morse potential

- b. Can you find a condition on E such that S_E is the empty set?
- c. Can you find a condition on E such that S_E is the whole real line?
- d. Based on the figure, which of the following statements are true?
 - $S_{E_1} = (r_2, r_3)$
 - $S_{E_1} = [r_2, r_3]$
 - $S_{E_1} = \{r_2, r_3\}$
- e. Based on the figure, which of the following statements are true?
 - $S_{E_2} = (r_1, +\infty)$
 - $S_{E_2} = [r_1, +\infty)$
 - $S_{E_2} = \{r_1, r_2, r_3\}$

Exercise 1.10 (Molecules as sets). In this exercise, a *molecule* is informally defined only by its chemical composition: the number of atoms of each element occurring in its molecular formula. Thus, H_2 and CO are two distinct molecules, while HCOOH and CO_2H_2 represent the same molecule, since both contain one C atom, two H atoms, and two O atoms.

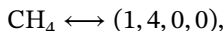
We ignore molecular geometry, bonding, isotopes, charge, and all questions of chemical stability. In particular, any formal combination of atoms is deemed a molecule, whether or not such a species could exist physically.

We restrict our attention to the four elements C, H, O, and N. Let M denote the set of all molecules that can be constructed from these elements. A molecule must contain at least one atom.

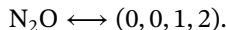
One useful way of describing a molecule is by the ordered quadruple

$$(n_C, n_H, n_O, n_N), \quad (1.1)$$

where each n_X is the number of atoms of element X in the molecule. For example,



and



- Explain why Equation 1.1 uniquely determines an element of M . Give 4-5 additional examples of molecules in terms of this representation.
- Express M in terms of $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. What is the cardinality of M ? Is it finite, countably infinite, or uncountably infinite? Justify your answer.
- Let

$$M_2 = \{m \in M \mid m \text{ contains exactly two atoms}\}.$$

Find $|M_2|$ and list all elements of M_2 .

- Let $S \subset M$ be the set

$$S = \{m \in M \mid m \text{ contains at least one H atom}\}.$$

Describe S in terms of the quadruple representation above. Is S finite, countably infinite, or uncountably infinite? Justify your answer.

- Describe the complement $M \setminus S$. What is its cardinality?

e. Define

$$C = \{m \in M \mid m \text{ contains at least one C atom}\}.$$

Describe the following sets both in words and in terms of the quadruple representation:

$$C \cap S, \quad C \cup S, \quad C \setminus S.$$

Determine the cardinality of each set.

f. For $n \geq 1$, define

$$M_n = \{m \in M \mid m \text{ contains exactly } n \text{ atoms}\}.$$

Thus, a molecule in M_n satisfies

$$n_{\text{C}} + n_{\text{H}} + n_{\text{O}} + n_{\text{N}} = n.$$

Show that

$$|M_n| = \binom{n+3}{3}.$$

Check that your formula agrees with your answer to part **b**.

g. Explain why

$$M = \bigcup_{n=1}^{\infty} M_n.$$

Are the sets M_n pairwise disjoint? Use this decomposition to give another argument that M is countably infinite.

h. Show that M and S have the same cardinality. Explain why this is possible even though S is a proper subset of M .

2 Cardinality and countability

Cardinality compares sets through bijections. The exercises cover countable sets, Cantor's diagonal argument, and the cardinality of the continuum.

Recall that the *cardinality* $|S|$ of a set S is the number of elements in S . For finite sets, this is intuitive, but what about infinite sets?

One says that A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$, i.e., f is one-to-one and onto. (Intuitively, you can draw lines between individual elements of A and B , only one line to/from each element. Precisely: Onto means, to every $b \in B$ there exists a $a \in A$ such that $f(a) = b$. One-to-one: $f(a) = f(a')$ implies $a = a'$.) One can think of the bijection as labelling each $b \in B$ by exactly one element $a \in A$, and that all labels from A are used up.

Let $\mathbb{N}_n = \{1, 2, \dots, n\}$. This set has cardinality n .

The set $\mathbb{N}$ has an infinite number of elements. By definition, $|\mathbb{N}| = \aleph_0$ (“aleph-nought”).

Exercises

Exercise 2.1 (Countability of the integers). Show that $|\mathbb{Z}| = \aleph_0$ by explicitly constructing a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$.

Exercise 2.2 (Countability of the natural number plane). Show that $|\mathbb{N} \times \mathbb{N}| = \aleph_0$. Hint: Draw a picture of $\mathbb{N} \times \mathbb{N}$, and try to draw a line through all the points in this set. How does this show the existence of a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$?

Exercise 2.3 (Countability of the rational numbers). Show that $|\mathbb{Q}| = \aleph_0$.

Exercise 2.4 (Interval vs. real numbers). Show that the interval $I = (-1, 1)$ has the same cardinality as $\mathbb{R}$. You must construct a function $f : (-1, 1) \rightarrow \mathbb{R}$ that is a bijection.

Exercise 2.5 (Uncountability of the real numbers). In this exercise, we show that $|\mathbb{R}| > \aleph_0$. The cardinality would be $\aleph_0$ if we could write a list of the real numbers. It suffices to find a list of the numbers in $I = [0, 1[$, i.e., infinite decimal expansions $0.d_1d_2d_3\cdots$, with $d_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

We will look for a contradiction. Consider a list $x_1, x_2, x_3, \dots$ of all real numbers in $[0, 1[$. This is essentially an infinite matrix of digits. Can you find a real number $x \in [0, 1]$ which is *not* in this list, i.e., $x \neq x_j$ for all j ? Hint: consider the diagonal of the matrix.

The cardinality of $\mathbb{R}$, “the continuum”, is written $2^{\aleph_0}$.

Exercise 2.6 (Constructing a one-to-one map). Show that $|[0, 1[\times [0, 1[| = |[0, 1[|$, by constructing a one-to-one map. Hint: Use decimal expansions. Conclude that also $|\mathbb{R}^2| = |\mathbb{C}| = |\mathbb{R}|$.

3 Summing over indices

Summing several things in a single expression occurs frequently in mathematics, and in particular in applications to physics and chemistry. The general syntax for a summation over an index is:

$$\sum_{i \in I} f(i),$$

where I is some countable set of indices, i is the index of summation, and $f(i)$ is the function/quantity being summed. The set I can be finite or infinite, and the index i can be a number, a vector, or even a more complicated object.

The most basic and common case is a sum over a finite or infinite set of integers, which is usually written as

$$\sum_{i=a}^b f(i)$$

where i is the index of summation, a is the lower limit of the index i , b is the upper limit, and $f(i)$ is the function/quantity being summed.

Here are some concrete examples:

1. A sum over a finite number of terms:

$$\sum_{i=1}^5 i = 1 + 2 + 3 + 4 + 5$$

2. An infinite series:

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

3. A series expansion. In this example, the exponential function is expressed as a Taylor series, summing over all non-negative integers:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

4. Vector notation. For example, a vector $\mathbf{r} \in \mathbb{R}^3$ can be expressed as a sum over its components:

$$\sum_{i=1}^3 r_i \mathbf{e}_i = \mathbf{r}$$

5. A matrix-vector product expressed as a sum over indices: The i -th component of the product of a matrix A and a vector $\mathbf{x}$ is given by the sum over the products of the corresponding elements of the matrix and vector:

$$(A\mathbf{x})_i = \sum_{j=1}^n A_{ij}x_j$$

Manipulating sums

The index in a sum is called a **summation index** or **dummy index**. Its name has no significance. For example,

$$\sum_{i=1}^n a_i = \sum_{j=1}^n a_j.$$

The index may therefore be renamed, as long as it is renamed consistently within the sum.

Summation is **linear**. Sums can be split or combined,

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i,$$

and factors that do not depend on the summation index can be taken outside the sum,

$$\sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i,$$

where it is important that c is independent of i .

Expressions may contain both summed and unsummed indices. An index that is not summed over is called a **free index**. For example, in the matrix-vector product

$$y_i = \sum_{j=1}^n A_{ij}x_j,$$

j is a dummy index, while i is a free index. The equation therefore represents one equation for each possible value of i .

Special symbols used in summation notation

The **Kronecker delta** δ_{ij} is a special symbol that is often used in summation notation. It is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Sometimes, if confusion can arise one inserts a comma between the two indices, writing $\delta_{i,j}$ instead of δ_{ij} .

The Kronecker delta forms the matrix elements of the identity matrix.

The **Levi-Civita symbol** ϵ_{ijk} is another special symbol that is often used in summation notation in the three-dimensional case, i.e., where $i, j, k \in \{1, 2, 3\}$. It is defined as

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0 & \text{if any two indices are equal.} \end{cases}$$

Exercises: Basic summation

These exercises are at a very basic level. If you are already comfortable with summation notation, you may skip them.

Exercise 3.1 (Very basic sums). Expand the following sums and calculate their values:

a) $\sum_{i=1}^4 i$

b) $\sum_{i=1}^4 i^2$

c) $\sum_{k=0}^3 2^k$

Exercise 3.2 (Simplifying as a sum). Write each expression using summation notation:

a) $x_1 + x_2 + x_3 + x_4 + x_5$

b) $1 + q + q^2 + q^3 + q^4$

c) $a_1 b_1 + a_2 b_2 + a_3 b_3$

Exercise 3.3 (Estimating $\sqrt{e}$ via a Taylor series). Use the first four terms of the series for e^x to estimate $\sqrt{e}$:

$$\sqrt{e} = e^{0.5} \approx \sum_{n=0}^3 \frac{(0.5)^n}{n!}.$$

Write out the four terms and add them. The estimate is not especially accurate; the point is to see how the index n controls both the power and the factorial.

Exercise 3.4 (Reconstruct a vector). The components of a vector are $r_1 = 2$, $r_2 = -1$, and $r_3 = 4$. Let $\mathbf{e}_i$ be the standard basis in $\mathbb{R}^3$.

a) Write out $\sum_{i=1}^3 r_i \mathbf{e}_i$ term by term.

b) Gather the result as a column vector

Exercise 3.5 (Summing in a matrix–vector product). Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Use

$$(\mathbf{Ax})_i = \sum_{j=1}^3 A_{ij} x_j$$

to write out and calculate $(\mathbf{Ax})_1$ and $(\mathbf{Ax})_2$. Collect the two answers in a column vector.

Exercise 3.6 (The summation index is a dummy index). Consider

$$S = \sum_{i=1}^4 i^2.$$

- a. Write out the sum explicitly and calculate its value.
- b. Write out and calculate

$$\sum_{j=1}^4 j^2.$$

- c. Explain why

$$\sum_{i=1}^4 i^2 = \sum_{j=1}^4 j^2.$$

- d. Rewrite

$$\sum_{k=1}^5 a_k b_k$$

using p as the summation index.

Exercise 3.7 (Linearity of sums). Let a_i and b_i , $i = 1, 2, 3$ be two sets of numbers and let c be a constant, i.e., just a number.

- a. Write out both sides of

$$\sum_{i=1}^3 (a_i + b_i) = \sum_{i=1}^3 a_i + \sum_{i=1}^3 b_i$$

and verify that they are equal.

- b. Write out both sides of

$$\sum_{i=1}^3 c a_i = c \sum_{i=1}^3 a_i$$

and verify that they are equal.

- c. Use these properties to rewrite

$$\sum_{i=1}^3 (2a_i - 3b_i)$$

as two separate sums (i.e., each has a single summation sign).

Exercise 3.8 (When something does not depend on the summation index). Let x be a number independent of the summation index i .

- a. Simplify

$$\sum_{i=1}^5 x.$$

- b. Simplify

$$\sum_{i=1}^5 ix.$$

- c. Explain why x can be taken outside the sum in

$$\sum_{i=1}^n ix = x \sum_{i=1}^n i.$$

Exercise 3.9 (Free and summed indices). Consider the following expression for the i -th component of a vector $\mathbf{y}$, being the product of a matrix A and a vector $\mathbf{x}$:

$$y_i = \sum_{j=1}^3 A_{ij} x_j.$$

- a. Which index is summed over?
- b. Which index is not summed over?
- c. For $i = 2$, write the right-hand side explicitly.
- d. If A has four rows and three columns, how many components does $\mathbf{y}$ have?
- e. Explain why replacing j by k ,

$$y_i = \sum_{k=1}^3 A_{ik} x_k,$$

does not change the equation.

Exercise 3.10 (Shifting the summation index). Consider

$$S = \sum_{i=1}^4 a_i.$$

- a. Write out S explicitly.
- b. Show that the same sum can be written as

$$S = \sum_{j=0}^3 a_{j+1}$$

by writing out the terms.

- c. Rewrite

$$\sum_{i=2}^5 a_i$$

as a sum whose index starts at $j = 0$.

- d. Rewrite

$$\sum_{i=0}^{n-1} a_{i+1}$$

as a sum with index j running from 1 to n .

Exercise 3.11 (The Kronecker delta: basic exercise). The **Kronecker delta** is defined in the introduction to this chapter. It is a special symbol that is often used in summation notation.

- a. Evaluate δ_{11} , δ_{12} , δ_{23} , and δ_{33} .
- b. Let $a_1 = 2$, $a_2 = 5$, and $a_3 = 7$. Write out all terms in the sum

$$\sum_{j=1}^3 \delta_{2,j} a_j$$

and evaluate it.

- c. Repeat (b) for

$$\sum_{j=1}^3 \delta_{1,j} a_j \quad \text{and} \quad \sum_{j=1}^3 \delta_{3,j} a_j.$$

- d. Based on your results, simplify the general expression

$$\sum_{j=1}^n \delta_{i,j} a_j.$$

- e. In the expression in (d), which index is a dummy index and which is a free index? What values can the free index take?

Exercise 3.12 (The Levi–Civita symbol: basic exercise). The **Levi–Civita symbol** ε_{ijk} is defined in the introduction to this chapter.

- Evaluate ε_{123} , ε_{132} , ε_{231} , ε_{321} , and ε_{223} .
- Find all triples (i, j, k) for which $\varepsilon_{ijk} = +1$.
- Find all triples (i, j, k) for which $\varepsilon_{ijk} = -1$.
- What happens to ε_{ijk} if two indices are interchanged? Illustrate with two examples.
- Let a_1 , a_2 , and a_3 be the components of a vector. Write out the sum

$$\sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{1jk} a_j b_k$$

and remove all terms that vanish because of the Levi–Civita symbol.

Exercise 3.13 (A molecular orbital in an atomic-orbital basis). In quantum chemistry, a molecular orbital $\psi_p(\mathbf{r})$ is commonly expanded in a finite set of atomic-orbital basis functions $\chi_\mu(\mathbf{r})$,

$$\psi_p(\mathbf{r}) = \sum_{\mu=1}^K C_{\mu p} \chi_\mu(\mathbf{r}).$$

Here, μ labels the basis functions, while p labels the molecular orbital.

Consider three basis functions and the coefficient matrix

$$C = \begin{bmatrix} 0.7 & 0.5 \\ 0.7 & -0.5 \\ 0.1 & 0.7 \end{bmatrix}.$$

- In the expression for $\psi_p(\mathbf{r})$, identify the dummy index and the free index.
- Write $\psi_1(\mathbf{r})$ explicitly as a linear combination of $\chi_1(\mathbf{r})$, $\chi_2(\mathbf{r})$, and $\chi_3(\mathbf{r})$, using the matrix elements of C .
- Do the same for $\psi_2(\mathbf{r})$.

d. Suppose that at a particular point $\mathbf{r}_0$,

$$\chi_1(\mathbf{r}_0) = 1.0, \quad \chi_2(\mathbf{r}_0) = 0.5, \quad \chi_3(\mathbf{r}_0) = 0.2.$$

Calculate $\psi_1(\mathbf{r}_0)$ and $\psi_2(\mathbf{r}_0)$.

- e. Explain why the summation index μ could be replaced by another symbol, such as i , without changing the meaning of the expression.

4 Combinatorics

Combinatorics is the part of mathematics concerned with *counting* in a broad sense, for example counting the number of ways to arrange a set of objects or the number of combinations that can be made from a set of objects. It is a fundamental part of discrete mathematics and has applications in many areas of mathematics, computer science, chemistry, and physics.

Exercises: Factorials

The *factorial* of a non-negative integer n is defined as the product of all positive integers less than or equal to n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdots 3 \cdot 2 \cdot 1, \quad n \geq 1.$$

It is also defined that $0! = 1$. The factorial appears in many mathematical contexts, including series expansions, and of course combinatorics. Its basic use is the following:

Factorial and counting

Given a set of n distinct objects, the number of ways to arrange them in a sequence (i.e., the number of permutations) is $n!$.

In one exercise, you will need the following approximation for large n , known as Stirling's approximation:

Stirling's approximation

For large n , the factorial $n!$ can be approximated as

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

In calculations, especially numerical calculations, it is often convenient to use the logarithm of the factorial. Using Stirling's approximation, we have

$$\log(n!) \approx \frac{1}{2} \log(2\pi n) + n \log(n) - n.$$

Exercise 4.1 (Basic factorials). Compute the factorials $0!$, $1!$, $2!$, $3!$, $4!$, and $5!$.

Exercise 4.2 (Lunch line). 42 people stand in a lunch line. How many different lunch line arrangements with 42 people are possible?

Exercise 4.3 (Chemist's shelf). Compute the number of arrangements for the following scenarios:

- A chemist has 10 different chemical samples. How many ways can the chemist arrange the samples on a shelf?
- A chemist has 10 different chemical samples. 4 of the samples are liquids, and the other 6 are solids. How many ways can the chemist arrange the samples on a shelf if all the liquids must be together?

Exercise 4.4 (Stirling's approximation). We study Stirling's approximation for the factorial.

- Use Stirling's approximation to estimate $5!$. Compare your result with the exact result

$$5! = 120.$$

- Use Stirling's approximation to estimate $10!$. Compare again with the exact number

$$10! = 3\,628\,800.$$

- The relative error of an approximation A to an exact value x is

$$\frac{|A - x|}{x}.$$

Calculate the relative error of Stirling's approximation for $5!$ and $10!$. For which value of n is the approximation better?

- d. Use Stirling's approximation to estimate $50!$. The exact value is approximately

$$50! \approx 3.041 \times 10^{64}.$$

- e. Based on your results above, what happens to the accuracy of Stirling's approximation as n increases?

Exercises: Binomial coefficients and Pascal's triangle

The *binomial coefficient* $\binom{n}{k}$ is defined as the number of ways to choose k elements from a set of n elements, without regard to the order of selection. It is given by the formula:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

Pascal's triangle is a triangular array of numbers where each number is the sum of the two numbers directly above it:

$$1 \tag{4.1}$$

$$1 \quad 1 \tag{4.2}$$

$$1 \quad 2 \quad 1 \tag{4.3}$$

$$1 \quad 3 \quad 3 \quad 1 \tag{4.4}$$

$$1 \quad 4 \quad 6 \quad 4 \quad 1 \tag{4.5}$$

$$\vdots \tag{4.6}$$

Fact: The n -th row of Pascal's triangle contains the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 4.5 (Basic binomial coefficients).

- Compute the binomial coefficients $\binom{5}{2}$, $\binom{6}{3}$, and $\binom{7}{4}$.
- In Norwegian Lotto, 7 numbers are drawn from a set of 34 numbers. How many different combinations of 7 numbers can be drawn?

Exercise 4.6 (Pascal's triangle). Compute the first 6 rows of Pascal's triangle. Verify that the numbers in the n -th row correspond to the binomial coefficients $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$.

Exercise 4.7 (Pascal's identity). The binomial coefficients satisfy **Pascal's identity**,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

- Explain why this relation is equivalent to the construction of Pascal's triangle, where each number is the sum of the two numbers directly above it.
- Write each binomial coefficient on the right-hand side in terms of factorials.
- Put the two terms over a common denominator and simplify the expression.
- Show that the resulting expression is

$$\frac{n!}{k!(n-k)!},$$

and hence prove Pascal's identity.

- Give a combinatorial explanation of Pascal's identity. *Hint:* Consider choosing k objects from a set of n objects. Pick one particular object and separate the choices into two cases: those that contain this object and those that do not.

Exercise 4.8 (Pascal's triangle and the Sierpinski triangle). Write a computer program to generate the first 2^n rows of Pascal's triangle, where $n = 1, 2, \dots$. Create a $2^n \times 2^n$ lower triangular matrix T , where $T_{k,n} = 0$ if $\binom{n}{k}$ is even, and $T_{k,n} = 1$ if the coefficient is odd, for $k \leq n$ (indexing from zero). Visualize the matrix. What pattern do you observe? This pattern is known as the Sierpinski triangle.

If you use Python and sympy, you can generate binomial factors with the function `sympy.binomial(n, k)`. Check for parity using the modulo operator `%`. For example, `sympy.binomial(n, k) % 2` will return 0 for even numbers and 1 for odd numbers.

Exercise 4.9 (The binomial theorem). The **binomial theorem** states that

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

- Use ordinary multiplication to expand $(x + y)^2$.
- Use the binomial theorem to expand $(x + y)^3$.
- Use the binomial theorem to expand $(x + y)^4$.
- Compare the coefficients in your expansion of $(x + y)^4$ with the row

$$1, \quad 4, \quad 6, \quad 4, \quad 1$$

of Pascal's triangle.

- Use the binomial theorem to expand $(1 + x)^5$.
- Without carrying out any multiplication, find the coefficient of x^3 in $(1 + x)^7$.

Exercise 4.10 (Coefficients in binomial expansions). Consider the binomial expansion

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

- Write out the complete expansion of $(a + b)^6$.

- b. Find the coefficient of a^4b^2 in $(a + b)^6$ without writing out the full expansion.
- c. Find the coefficient of x^4 in $(2 + x)^7$.
- d. Find the coefficient of x^3 in $(1 - 2x)^5$.
- e. Suppose that each of n electrons can independently be assigned either label α or label β . Explain why the number of assignments containing exactly k electrons labelled β is $\binom{n}{k}$.
- f. Use the binomial theorem with $a = b = 1$ to show that

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Interpret this result in terms of the number of possible assignments of the two labels α and β to n electrons.

5 Basic mathematical proofs

Mathematics is a way of establishing, with certainty, that a statement follows from a set of assumptions. A *proof* is a sequence of logical steps that makes this connection explicit. In quantum chemistry, proofs are useful even when the final goal is numerical. They help us understand why an algorithm works, which properties of matrices and operators are guaranteed, and which conclusions depend on particular assumptions.

This chapter introduces a small collection of proof strategies that occur repeatedly in mathematics and mathematical physics. We consider direct proof, disproof by counterexample, proof by contrapositive, proof by contradiction, the pigeonhole principle, and proof by induction. The beginner exercises are deliberately elementary: their purpose is to practise the logical structure of each method without making the algebra difficult. The intermediate exercises apply the same ideas to matrices, sums, and simple settings related to quantum mechanics and quantum chemistry.

Some basic proof strategies

Direct proof and counterexamples

In a **direct proof**, we start from the assumptions and use definitions, algebra, and previously established facts until we reach the desired conclusion. For example, to prove that the sum of two even integers is even, write the integers as $2m$ and $2n$. Their sum is $2(m + n)$ and is therefore even.

A universal statement such as “for every x , property $P(x)$ holds” can be disproved by finding a single **counterexample**: one value of x for which $P(x)$ is false. Counterexamples are often the quickest way to test whether a proposed theorem can possibly be correct.

Proof by contrapositive

A statement of the form

$$P \implies Q$$

is logically equivalent to its **contrapositive**

$$\neg Q \implies \neg P.$$

Sometimes the contrapositive is much easier to prove than the original statement. Notice that the contrapositive is not the same as the converse $Q \implies P$.

Proof by contradiction

In a **proof by contradiction**, we assume that the statement we want to prove is false. We then show that this assumption leads to an impossibility: for example, a statement that contradicts an assumption or a known fact. We conclude that the original statement must be true.

This method is particularly natural for statements asserting that something cannot exist. It also occurs frequently in proofs about linear algebra and quantum mechanics.

The pigeonhole principle

The **pigeonhole principle** says that if more than n objects are placed into n boxes, at least one box must contain at least two objects. The idea is elementary, but surprisingly powerful. More generally, if N objects are distributed among n boxes, at least one box contains at least $\lceil N/n \rceil$ objects.

Proof by induction

Mathematical induction is used to prove statements indexed by an integer, usually $n = 1, 2, 3, \dots$. A proof has two main steps:

1. **Base case:** prove the statement for the first value of n .
2. **Induction step:** assume the statement is true for some n (the **induction hypothesis**) and use this assumption to prove that it is true for $n + 1$.

If both steps succeed, the statement follows for every integer covered by the induction.

Exercises

Exercise 5.1 (A first direct proof). An integer n is *even* if it can be written as $n = 2k$ for some integer k .

- a. Let m and n be even integers. Write m and n using the definition of an even integer.
- b. Use your expressions to show that $m + n$ can be written as 2 times an integer.
- c. Conclude that the sum of two even integers is even.
- d. Which part of your argument would fail if one of the two integers were odd?

Exercise 5.2 (Learning to use counterexamples). For each statement below, decide whether it is true. If it is false, disprove it by giving one counterexample.

- a. For every real number x , $x^2 \geq 0$.
- b. For every real number x , $x^2 > x$.
- c. For all real numbers x and y , $(x + y)^2 = x^2 + y^2$.
- d. Every diagonal 2×2 matrix is invertible.
- e. If A is a real matrix, then every entry of $A^T A$ is positive.

Exercise 5.3 (A first proof by contrapositive). Consider the statement: *If n^2 is even, then n is even*, where n is an integer.

- Write down the contrapositive of the statement.
- An odd integer can be written as $n = 2k + 1$. Expand $(2k + 1)^2$.
- Show that the square of an odd integer is odd.
- Use the contrapositive to conclude that if n^2 is even, then n is even.

Exercise 5.4 (A first proof by contradiction). Prove that there is no largest integer.

- Suppose, for contradiction, that there is a largest integer. Call it N .
- Consider the integer $N + 1$. How does it compare with N ?
- Explain why this contradicts the assumption that N is the largest integer.
- State the conclusion.

Exercise 5.5 (The pigeonhole principle).

- A tutorial group contains 13 students. Show that at least two students were born in the same month.
- Identify explicitly the “pigeons” and the “pigeonholes” in part a.
- A basis set contains 21 basis functions, each assigned one of four angular-momentum labels: s , p , d , or f . Show that at least six basis functions have the same label.
- Which version of the pigeonhole principle do you need for part c?

Exercise 5.6 (Your first induction proof). Prove by induction that

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

for every positive integer n .

- Check the formula for $n = 1$.
- Assume that the formula is true for some positive integer n . Write down this induction hypothesis explicitly.
- Add $n + 1$ to both sides of the induction hypothesis.
- Simplify the right-hand side and show that you obtain $\frac{(n+1)(n+2)}{2}$.

- e. Explain why the base case and induction step prove the formula for every positive integer n .

Exercise 5.7 (Direct proof with a matrix). Let A be a real $m \times n$ matrix and $x \in \mathbb{R}^n$. Prove directly that

$$x^T A^T A x \geq 0.$$

- Introduce the vector $y = Ax$ and rewrite the expression in terms of y .
- Write $y^T y$ as a sum over components.
- Complete the proof.
- Show that $A^T A$ is symmetric.
- Explain why these results show that $A^T A$ is positive semidefinite.

Exercise 5.8 (Invertibility by contrapositive). Let A and B be square matrices of the same dimension. Prove the statement

If AB is invertible, then both A and B are invertible.

Use a contrapositive argument for each factor.

- Suppose B is not invertible. Show that there is a nonzero vector x such that $Bx = 0$.
- Show that $ABx = 0$ and conclude that AB cannot be invertible.
- Suppose instead that A is not invertible. You may use the fact that then there exists a nonzero row vector $y^\dagger$ such that $y^\dagger A = 0$. Show that $y^\dagger AB = 0$.
- Combine the two arguments to prove the original statement.

Exercise 5.9 (Orthogonal eigenvectors from a contradiction). Let A be a Hermitian matrix, and let

$$Au = \lambda u, \quad Av = \mu v,$$

where $\lambda \neq \mu$.

- Using $Au = \lambda u$, show that $v^\dagger Au = \lambda v^\dagger u$.

- b. Using Hermiticity and $Av = \mu v$, show that the same quantity is $v^\dagger Au = \mu v^\dagger u$.
- c. Suppose, for contradiction, that $v^\dagger u \neq 0$. Show that this would imply $\lambda = \mu$.
- d. Conclude that eigenvectors of a Hermitian matrix belonging to distinct eigenvalues are orthogonal.
- e. Why is this result useful when eigenvectors represent quantum states or molecular orbitals?

Exercise 5.10 (Pigeonholes and orbital occupations). Consider N electrons distributed among K spatial orbitals. Assume that each spatial orbital can contain at most two electrons.

- a. Use the pigeonhole principle to show that if $N > K$, at least one spatial orbital must be doubly occupied or the assumptions are inconsistent.
- b. If $N = 11$ electrons occupy $K = 6$ spatial orbitals, what is the minimum possible number of doubly occupied orbitals?
- c. Generalize part b: for $K \leq N \leq 2K$, prove that at least $N - K$ orbitals must be doubly occupied.
- d. Explain why the result is a counting statement and does not depend on the energies of the orbitals.

Exercise 5.11 (Induction and a geometric sum). Let r be a number with $r \neq 1$. Prove by induction that

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$

for every integer $n \geq 0$.

- a. Verify the base case $n = 0$.
- b. Write down the induction hypothesis.
- c. Add r^{n+1} and simplify the resulting expression.
- d. Complete the induction proof.
- e. For $|r| < 1$, what does the formula suggest as $n \rightarrow \infty$?

Exercise 5.12 (A finite-dimensional quantum-mechanical impossibility). In quantum mechanics, position and momentum satisfy the canonical commutation relation

$$[\hat{x}, \hat{p}] = i\hbar I.$$

Here $[A, B] = AB - BA$. Show that **no pair of finite-dimensional matrices** X and P can satisfy

$$[X, P] = i\hbar I$$

when $\hbar \neq 0$.

- Assume, for contradiction, that $n \times n$ matrices X and P satisfying this equation exist.
- Prove that $\text{tr}(XP) = \text{tr}(PX)$.
- Hence compute $\text{tr}([X, P])$.
- On the other hand, compute $\text{tr}(i\hbar I)$.
- Derive a contradiction.
- Explain what this tells us about representing the exact position and momentum operators by finite matrices. Does it prevent us from using finite matrix approximations in practical quantum chemistry?

Solutions

Solution for Exercise 5.1

- Since m and n are even, there are integers p and q such that $m = 2p$ and $n = 2q$.
- Their sum is

$$m + n = 2p + 2q = 2(p + q).$$

Since $p + q$ is an integer, $m + n$ is twice an integer.

- c. Therefore $m + n$ is even.
- d. If, for example, n were odd, we could not write it as $2q$ with q an integer. The first step of the proof would therefore no longer apply.

Solution for Exercise 5.2

- a. True. The square of every real number is nonnegative.
- b. False. For example, $x = 1$ gives $x^2 = x$, not $x^2 > x$. Also $x = 1/2$ gives $x^2 < x$.
- c. False. Take $x = y = 1$. Then $(x + y)^2 = 4$, whereas $x^2 + y^2 = 2$.
- d. False. For example,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

is diagonal but not invertible.

- e. False. Take

$$A = \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Then

$$A^T A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

which has negative off-diagonal entries. What is true is that $A^T A$ is positive semidefinite; this does not mean that every matrix element is positive.

Solution for Exercise 5.3

- a. The contrapositive is: *If n is odd, then n^2 is odd.*
- b. Write $n = 2k + 1$. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

- c. Since $2k^2 + 2k$ is an integer, n^2 has the form $2m + 1$ and is therefore odd.
- d. We have proved the contrapositive. Since a statement and its contrapositive are logically equivalent, it follows that if n^2 is even, then n is even.

Solution for Exercise 5.4

- a. Assume that a largest integer N exists.
- b. The number $N + 1$ is also an integer, and $N + 1 > N$.
- c. This contradicts the assumption that no integer is larger than N .
- d. Therefore there is no largest integer.

Solution for Exercise 5.5

- a. There are 13 students but only 12 months. If every month contained at most one birthday, there could be at most 12 students. Therefore at least two students were born in the same month.
- b. The students are the pigeons and the 12 months are the pigeonholes.
- c. There are 21 basis functions and four labels. Since

$$\left\lceil \frac{21}{4} \right\rceil = 6,$$

at least one label must occur at least six times.

- d. We use the generalized pigeonhole principle: distributing N objects among n boxes forces some box to contain at least $\lceil N/n \rceil$ objects.

Solution for Exercise 5.6

- a. For $n = 1$,

$$1 = \frac{1(1+1)}{2} = 1,$$

so the base case is true.

- b. Assume that for some positive integer n ,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

- c. Adding $n + 1$ gives

$$1 + 2 + \cdots + n + (n + 1) = \frac{n(n+1)}{2} + (n + 1).$$

- d. Factor and simplify:

$$\frac{n(n+1)}{2} + (n+1) = (n+1)\left(\frac{n}{2} + 1\right) = \frac{(n+1)(n+2)}{2}.$$

This is precisely the claimed formula with n replaced by $n + 1$.

- e. The statement is true for $n = 1$, and whenever it is true for n it is true for $n + 1$. By induction, it is true for every positive integer n .

Solution for Exercise 5.7

- a. Let $y = Ax$. Then

$$x^T A^T A x = (Ax)^T (Ax) = y^T y.$$

- b. In components,

$$y^T y = \sum_i y_i^2.$$

c. Every y_i^2 is nonnegative, so

$$x^T A^T A x = \sum_i y_i^2 \geq 0.$$

d. Using $(BC)^T = C^T B^T$,

$$(A^T A)^T = A^T (A^T)^T = A^T A.$$

Thus $A^T A$ is symmetric.

e. A real symmetric matrix M is positive semidefinite if $x^T M x \geq 0$ for every x . We have proved exactly this property for $M = A^T A$.

Solution for Exercise 5.8

a. If B is not invertible, its null space contains a nonzero vector x , so $Bx = 0$.

b. Then

$$ABx = A(Bx) = A0 = 0.$$

Thus AB has a nonzero vector in its null space and cannot be invertible. We have proved: if B is not invertible, then AB is not invertible. This is the contrapositive of “if AB is invertible, then B is invertible.”

c. If A is not invertible, choose a nonzero $y^\dagger$ with $y^\dagger A = 0$. Then

$$y^\dagger AB = (y^\dagger A)B = 0.$$

Therefore AB cannot be invertible. This proves the corresponding contrapositive for A .

d. If AB is invertible, neither A nor B can be noninvertible. Hence both A and B are invertible.

Solution for Exercise 5.9

- a. Since $Au = \lambda u$,

$$v^\dagger Au = \lambda v^\dagger u.$$

- b. Hermiticity gives $A^\dagger = A$. From $Av = \mu v$, and since the eigenvalues of a Hermitian matrix are real,

$$v^\dagger A = \mu v^\dagger.$$

Therefore

$$v^\dagger Au = \mu v^\dagger u.$$

- c. Comparing the two expressions,

$$(\lambda - \mu)v^\dagger u = 0.$$

Suppose $v^\dagger u \neq 0$. We could then divide by $v^\dagger u$ and obtain $\lambda = \mu$, contradicting the assumption $\lambda \neq \mu$.

- d. Therefore $v^\dagger u = 0$: the eigenvectors are orthogonal.
- e. Orthogonality allows distinct eigenstates or molecular orbitals of a Hermitian operator to be used as mutually orthogonal basis vectors. It is one of the central structural properties behind spectral expansions in quantum mechanics.

Solution for Exercise 5.10

- a. If $N > K$ and every orbital contained at most one electron, at most K electrons could be accommodated. Therefore at least one orbital must contain two electrons, unless the assumed occupation rules cannot be satisfied.

- b. With 11 electrons in 6 orbitals, first place one electron in each orbital. This accounts for 6 electrons. The remaining 5 electrons must enter already occupied orbitals, so at least 5 orbitals are doubly occupied. This arrangement is possible, so the minimum is 5.
- c. Place at most one electron in each of the K orbitals. This accommodates K electrons. The remaining $N - K$ electrons must each be placed in an already occupied orbital. Since no orbital can hold more than two electrons, these require $N - K$ distinct orbitals. Hence at least $N - K$ orbitals are doubly occupied.
- d. The argument uses only the numbers of electrons, orbitals, and allowed occupants. Orbital energies do not enter.

Solution for Exercise 5.11

- a. For $n = 0$, the left-hand side is 1, while

$$\frac{1 - r^{0+1}}{1 - r} = \frac{1 - r}{1 - r} = 1.$$

- b. Assume that

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

- c. Add r^{n+1} :

$$1 + r + \cdots + r^n + r^{n+1} = \frac{1 - r^{n+1}}{1 - r} + r^{n+1}.$$

Putting the right-hand side over a common denominator gives

$$\frac{1 - r^{n+1} + r^{n+1}(1 - r)}{1 - r} = \frac{1 - r^{n+2}}{1 - r}.$$

- d. This is the required formula with n replaced by $n + 1$. Together with the base case, induction proves the result for all $n \geq 0$.
- e. If $|r| < 1$, then $r^{n+1} \rightarrow 0$, suggesting the infinite geometric series

$$1 + r + r^2 + \cdots = \frac{1}{1-r}.$$

Solution for Exercise 5.12

- a. Assume that finite $n \times n$ matrices X and P exist such that

$$[X, P] = XP - PX = i\hbar I.$$

- b. For finite matrices, the trace is cyclic:

$$\text{tr}(XP) = \text{tr}(PX).$$

For example, in components,

$$\text{tr}(XP) = \sum_{ij} X_{ij} P_{ji} = \sum_{ji} P_{ji} X_{ij} = \text{tr}(PX).$$

- c. It follows that

$$\text{tr}([X, P]) = \text{tr}(XP) - \text{tr}(PX) = 0.$$

- d. But the assumed commutation relation gives

$$\text{tr}([X, P]) = \text{tr}(i\hbar I) = i\hbar \text{tr}(I) = i\hbar n.$$

- e. Since $n > 0$ and $\hbar \neq 0$, we have $i\hbar n \neq 0$. Thus the same trace would have to be both 0 and $i\hbar n$, which is a contradiction. Therefore no finite-dimensional matrices can satisfy the canonical commutation relation exactly.
- f. Exact position and momentum operators therefore cannot both be represented by finite-dimensional matrices while preserving the exact canonical commutator. In quantum mechanics they act naturally on infinite-dimensional spaces. This does **not** prevent finite matrix approximations: quantum chemistry routinely projects operators onto finite basis sets. Such representations can be extremely accurate for the quantities of interest, but they cannot preserve every exact infinite-dimensional operator identity.

Part III

Linear algebra

6 Vectors and matrices

The vector space $\mathbb{R}^n$ consists of all column vectors

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad x_i \in \mathbb{R},$$

with the usual vector addition and scalar multiplication. Similarly, $\mathbb{C}^n$ consists of column vectors with complex components. More generally, a vector space V over a field $\mathbb{F}$ is a set equipped with vector addition and scalar multiplication such that, for all $x, y, z \in V$ and $\alpha, \beta \in \mathbb{F}$,

1. **Commutativity of addition:** $x + y = y + x$.
2. **Associativity of addition:** $x + (y + z) = (x + y) + z$.
3. **Additive identity:** There exists a zero vector $0 \in V$ such that $x + 0 = x$.
4. **Additive inverse:** For every $x \in V$, there exists $-x \in V$ such that $x + (-x) = 0$.
5. **Distributivity over vector addition:** $\alpha(x + y) = \alpha x + \alpha y$.
6. **Distributivity over scalar addition:** $(\alpha + \beta)x = \alpha x + \beta x$.
7. **Associativity of scalar multiplication:** $(\alpha\beta)x = \alpha(\beta x)$.
8. **Scalar identity:** $1x = x$.

For a matrix $A \in \mathbb{R}^{m \times n}$ and a column vector $x \in \mathbb{R}^n$, the matrix-vector product $y = Ax \in \mathbb{R}^m$ is defined componentwise by

$$y_i = \sum_{j=1}^n A_{ij}x_j.$$

For matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, the matrix product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}.$$

The same definitions of matrix-vector and matrix-matrix multiplication apply over $\mathbb{C}$.

Exercises: Basic vector and matrix calculations

Exercise 6.1 (Basic vector operations). Consider the vectors

$$\mathbf{x} = (1, 2, -1), \quad \mathbf{y} = (3, -1, 2), \quad \mathbf{z} = (0, 2, 1)$$

in the vector space $\mathbb{R}^3$. (We here write the vectors as an ordered tuple instead of a column.)

- Compute $\mathbf{x} + \mathbf{y}$.
- Compute $\mathbf{x} - \mathbf{y}$.
- Compute $2\mathbf{x}$ and $-3\mathbf{y}$.
- Compute the linear combination $2\mathbf{x} + 3\mathbf{y}$.
- Compute $\mathbf{x} + 2\mathbf{y} - \mathbf{z}$.
- Find numbers α and β such that $\alpha\mathbf{x} + \beta\mathbf{y} = (7, 0, 3)$.
- Verify by direct calculation that $2(\mathbf{x} + \mathbf{y}) = 2\mathbf{x} + 2\mathbf{y}$. Which vector-space axiom does this illustrate?
- Is $(1, 2)$ an element of this vector space? Is $(0, 0, 0)$ an element? Explain briefly.

Exercise 6.2 (Basic matrix-vector products). Consider the matrices and vectors

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}.$$

- Compute $A\mathbf{x}$.
- Compute $B\mathbf{y}$.
- Compute $A(2\mathbf{x})$ and $2(A\mathbf{x})$. Verify that they are equal.
- Can the products $A\mathbf{y}$ and $B\mathbf{x}$ be formed? Explain your answer in terms of the dimensions of the matrices and vectors.

Exercise 6.3 (Matrix-matrix products). Consider the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ -1 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 2 \end{pmatrix}.$$

- Compute AB .
- Compute BA . Is $AB = BA$?
- Compute AC . What are the dimensions of the resulting matrix?
- Can the product CA be formed? Can the product BC be formed? Explain your answers using the dimensions of the matrices.

Exercise 6.4 (Linear independence and basis). We work in the space $V = \mathbb{R}^2$, plane vectors. We define two vectors,

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

- Show by elementary means that the two vectors are linearly independent, that is: If $x_1\mathbf{b}_1 + x_2\mathbf{b}_2 = \mathbf{0}$, then $x_1 = x_2 = 0$ must hold.
- The vectors form a basis since they are linearly independent. Explain in your own words what a basis is.
- Make a 2D drawing or plot of them as arrows from the origin.

Exercise 6.5 (Gaussian elimination and vector decomposition). Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Using Gaussian elimination, compute the coefficients x_1 and x_2 such that

$$\mathbf{v} = x_1\mathbf{b}_1 + x_2\mathbf{b}_2.$$

Illustrate this decomposition in the graph from the previous exercise, using the parallelogram rule: The sum of two vectors is obtained by placing the origin of one at the end of the second, or vice versa.

Exercise 6.6 (Office coordinates). A person walks from home. First, they walk to the subway station. This is 1 km north and 500 m east as the crow flies. The subway takes the person to the city centre, which 3 km east and 2 km south of the subway station, as the crow flies. The person then walks 200 m south and 400 m west to their office.

What are the coordinates of the office, relative to the person's home?

What is the distance to home, as the crow flies?

Exercise 6.7 (Boat sailing on the ocean). A boat is sailing on the ocean, at a constant speed 20 km/h relative to the water surface, in the northeast direction. However, the water surface is moving too, at a constant velocity. After two hours, the boat is located 20 km north and 10 km east of the starting point. What is the speed and direction of ocean drift?

Exercise 6.8 (An exercise by Robert Beezer). A three-digit number has two properties. The tens-digit and the ones-digit add up to 5. If the number is written with the digits in the reverse order, and then subtracted from the original number, the result is 792. Use a system of equations to find all of the three-digit numbers with these properties.

Exercises: More vectors and matrices

Exercise 6.9 (Compute the action of a matrix on vectors). Let A be the $n \times n$ matrix defined by $A_{i,i+1} = i^2$ for $1 \leq i < n$, and zero otherwise. Compute the action of A and A^T on the standard basis for $\mathbb{R}^n$. Compute the action of A and A^T on an arbitrary vector $\mathbf{x} \in \mathbb{R}^n$.

Exercise 6.10 (Matrix-matrix products, rows and columns). A useful way to understand matrix-matrix multiplication is to think of a matrix as a collection of vectors. This gives two complementary interpretations of the product AB :

- A acts separately on each **column** of B .
- B acts separately on each **row** of A , when the rows are regarded as row vectors.

These viewpoints are useful both for understanding matrix multiplication and for computations. For example, if the columns of B represent several vectors, then AB applies the linear transformation A to *all of these vectors at once*.

- a. Consider

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \end{bmatrix}.$$

Write B as a collection of its three column vectors,

$$B = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3].$$

Write down $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ explicitly.

- b. Compute $A\mathbf{b}_1$, $A\mathbf{b}_2$, and $A\mathbf{b}_3$ separately. Put the three resulting vectors next to each other as the columns of a new matrix:

$$[A\mathbf{b}_1, A\mathbf{b}_2, A\mathbf{b}_3].$$

- c. Compute AB directly using ordinary matrix multiplication. Compare your answer with part b.
- d. Now consider the general case. Let

$$A \in M(n, m, \mathbb{F}), \quad B \in M(m, o, \mathbb{F}),$$

and write the columns of B as

$$B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_o].$$

Explain why

$$AB = [A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_o].$$

In other words, explain why the j th column of AB is $A\mathbf{b}_j$.

- e. There is a similar interpretation in terms of rows. Write A using its row vectors,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}.$$

Show that

$$AB = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \vdots \\ \mathbf{a}_n^T B \end{bmatrix}.$$

Thus, the i th row of AB is the row vector $\mathbf{a}_i^T B$.

- f. Suppose the columns of B are several vectors that we want to transform using the same linear transformation A . Explain why the identity from part d means that we can collect all the vectors into one matrix B and perform a single matrix-matrix multiplication AB .

Exercise 6.11. Show that each column of $C = AB$ is a linear combination of the columns of A with coefficients determined by B ,

$$\mathbf{c}_i = \sum_j \mathbf{a}_j B_{ji}.$$

Show a similar result for the rows of C .

7 Vectors in 3D space

A vector in three-dimensional Euclidean space is specified by three components. We write

$$\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3.$$

Geometrically, $\mathbf{x}$ can be regarded as a directed displacement in space. Its Euclidean length is

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + x_3^2}.$$

Vectors can be added componentwise and multiplied by scalars, exactly as for vectors in $\mathbb{R}^n$.

Two particularly useful operations are available for vectors in three dimensions. The **dot product** of $\mathbf{x}$ and $\mathbf{y}$ is the scalar

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3 = \sum_{i=1}^3 x_i y_i.$$

It contains geometric information about the relative directions of the vectors:

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta,$$

where θ is the angle between them. In particular, two nonzero vectors are perpendicular if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

The **cross product** $\mathbf{x} \times \mathbf{y}$ is instead a vector. It is perpendicular to both $\mathbf{x}$ and $\mathbf{y}$, with direction determined by the right-hand rule, and its length is

$$\|\mathbf{x} \times \mathbf{y}\| = \|\mathbf{x}\| \|\mathbf{y}\| \sin \theta.$$

In components,

$$\mathbf{x} \times \mathbf{y} = (x_2 y_3 - x_3 y_2, x_3 y_1 - x_1 y_3, x_1 y_2 - x_2 y_1).$$

For manipulating vector expressions, it is often more convenient to work with components rather than with this explicit formula. The **Kronecker delta**

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

acts as an identity for indices, while the **Levi-Civita symbol** ε_{ijk} encodes the orientation of three-dimensional space. It is +1 for even permutations of (1, 2, 3), -1 for odd permutations, and zero whenever two indices are equal. With this notation the cross product has the compact representation

$$(\mathbf{x} \times \mathbf{y})_i = \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} x_j y_k.$$

This index notation is especially useful for deriving identities involving several dot and cross products: complicated-looking vector expressions can be reduced to systematic manipulations of sums, Kronecker deltas, and Levi-Civita symbols.

Exercises

Exercise 7.1 (The vector triple product and the Levi-Civita symbol). It is recommended that you also look at Exercise 3.12 before doing this exercise, and also Exercise 3.11 on the Kronecker delta.

Recall that the i th Cartesian component of the cross product of two vectors $\mathbf{a}$ and $\mathbf{b}$ can be written as

$$(\mathbf{a} \times \mathbf{b})_i = \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} a_j b_k,$$

where ε_{ijk} is the Levi-Civita symbol.

You may also use the contraction identity

$$\sum_{i=1}^3 \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km},$$

where δ_{ij} is the Kronecker delta.

- a. Write the i th component of $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ using two Levi-Civita symbols. Be careful to use different dummy indices for the two cross products.
- b. Use the cyclic property

$$\varepsilon_{ijk} = \varepsilon_{kij}$$

and the contraction identity above to show that

$$\sum_{k=1}^3 \varepsilon_{ijk} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}.$$

- c. Use the Kronecker deltas to carry out the remaining sums and show that

$$[\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_i = b_i \sum_{j=1}^3 a_j c_j - c_i \sum_{j=1}^3 a_j b_j.$$

- d. Hence prove the vector triple-product identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}).$$

This identity is sometimes remembered as “BAC minus CAB”.

8 Hybridization and orbital geometry

Hybrid orbitals provide a useful example of how vectors, matrices, geometry, and chemistry fit together. In this chapter we use the familiar atomic-orbital labels s , p_x , p_y , and p_z as an orthonormal basis for a four-dimensional vector space. A hybrid orbital is then simply a linear combination of these basis functions,

$$h = as + b_x p_x + b_y p_y + b_z p_z.$$

It is convenient to represent this orbital by its coefficient vector

$$\mathbf{h} = \begin{bmatrix} a \\ b_x \\ b_y \\ b_z \end{bmatrix}.$$

The coefficients multiplying p_x , p_y , and p_z also have a geometrical interpretation. If

$$\mathbf{n} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}, \quad \|\mathbf{n}\| = 1,$$

is a direction in ordinary three-dimensional space, then the linear combination

$$p_{\mathbf{n}} = n_x p_x + n_y p_y + n_z p_z$$

is a p orbital oriented along $\mathbf{n}$. A hybrid orbital pointing in this direction may therefore be written

$$h = as + b p_{\mathbf{n}} = as + b(n_x p_x + n_y p_y + n_z p_z).$$

In this exercise we use normalization, orthogonality, dot products, and matrix transformations to construct sp , sp^2 , and sp^3 hybrid orbitals. The emphasis is on the linear-algebraic structure of hybridization. The orbitals used for visualization are simple analytic model functions intended to show their shapes and orientations, rather than quantitatively accurate atomic orbitals.

Exercises

Exercise 8.1 (Hybrid orbitals from vectors and matrices). Assume that the four orbitals

$$s, \quad p_x, \quad p_y, \quad p_z$$

form an orthonormal basis.

A hybrid orbital directed along a unit vector

$$\mathbf{n} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

will be written as

$$h = as + b(n_x p_x + n_y p_y + n_z p_z).$$

- a. Consider two sp hybrids directed along the positive and negative x axes,

$$\mathbf{n}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{n}_2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}.$$

Assume that the two hybrids have the form

$$h_i = as + b p_{\mathbf{n}_i}.$$

Require both hybrids to be normalized and mutually orthogonal. Determine a and b , and write the two normalized sp hybrids explicitly in terms of s and p_x .

b. Now consider three directions in the xy plane separated by 120° ,

$$\mathbf{n}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{n}_2 = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}, \quad \mathbf{n}_3 = \begin{bmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}.$$

Verify using dot products that

$$\mathbf{n}_i^T \mathbf{n}_j = -\frac{1}{2}, \quad i \neq j.$$

Assume three hybrids of the form

$$h_i = as + b p_{\mathbf{n}_i}.$$

Use normalization and mutual orthogonality to determine a and b . Write the three sp^2 hybrids explicitly as linear combinations of s , p_x , and p_y .

c. Consider the four tetrahedral directions

$$\mathbf{n}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{n}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix},$$

$$\mathbf{n}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{n}_4 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

Verify that every $\mathbf{n}_i$ is a unit vector and that

$$\mathbf{n}_i^T \mathbf{n}_j = -\frac{1}{3}, \quad i \neq j.$$

Use the dot product to calculate the angle between any two tetrahedral directions.

- d. Assume four sp^3 hybrids of the form

$$h_i = as + b p_{\mathbf{n}_i}.$$

Require the hybrids to be normalized and mutually orthogonal. Determine a and b , and show that the four hybrid orbitals can be written

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{bmatrix} = U \begin{bmatrix} s \\ p_x \\ p_y \\ p_z \end{bmatrix},$$

where

$$U = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

- e. Show by direct matrix multiplication that

$$U^T U = I.$$

What does this tell you about the transformation between the original atomic-orbital basis and the hybrid-orbital basis? Find U^{-1} and use it to express s , p_x , p_y , and p_z in terms of the four sp^3 hybrids.

- f. Use Python to visualize the tetrahedral directions as arrows from the origin. Confirm visually that the four vectors point toward the corners of a tetrahedron. A possible starting point is

```
import numpy as np
import plotly.graph_objects as go

directions = np.array([
    [ 1,  1,  1],
    [ 1, -1, -1],
    [-1,  1, -1],
    [-1, -1,  1]
], dtype=float)
```

```

directions /= np.linalg.norm(directions, axis=1)[: , None]

# Build a Plotly figure and add one line and one cone
# for each direction vector.
#
# Hint:
#   go.Scatter3d(...) can be used for the lines.
#   go.Cone(...) can be used for arrowheads.
#
# Make the x, y, and z axis scales equal.

```

- g. Finally, visualize the four sp^3 hybrid orbitals as isosurfaces. For the visualization, use the simple model functions

$$s(\mathbf{r}) = e^{-\alpha r}, \quad p_x(\mathbf{r}) = x e^{-\alpha r}, \quad p_y(\mathbf{r}) = y e^{-\alpha r}, \quad p_z(\mathbf{r}) = z e^{-\alpha r},$$

where $r = \sqrt{x^2 + y^2 + z^2}$ and $\alpha > 0$ controls the radial decay.

Construct the four hybrids by applying the matrix U to these four functions point by point on a three-dimensional grid. Plot positive and negative isosurfaces for each hybrid using Plotly or any tool you like. The following skeleton may be used as a starting point:

```

import numpy as np
import plotly.graph_objects as go

alpha = 1.0
L = 4.0
n = 50

x = np.linspace(-L, L, n)
y = np.linspace(-L, L, n)
z = np.linspace(-L, L, n)
X, Y, Z = np.meshgrid(x, y, z, indexing="ij")

R = np.sqrt(X**2 + Y**2 + Z**2)

s = np.exp(-alpha * R)
px = X * np.exp(-alpha * R)

```

```

py = Y * np.exp(-alpha * R)
pz = Z * np.exp(-alpha * R)

U = 0.5 * np.array([
    [1, 1, 1, 1],
    [1, 1, -1, -1],
    [1, -1, 1, -1],
    [1, -1, -1, 1]
])

basis = np.stack([s, px, py, pz])

# Apply U to the first axis of `basis`.
# One convenient possibility is np.tensordot.
hybrids = ...

# Select one hybrid, for example:
h = hybrids[0]

# Choose an isovalue. It is often convenient to define it
# as a fraction of max(abs(h)).
isovalue = ...

fig = go.Figure()

# Add one isosurface for +isovalue and one for -isovalue.
# Plotly's go.Isosurface expects flattened coordinate and
# value arrays.
#
# fig.add_trace(go.Isosurface(
#     x=X.ravel(),
#     y=Y.ravel(),
#     z=Z.ravel(),
#     value=h.ravel(),
#     isomin=...,
#     isomax=...,
#     surface_count=1,
#     caps=dict(x_show=False, y_show=False, z_show=False)
# ))
#

```

```
# Repeat for the opposite sign.  
#  
# Then repeat for h = hybrids[1], hybrids[2], hybrids[3],  
# either in separate figures or by arranging the results  
# in a layout of your choice.  
  
fig.show()
```

Compare the directions of the large lobes in your plots with the four vectors $\mathbf{n}_1, \dots, \mathbf{n}_4$. Explain how the coefficients of p_x , p_y , and p_z determine the spatial orientation of each hybrid.

9 Eigenvalues and eigenvectors

Eigenvalue problems are among the most important problems in linear algebra, and they occur throughout quantum chemistry. In an eigenvalue problem, we are given a square matrix $A \in \mathbb{C}^{n \times n}$ and seek a number $\lambda \in \mathbb{C}$ and a nonzero vector $\mathbf{x} \in \mathbb{C}^n$ such that

$$A\mathbf{x} = \lambda\mathbf{x}.$$

The number λ is called an *eigenvalue* of A , and the nonzero vector $\mathbf{x}$ is called an *eigenvector* corresponding to λ . In column-vector notation,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

The equation $A\mathbf{x} = \lambda\mathbf{x}$ says that multiplication by A has a particularly simple effect on an eigenvector: it only multiplies the vector by a number. Thus, the direction represented by $\mathbf{x}$ is left unchanged, apart from multiplication by the scalar λ . Notice that $\mathbf{x} = \mathbf{0}$ is excluded from the definition, since $A\mathbf{0} = \lambda\mathbf{0}$ holds for every λ and therefore tells us nothing about the matrix.

The eigenvalue equation can equivalently be written as

$$(A - \lambda I)\mathbf{x} = \mathbf{0},$$

where I is the identity matrix. Since we require $\mathbf{x} \neq \mathbf{0}$, the matrix $A - \lambda I$ must have a nontrivial null space and therefore cannot be invertible. For a finite-dimensional square matrix, this is equivalent to

$$\det(A - \lambda I) = 0.$$

This equation is called the **characteristic equation** or **secular equation** of A . Its solutions are the eigenvalues of A . Once an eigenvalue λ has been found, the corresponding eigenvectors are obtained by solving the homogeneous linear system $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

An especially important situation occurs when A has enough linearly independent eigenvectors to form a basis of $\mathbb{C}^n$. Suppose these eigenvectors are

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n,$$

where the subscript labels the different vectors. Thus,

$$\mathbf{x}_j = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}$$

denotes the j 'th eigenvector, while x_{ij} denotes its i th component. If $\mathbf{x}_j$ has eigenvalue λ_j , then

$$A\mathbf{x}_j = \lambda_j\mathbf{x}_j, \quad j = 1, \dots, n.$$

We can collect these eigenvectors as the columns of a matrix

$$X = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n]$$

and collect their corresponding eigenvalues in the diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

The n separate eigenvalue equations can then be written as the single matrix equation

$$AX = XD.$$

Because the eigenvectors form a basis, X is invertible. Consequently,

$$A = XDX^{-1}.$$

A matrix for which such an eigenvector basis exists is called *diagonalizable*. Diagonalization is useful because the action of a diagonal matrix is particularly simple: it acts independently on each component. In a basis of eigenvectors, the matrix A therefore has its simplest possible form. This idea is fundamental in quantum mechanics, where many important matrices, in particular Hermitian matrices, possess an orthonormal basis of eigenvectors.

In the exercises below, we study how eigenvalues and eigenvectors are found, when a matrix can be diagonalized, and why Hermitian and, more generally, normal matrices have especially useful eigenvalue decompositions.

Exercises

Exercise 9.1 (Checking candidate eigenvectors). Consider the real symmetric matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Recall that a nonzero vector $\mathbf{x}$ is an eigenvector of A if

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some number λ . In other words, after multiplication by A , the vector must point in the same or exactly opposite direction: it may be stretched, compressed, or multiplied by a negative number, but its direction cannot otherwise change.

For each of the following candidate vectors, compute $A\mathbf{x}$ and decide whether $\mathbf{x}$ is an eigenvector of A . If it is, find the corresponding eigenvalue λ .

- a. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Compute $A\mathbf{x}$. Can the result be written as $\lambda\mathbf{x}$? If so, what is λ ?

b. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Compute $A\mathbf{x}$ and determine whether $\mathbf{x}$ is an eigenvector. If it is, find its eigenvalue.

c. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute $A\mathbf{x}$. Is the result a scalar multiple of $\mathbf{x}$? Is $\mathbf{x}$ an eigenvector?

d. Consider

$$\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Without doing the full matrix-vector multiplication at first, compare this vector with the vector in part a. Do you expect it to be an eigenvector? Check your answer by computing $A\mathbf{x}$.

- e. The vectors in parts a and d are different vectors, but they point in the same direction. What eigenvalue do they have? What does this tell you about scalar multiples of an eigenvector?
- f. Compare the two eigenvectors found in parts a and b. Compute their ordinary dot product. What do you observe?

Exercise 9.2 (Eigenvalues in chemical kinetics). Consider the reversible reaction



The positive number k is called the *rate constant*. It tells us how rapidly the reaction takes place. Here we use the same rate constant for the forward reaction $A \rightarrow B$ and the backward reaction $B \rightarrow A$, to keep the example simple.

Let $a(t)$ and $b(t)$ denote the concentrations of A and B. For a first-order reaction, the reaction rate is proportional to the concentration of the reacting species. Thus, A is converted into B at a rate ka , while B is converted into A at a rate kb . The larger k is, the faster the reaction proceeds. If concentration is measured in mol L^{-1} and time in seconds, k has units of s^{-1} .

The concentration of A therefore decreases due to the forward reaction at a rate ka and increases due to the backward reaction at a rate kb . Similarly, the concentration of B increases at a rate ka and decreases at a rate kb . This gives the rate equations

$$\frac{da}{dt} = -ka + kb, \quad \frac{db}{dt} = ka - kb.$$

If we collect the concentrations in the vector

$$\mathbf{x}(t) = \begin{bmatrix} a(t) \\ b(t) \end{bmatrix},$$

the rate equations can be written as

$$\frac{d\mathbf{x}}{dt} = K\mathbf{x}, \quad K = \begin{bmatrix} -k & k \\ k & -k \end{bmatrix}.$$

We will see that the eigenvectors of K describe two particularly simple ways in which the concentrations can change.

- a. Consider the vector

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Calculate $K\mathbf{x}_1$ and show that $\mathbf{x}_1$ is an eigenvector of K . What is its eigenvalue? Explain chemically why nothing changes when $a = b$.

- b. Now consider

$$\mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Calculate $K\mathbf{x}_2$ and show that it is also an eigenvector. What is its eigenvalue?

- c. Suppose that the concentration vector has the form

$$\mathbf{x}(t) = c(t)\mathbf{x}_2.$$

Insert this into the rate equation and use your result from part b to show that

$$\frac{dc}{dt} = -2kc.$$

Thus $c(t) = c(0)e^{-2kt}$. What happens to this contribution as time increases? What happens more rapidly if the rate constant k is increased? Explain why the two eigenvectors $\mathbf{x}_1$ and $\mathbf{x}_2$ can be interpreted as an *equilibrium mode* and a *relaxation mode*, respectively.

Exercise 9.3 (Eigenvalues and diagonalization of a Hermitian matrix). Consider the matrix

$$A = \begin{bmatrix} 2 & -i \\ i & 2 \end{bmatrix}.$$

- a. Show that A is Hermitian, that is, verify that

$$A^H = A.$$

- b. The eigenvalues of A are the values of λ for which

$$\det(A - \lambda I) = 0.$$

Set up the matrix $A - \lambda I$, calculate its determinant, and find the two eigenvalues λ_1 and λ_2 . Verify that both eigenvalues are real. (The determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$.)

- c. For each eigenvalue λ_j , solve

$$(A - \lambda_j I)\mathbf{x}_j = \mathbf{0}$$

to find a corresponding eigenvector $\mathbf{x}_j$. Normalize the eigenvectors so that

$$\mathbf{x}_j^H \mathbf{x}_j = 1.$$

- d. Verify explicitly that your two normalized eigenvectors are orthogonal,

$$\mathbf{x}_1^H \mathbf{x}_2 = 0.$$

Thus, the eigenvectors form an orthonormal basis of $\mathbb{C}^2$.

- e. Form the matrix U whose columns are the normalized eigenvectors,

$$U = [\mathbf{x}_1 \ \mathbf{x}_2].$$

Show explicitly that U is unitary by calculating $U^H U$.

- f. Form the diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix},$$

where the ordering of the eigenvalues must agree with the ordering of the eigenvectors in U .

- g. Verify by direct matrix multiplication that

$$A = U D U^H.$$

Equivalently, verify that $U^H A U = D$. Explain briefly why $U^{-1} = U^H$ in this diagonalization.

Exercise 9.4 (A matrix that cannot be diagonalized). Consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

- Find the eigenvalues of A by solving the characteristic equation $\det(A - \lambda I) = 0$. How many distinct eigenvalues does A have?
- Find all eigenvectors corresponding to the eigenvalue you found. How many linearly independent eigenvectors does A have?
- To diagonalize a 2×2 matrix, we need two linearly independent eigenvectors, which can be used as the columns of an invertible matrix U . Explain why this is impossible for A , and conclude that A is not diagonalizable.

Exercise 9.5 (A reversible reaction with different rate constants). It is recommended that you first complete Exercise 9.2.

Consider now the more general reversible reaction



where $k_f > 0$ is the rate constant for the forward reaction $A \rightarrow B$ and $k_b > 0$ is the rate constant for the backward reaction $B \rightarrow A$. The concentrations satisfy

$$\frac{d\mathbf{x}}{dt} = K\mathbf{x}, \quad \mathbf{x}(t) = \begin{bmatrix} a(t) \\ b(t) \end{bmatrix}, \quad K = \begin{bmatrix} -k_f & k_b \\ k_f & -k_b \end{bmatrix}.$$

- Find the two eigenvalues of K and a corresponding eigenvector for each eigenvalue. Show that one eigenvalue is zero and that the other is negative.
- The eigenvector with eigenvalue zero describes an equilibrium concentration vector. Use it to show that at equilibrium

$$\frac{b}{a} = \frac{k_f}{k_b}.$$

Explain why this result also follows directly by requiring the forward and backward reaction rates to be equal.

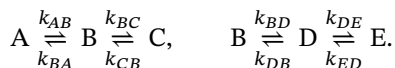
- The other eigenvector describes relaxation toward equilibrium. Show that its time dependence contains the factor

$$e^{-(k_f+k_b)t}.$$

What does this tell you about the rate at which the system approaches equilibrium?

Exercise 9.6 (Eigenvalues of a reaction network). It is recommended that you first complete or at least read through Exercise 9.2 and Exercise 9.5.

Consider the reaction network



All reactions are first order. Let

$$\mathbf{x}(t) = \begin{bmatrix} a(t) \\ b(t) \\ c(t) \\ d(t) \\ e(t) \end{bmatrix}$$

contain the concentrations of the five species. Use the rate constants

$$k_{AB} = 1.0, \quad k_{BA} = 0.4, \quad k_{BC} = 0.7, \quad k_{CB} = 0.2, \quad k_{BD} = 0.5, \quad k_{DB} = 0.3, \quad k_{DE} = 0.$$

in units of s^{-1} .

- Write the five coupled rate equations and show that they can be written as

$$\frac{d\mathbf{x}}{dt} = K\mathbf{x}.$$

Construct the 5×5 rate matrix K . Check that the sum of the components of $d\mathbf{x}/dt$ is zero. What quantity is therefore conserved?

- Use Python to compute the eigenvalues and eigenvectors of K . You may use, for example,

```
import numpy as np

eigenvalues, eigenvectors = np.linalg.eig(K)
```

Identify the eigenvalue that is zero, up to numerical precision. Verify that all the other eigenvalues have negative real parts. What do the zero eigenvalue and the negative eigenvalues mean physically?

- c. Scale the eigenvector belonging to the zero eigenvalue so that its components sum to 1. Interpret the resulting components in terms of fractions of A, B, C, D, and E.
- d. Start with all material in A,

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Compute

$$\mathbf{x}(t) = e^{Kt}\mathbf{x}(0)$$

numerically for a suitable range of times. You may use `scipy.linalg.expm` to evaluate the matrix exponential. Plot all five concentrations as functions of time.

Verify from your plot and numerical results that the total concentration remains constant and that the concentration vector approaches the equilibrium composition found in part c.

- e. Sort the nonzero eigenvalues according to the magnitude of their real parts. The eigenvalue closest to zero describes the slowest decaying mode of the reaction network. Determine its characteristic relaxation time

$$\tau = -\frac{1}{\operatorname{Re} \lambda}.$$

Compare this value with the time scale over which your numerical solution approaches equilibrium. Explain how the eigenvalues of the rate matrix separate the dynamics of a complicated reaction network into different relaxation time scales.

- f. The eigenvalue analysis predicts not only that the system approaches equilibrium, but also how rapidly it does so. Let $\mathbf{x}_{\text{eq}}$ denote the equilibrium concentration vector found in part c, and define the distance from equilibrium by

$$d(t) = \|\mathbf{x}(t) - \mathbf{x}_{\text{eq}}\|_2.$$

Using your numerical solution from part d, plot $d(t)$ as a function of time using a logarithmic scale for the vertical axis. You can use `plt.semilogy` for this.

At sufficiently long times, the slowest decaying eigenmode should dominate. If λ_{slow} is the nonzero eigenvalue closest to zero, we therefore expect approximately

$$d(t) \propto e^{\lambda_{\text{slow}} t}.$$

On the same plot, add a curve proportional to $e^{\lambda_{\text{slow}} t}$, choosing its prefactor so that it can easily be compared with $d(t)$. Do the two curves become parallel at long times? Explain why a straight line on this logarithmic plot provides a direct visualization of the slowest relaxation mode of the reaction network.

Solution for Exercise 9.6

- a. For species A, material is lost through $A \rightarrow B$ and gained through $B \rightarrow A$, giving

$$\frac{da}{dt} = -k_{AB}a + k_{BA}b.$$

Similarly, the remaining rate equations are

$$\frac{db}{dt} = k_{AB}a - (k_{BA} + k_{BC} + k_{BD})b + k_{CB}c + k_{DB}d,$$

$$\frac{dc}{dt} = k_{BC}b - k_{CB}c,$$

$$\frac{dd}{dt} = k_{BD}b - (k_{DB} + k_{DE})d + k_{ED}e,$$

and

$$\frac{de}{dt} = k_{DE}d - k_{ED}e.$$

Thus,

$$\frac{d\mathbf{x}}{dt} = K\mathbf{x},$$

with

$$K = \begin{bmatrix} -k_{AB} & k_{BA} & 0 & 0 & 0 \\ k_{AB} & -(k_{BA} + k_{BC} + k_{BD}) & k_{CB} & k_{DB} & 0 \\ 0 & k_{BC} & -k_{CB} & 0 & 0 \\ 0 & k_{BD} & 0 & -(k_{DB} + k_{DE}) & k_{ED} \\ 0 & 0 & 0 & k_{DE} & -k_{ED} \end{bmatrix}.$$

With the numerical rate constants, this becomes

$$K = \begin{bmatrix} -1.0 & 0.4 & 0 & 0 & 0 \\ 1.0 & -1.6 & 0.2 & 0.3 & 0 \\ 0 & 0.7 & -0.2 & 0 & 0 \\ 0 & 0.5 & 0 & -1.1 & 0.1 \\ 0 & 0 & 0 & 0.8 & -0.1 \end{bmatrix}.$$

Adding the five rate equations gives

$$\frac{d}{dt}(a + b + c + d + e) = 0.$$

All reaction terms cancel pairwise. Therefore the total concentration

$$a + b + c + d + e$$

is conserved.

This is also visible from K : every column of K sums to zero.

b. We construct the matrix and diagonalize it numerically.

```

import numpy as np

kAB = 1.0
kBA = 0.4
kBC = 0.7
kCB = 0.2
kBD = 0.5
kDB = 0.3
kDE = 0.8
kED = 0.1

K = np.array([
    [-kAB, kBA, 0.0, 0.0, 0.0],
    [kAB, -(kBA + kBC + kBD), kCB, kDB, 0.0],
    [0.0, kBC, -kCB, 0.0, 0.0],
    [0.0, kBD, 0.0, -(kDB + kDE), kED],
    [0.0, 0.0, 0.0, kDE, -kED]
])

print(K)

# Check conservation: the columns should sum to zero
print("Column sums:", K.sum(axis=0))

[[-1.  0.4  0.  0.  0. ]
 [ 1. -1.6  0.2  0.3  0. ]
 [ 0.  0.7 -0.2  0.  0. ]
 [ 0.  0.5  0. -1.1  0.1]
 [ 0.  0.  0.  0.8 -0.1]]
Column sums: [ 0.00000000e+00 -2.22044605e-16  0.00000000e+00  0.00000000e+00  0.00000000e+00]

```

We now find the eigenvalues and eigenvectors. We sort the eigenvalues from largest to smallest real part so that the eigenvalue closest to zero appears first.

```

eigenvalues, eigenvectors = np.linalg.eig(K)

order = np.argsort(eigenvalues.real)[::-1]
eigenvalues = eigenvalues[order]
eigenvectors = eigenvectors[:, order]

```

```
print("Eigenvalues:")
for value in eigenvalues:
    print(value)
```

Eigenvalues:
-3.5641539813319436e-17
-0.07786195015156719
-0.6247369441879431
-1.1352737121412337
-2.1621273935192553

Numerically, the eigenvalues are approximately

$$\lambda = 0, -0.07786, -0.62474, -1.13527, -2.16213 \quad \text{s}^{-1}.$$

The eigenvalue reported by the computer as a number of order 10^{-17} should be interpreted as zero; the small deviation from exactly zero is due to floating-point arithmetic.

The zero eigenvalue represents an unchanging mode. This is the equilibrium mode of the reaction network. If

$$K\mathbf{x}_{\text{eq}} = \mathbf{0},$$

then

$$\frac{d\mathbf{x}_{\text{eq}}}{dt} = \mathbf{0}.$$

All the other eigenvalues are negative. A mode with eigenvalue $\lambda < 0$ behaves as

$$e^{\lambda t},$$

and therefore decays with time. These eigenvalues describe the different ways in which deviations from equilibrium relax.

- c. We identify the eigenvector belonging to the eigenvalue closest to zero and normalize it so that its components sum to one.

```
i0 = np.argmin(np.abs(eigenvalues))

x_eq = eigenvectors[:, i0].real

# Eigenvectors can have an arbitrary overall sign
if x_eq.sum() < 0:
    x_eq = -x_eq

# Normalize to total concentration 1
x_eq = x_eq / x_eq.sum()

print("Equilibrium fractions:")
for species, value in zip(["A", "B", "C", "D", "E"], x_eq):
    print(f"{species}: {value:.6f}")

print("Sum:", x_eq.sum())
```

```
Equilibrium fractions:
A: 0.020101
B: 0.050251
C: 0.175879
D: 0.083752
E: 0.670017
Sum: 1.0
```

This gives approximately

$$\mathbf{x}_{\text{eq}} = \begin{bmatrix} 0.02010 \\ 0.05025 \\ 0.17588 \\ 0.08375 \\ 0.67002 \end{bmatrix}.$$

Thus the equilibrium mixture contains approximately

2.0% A, 5.0% B, 17.6% C, 8.4% D, 67.0% E.

The equilibrium vector can also be understood directly from the individual reversible reactions. At equilibrium,

$$k_{AB}a = k_{BA}b, \quad k_{BC}b = k_{CB}c, \quad k_{BD}b = k_{DB}d, \quad k_{DE}d = k_{ED}e.$$

For example,

$$\frac{b}{a} = \frac{k_{AB}}{k_{BA}} = \frac{1.0}{0.4} = 2.5,$$

and the other equilibrium ratios follow similarly.

- d. We now start with all material in A,

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

and evaluate

$$\mathbf{x}(t) = e^{Kt}\mathbf{x}(0).$$

Because the slowest relaxation time will turn out to be around 13 s, a time interval of about 0–80 s is sufficient to see the system approach equilibrium.

```
import matplotlib.pyplot as plt
from scipy.linalg import expm

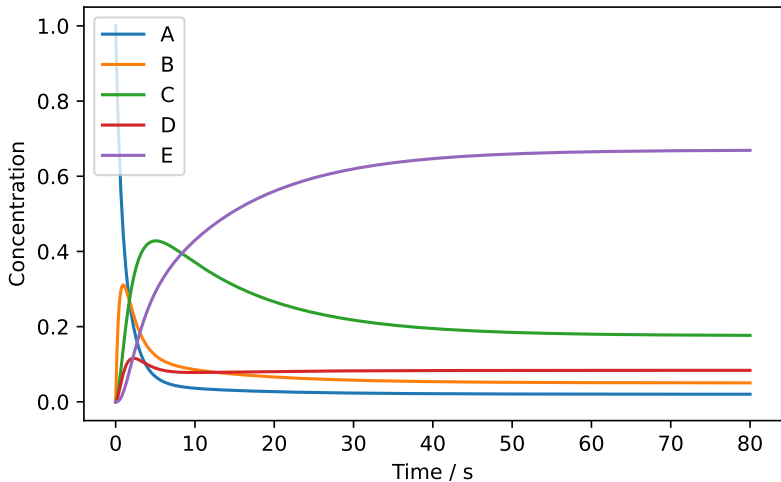
x0 = np.array([1.0, 0.0, 0.0, 0.0, 0.0])

times = np.linspace(0.0, 80.0, 500)

x_t = np.array([
    expm(K * t) @ x0
    for t in times
])
```

```
plt.figure()
for i, species in enumerate(["A", "B", "C", "D", "E"]):
    plt.plot(times, x_t[:, i], label=species)

plt.xlabel("Time / s")
plt.ylabel("Concentration")
plt.legend()
plt.tight_layout()
plt.show()
```



The plot shows the initial A concentration decreasing while the other species are populated. Eventually all five concentrations become constant and approach the equilibrium fractions found in part c.

We can also check conservation numerically.

```
total = x_t.sum(axis=1)

print("Minimum total concentration:", total.min())
print("Maximum total concentration:", total.max())
print("Final concentration vector:")
print(x_t[-1])
print("Equilibrium vector:")
print(x_eq)
```

```

Minimum total concentration: 0.9999999999999933
Maximum total concentration: 1.0000000000000058
Final concentration vector:
[0.02016436 0.05039847 0.17672309 0.08372369 0.66899039]
Equilibrium vector:
[0.0201005 0.05025126 0.1758794 0.08375209 0.67001675]

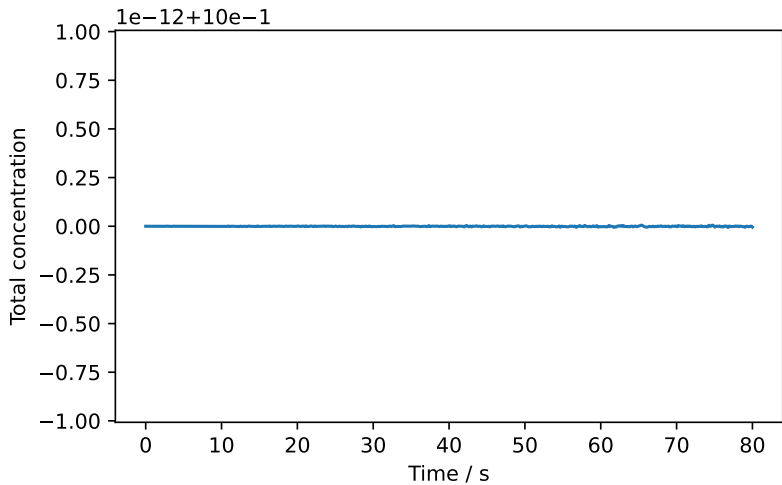
```

A useful visualization of conservation is

```

plt.figure()
plt.plot(times, total)
plt.xlabel("Time / s")
plt.ylabel("Total concentration")
plt.tight_layout()
plt.show()

```



Apart from tiny floating-point errors, the total concentration remains exactly 1.

- e. The nonzero eigenvalue closest to zero is

$$\lambda_{\text{slow}} \approx -0.07786 \text{ s}^{-1}.$$

Its characteristic relaxation time is therefore

$$\tau = -\frac{1}{\lambda_{\text{slow}}} \approx \frac{1}{0.07786} \text{ s} \approx 12.84 \text{ s}.$$

We can obtain all the characteristic relaxation times numerically.

```
nonzero = eigenvalues[np.abs(eigenvalues) > 1e-10]

# Sort from slowest to fastest decay
nonzero = nonzero[np.argsort(np.abs(nonzero.real))]

print("Relaxation modes:")
for lam in nonzero:
    tau = -1.0 / lam.real
    print(f"lambda = {lam.real: .6f} 1/s, tau = {tau:.3f} s")
```

```
Relaxation modes:
lambda = -0.077862 1/s, tau = 12.843 s
lambda = -0.624737 1/s, tau = 1.601 s
lambda = -1.135274 1/s, tau = 0.881 s
lambda = -2.162127 1/s, tau = 0.463 s
```

The relaxation times are approximately

$$12.84, \quad 1.60, \quad 0.88, \quad 0.46 \text{ s}.$$

Thus the reaction network contains several distinct time scales. Some re-distribution takes place very rapidly, while one mode relaxes considerably more slowly.

After a few multiples of the shorter relaxation times, the fast modes have essentially disappeared. The remaining deviation from equilibrium is then dominated by the slow mode,

$$\mathbf{x}(t) - \mathbf{x}_{\text{eq}} \approx c_{\text{slow}} e^{\lambda_{\text{slow}} t} \mathbf{v}_{\text{slow}}.$$

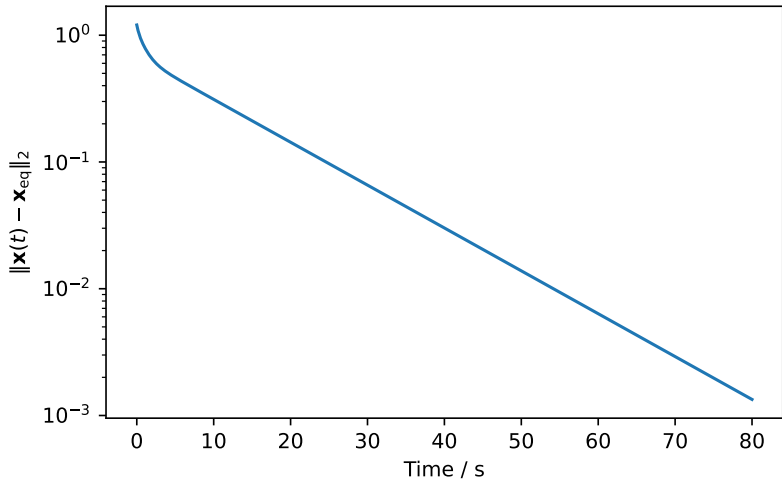
This illustrates an important use of eigenvalue analysis: a complicated set of coupled rate equations can be decomposed into independent modes with characteristic time scales determined by the eigenvalues.

f. We now calculate the Euclidean distance from equilibrium,

$$d(t) = \|\mathbf{x}(t) - \mathbf{x}_{\text{eq}}\|_2.$$

```
distance = np.linalg.norm(x_t - x_eq, axis=1)

plt.figure()
plt.semilogy(times, distance)
plt.xlabel("Time / s")
plt.ylabel(r"$\|\mathbf{x}(t) - \mathbf{x}_{\mathrm{eq}}\|_2$")
plt.tight_layout()
plt.show()
```



The curve becomes approximately straight at long times. To understand this, suppose that only the slowest mode remains. Then

$$d(t) \propto e^{\lambda_{\text{slow}} t}.$$

Taking the logarithm gives

$$\log d(t) = \text{constant} + \lambda_{\text{slow}} t.$$

Thus an exponential decay becomes a straight line on a semilogarithmic plot, with slope λ_{slow} .

We can compare the numerical distance directly with the predicted slow exponential. We choose a reference time after the faster modes have mostly decayed and set the prefactor so that the two curves agree at that time.

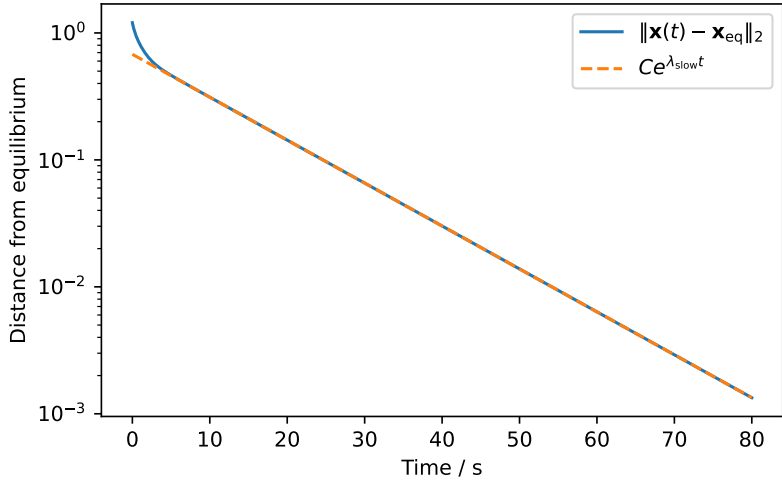
```
lambda_slow = nonzero[0].real

# Choose a reference point in the long-time regime
t_ref = 30.0
i_ref = np.argmin(np.abs(times - t_ref))

# Choose the prefactor so the exponential agrees with d(t)
# at the reference time
prefactor = distance[i_ref] / np.exp(lambda_slow * times[i_ref])
slow_decay = prefactor * np.exp(lambda_slow * times)

plt.figure()
plt.semilogy(
    times,
    distance,
    label=r"$\|\mathbf{x}(t)-\mathbf{x}_{\mathrm{eq}}\|_2$"
)
plt.semilogy(
    times,
    slow_decay,
    "--",
    label=r"$C e^{\lambda_{\mathrm{slow}}t}$"
)

plt.xlabel("Time / s")
plt.ylabel("Distance from equilibrium")
plt.legend()
plt.tight_layout()
plt.show()
```



At early times the two curves need not agree because several eigenmodes contribute simultaneously:

$$\mathbf{x}(t) - \mathbf{x}_{\text{eq}} = \sum_j c_j e^{\lambda_j t} \mathbf{v}_j.$$

The modes with more negative eigenvalues decay most rapidly. At sufficiently long times, they become negligible compared with the mode whose eigenvalue is closest to zero. Consequently,

$$\mathbf{x}(t) - \mathbf{x}_{\text{eq}} \approx c_{\text{slow}} e^{\lambda_{\text{slow}} t} \mathbf{v}_{\text{slow}},$$

and the two curves become parallel on the logarithmic plot.

The numerical dynamics therefore provide a direct visualization of the eigenvalue analysis: the smallest-magnitude nonzero eigenvalue controls the long-time approach to equilibrium.

10 General vector spaces

A vector space needs vector addition and scalar multiplication, but it need not resemble Euclidean coordinate space.

You may be very familiar with Euclidean space $\mathbb{R}^n$ as a vector space, and quite used to the notion that we have an inner product and a corresponding distance measure on this space. However, linear spaces are much more general. An inner product or norm is not part of the definition of a general vector space. Only the a “linear structure” matters: Consider for example a general vector space over $\mathbb{R}$: An abstract set V with operations such that one can *add vectors*,

$$z = x + y \in V,$$

and *multiply vectors by constants* $c \in \mathbb{R}$,

$$z = cx \in V$$

where $c \in \mathbb{R}$. The full set of axioms is as follows. We focus on vector spaces over the field $\mathbb{R}$, but the definition can be generalized to vector spaces over any field such as $\mathbb{C}$.

Definition: Vector space. A *vector space over the field* $\mathbb{R}$ is a set V together with a binary *vector addition* $+: V \times V \rightarrow V$ and *scalar multiplication* $\cdot: \mathbb{R} \times V \rightarrow V$ such that, for all $x, y, z \in V$ and all $\alpha, \beta \in \mathbb{R}$, the following axioms are true:

1. There exists a $0 \in V$ such that $0 + x = x$ for all $x \in V$
2. $x + (y + z) = (x + y) + z$
3. $x + y = y + x$
4. For every $x \in V$, there exists $x' \in V$ such that $x + x' = 0$
5. $(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$
6. $1 \cdot x = x$
7. $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$

$$8. \alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$$

In this exercise, we will train our ability to think abstractly, and also try to get acquainted with general vector spaces.

Example: Somerset Apple Cake. Ingredients:

- 170 g butter
- 170 g light brown sugar
- 3 eggs
- 1 tablespoon liquid honey
- 1.5 teaspoons cinnamon
- 0.5 teaspoon cloves
- 0.5 teaspoon nutmeg
- 1 teaspoon baking powder
- 240 g wheat flour
- 700 g apples, diced
- 100 ml apple cider, apple juice, or milk
- 100 g light sultana raisins (optional)

Instructions:

- Cream the room-temperature butter with the sugar until light and fluffy.
- Beat in the eggs and honey.
- Sift the flour, baking powder, and spices. Add them to the mixture and stir until the batter is smooth and lump-free.
- Stir in the milk or apple cider (and the raisins, if you choose to use them; they can be soaked in cider for a couple of hours beforehand, if desired).
- Finally, fold the apple pieces into the batter.
- Pour the batter into a greased round tin with parchment paper at the bottom, or use a deep, round ovenproof dish (24 cm in diameter).

- Bake the cake in the middle of the oven at 160°C for 1.5 hours (check with a cake tester to ensure it's fully baked).
- Serve warm or cold with a dollop of whipped cream, clotted cream, vanilla ice cream, or whatever you like.

Exercises

Exercise 10.1 (Cooking with vectors 1). We can think of an ingredient list as a vector. For example, suppose that our universe of ingredients consists only of

(butter, sugar, eggs, flour, apples).

After choosing suitable units for each ingredient, an ingredient list can then be represented by a vector

$$x = (x_1, x_2, x_3, x_4, x_5),$$

where each component gives the amount of one ingredient.

- Using grams for butter, sugar, flour, and apples, and number of eggs for eggs, write the corresponding five-component vector for the Somerset Apple Cake recipe.
- What would the vector $(100, 0, 2, 300, 0)$ mean? What does a zero component mean?
- What is the dimension of this vector space? Give a natural set of basis vectors and explain what each basis vector represents.
- Suppose that x and y are the ingredient vectors for two recipes. What does the vector $x + y$ represent?
- What does the vector $2x$ represent? What about $\frac{1}{2}x$? Does the scalar 2 have a physical unit?
- Our choice of units is part of the way we represent the recipe. Suppose milk were another component. Explain why we should not represent 1 litre of milk as 1 in one recipe and 1000 ml of milk as 1000 in another without taking account of the different units.

- g. Typical recipes contain ingredients that are not among our five chosen ingredients. How could we enlarge our vector space to accommodate more recipes? What happens to the dimension of the space?

Exercise 10.2 (Cooking with vectors 2). In Exercise 10.1, we represented recipes using a fixed list of possible ingredients. If you did not do this exercise already, please do so first. In this exercise, we make this construction more general.

Let S be the set of all possible ingredients. An ingredient list can be represented by a function

$$f : S \rightarrow \mathbb{R},$$

where $f(s)$ gives the amount of ingredient s . We assume that $f(s) = 0$ for all but finitely many ingredients. The set L of all such functions is a vector space, with addition and scalar multiplication defined in the usual way.

- Explain how this description generalizes the finite-dimensional ingredient vectors used in Exercise 10.1. If S contains infinitely many ingredients, what can you say about the dimension of L ?
- Mathematically, L contains vectors with negative components. Can such vectors be interpreted as ordinary ingredient lists? What about the zero vector?
- Define the subset

$$L_+ = \{f \in L \mid f(s) \geq 0 \text{ for every } s \in S\}.$$

Is L_+ a vector space? Which vector-space properties fail? Is L_+ closed under addition? Is it closed under multiplication by *nonnegative* scalars?

- Even if $f, g \in L_+$ are ingredient lists for two perfectly good recipes, $f + g$ need not describe a sensible recipe. What information is missing from an ingredient vector?
- We might try to describe the steps of a recipe by operators acting on L . Consider, for example, the instruction “cream the butter and sugar”. Can this operation be represented naturally as a linear map $T : L \rightarrow L$ on our present ingredient space? Explain the challenges one encounters.

- f. What additional information might have to be included in our mathematical description before operations such as mixing, creaming, heating, and baking could be represented?
- g. In quantum mechanics, we similarly choose a mathematical space to represent physical states and operators to represent physical quantities and transformations. Why is the choice of the space important for determining what the mathematical model can describe?

Solutions

Solution for Exercise 10.1

- a. With the ordering

(butter, sugar, eggs, flour, apples),

the Somerset Apple Cake recipe is represented by

$$x = (170, 170, 3, 240, 700).$$

The components do not all have the same units: the first, second, fourth, and fifth components are measured in grams, while the third is a number of eggs. This is not a problem as long as the meaning and unit of each component are fixed.

- b. The vector

$$(100, 0, 2, 300, 0)$$

represents an ingredient list containing 100 g butter, no sugar, 2 eggs, 300 g flour, and no apples.

A zero component means that the corresponding ingredient is absent.

- c. The vector space has dimension 5. A natural basis is

$$e_1 = (1, 0, 0, 0, 0), \quad e_2 = (0, 1, 0, 0, 0), \quad \dots, \quad e_5 = (0, 0, 0, 0, 1).$$

Each basis vector represents one unit of one ingredient and zero of all the others. For example, e_1 represents 1 g of butter and e_3 represents 1 egg.

Any ingredient vector can be written as a linear combination of these basis vectors. For example,

$$(170, 170, 3, 240, 700) = 170e_1 + 170e_2 + 3e_3 + 240e_4 + 700e_5.$$

- d. Addition is performed component by component. If

$$x = (x_1, \dots, x_5), \quad y = (y_1, \dots, y_5),$$

then

$$x + y = (x_1 + y_1, \dots, x_5 + y_5).$$

Thus, $x + y$ represents putting together the ingredient quantities from the two lists. If both recipes require flour, for example, the flour amounts are added.

The resulting vector is certainly a valid ingredient vector, although it need not correspond to a sensible cake recipe.

- e. Multiplying by 2 doubles the amount of every ingredient:

$$2x = (2x_1, \dots, 2x_5).$$

This can be interpreted as making twice as much of the recipe. Similarly,

$$\frac{1}{2}x$$

corresponds to making half as much.

The scaling factor is dimensionless. It simply tells us by what numerical factor all ingredient quantities are changed.

- f. The numerical value of a component only has meaning together with the unit chosen for that component. If the milk coordinate is defined to be measured in millilitres, then

$$1 \text{ litre} = 1000 \text{ ml}$$

must be represented by the number 1000.

If we sometimes used litres and sometimes millilitres without converting them, then the same physical amount could be represented by different vectors. We therefore choose one standard unit for each component.

- g. We can add more components, one for each new ingredient. For example, if cinnamon and milk are added to our list of possible ingredients, the vectors would have seven components instead of five.

Each additional independent ingredient increases the dimension by one. If we allow n possible ingredients, the corresponding vector space has dimension n .

11 The complex numbers as a real vector space

Identifying $x + iy$ with (x, y) turns the complex plane into a two-dimensional real vector space.

The geometrical interpretation of the $\mathbb{C}$ as the plane $\mathbb{R}^2$ indicates that $\mathbb{C}$ can be viewed as a two-dimensional real vector space. Indeed, addition and multiplication with *real* scalars are compatible with the axioms for Euclidean space $\mathbb{R}^2$.

Exercises

Exercise 11.1 (Complex numbers as a real vector space).

- Show that $\mathbb{C}$ regarded can be regarded as the Euclidean plane: Check axioms for Euclidean space and check that vector addition and multiplication with real scalars are the same in the two spaces. Check also that the Euclidean norm is the modulus of the complex number. What are the complex numbers that correspond to the standard basis in $\mathbb{R}^2$? Conclude that $\mathbb{C}$ can be regarded as a *real* Euclidean vector space of dimension 2.
- Show that multiplication with a *complex* number z is a linear operator on $\mathbb{R}^2$, and find its matrix in the standard basis. What is the matrix of multiplication with i ?
- Show that multiplication with a complex number of modulus 1 is a rotation in $\mathbb{R}^2$.
- Show that the map $z \mapsto \bar{z}$ is a linear transformation in $\mathbb{R}^2$. What kind of linear transformation is this? Is it a linear transformation in $\mathbb{C}$?

12 Gaussian elimination

Exercises

Exercise 12.1 (Gaussian elimination and overlap matrix). Consider the two vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

- a. Compute the overlap matrix S with elements

$$S_{ij} = \mathbf{v}_i^T \mathbf{v}_j.$$

- b. Verify that S is symmetric.
c. Use Gaussian elimination on the augmented matrix $[S | I]$ to compute S^{-1} .
d. Verify your result by computing SS^{-1} .

Exercise 12.2. Write down the dual basis of $B = \{\mathbf{b}_1, \mathbf{b}_2\}$.

Exercise 12.3. (Robert Beezer) Consider the following system of equations:

$$\begin{aligned} 2x_1 - 3x_2 + x_3 + 7x_4 &= 14 \\ 2x_1 + 8x_2 - 4x_3 + 5x_4 &= -1 \\ x_1 + 3x_2 - 3x_3 &= 4 \\ -5x_1 + 2x_2 + 3x_3 + 4x_4 &= -19 \end{aligned}$$

Use Gaussian elimination to find all possible solutions. Write the solution set using set notation.

Exercise 12.4. (Robert Beezer) Find all possible solutions of the linear system

$$3x_1 + 4x_2 - x_3 + 2x_4 = 6$$

$$x_1 - 2x_2 + 3x_3 + x_4 = 2$$

$$10x_2 - 10x_3 - x_4 = 1$$

Write down the solution set using set notation.

Exercise 12.5. (Robert Beezer) Find all possible solutions of the linear system

$$2x_1 + 4x_2 + 5x_3 + 7x_4 = -26$$

$$x_1 + 2x_2 + x_3 - x_4 = -4$$

$$-2x_1 - 4x_2 + x_3 + 11x_4 = -10$$

Write down the solution set using set notation.

Exercise 12.6. Let $\mathbf{b}_i, i = 1, 2, \dots, n$ be a basis for $\mathbb{R}^n$. Let

$$B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n]$$

be the *basis matrix*, whose columns are precisely the $\mathbf{b}_i$. Explain that if $\mathbf{y} = x_1\mathbf{b}_1 + \dots + x_n\mathbf{b}_n$, then

$$B\mathbf{x} = \mathbf{y}.$$

Solve for $\mathbf{x}$ using B in symbols.

Exercise 12.7. Let n linearly independent vectors $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of $\mathbb{F}^n$ be given. Write down a formula for the dual basis in terms of the overlap matrix.

13 Bra-ket notation

Bra-ket notation, introduced by Dirac, is a compact way of writing the linear algebra of quantum mechanics. A vector in a Hilbert space is written as a *ket*,

$$|\psi\rangle,$$

and its Hermitian adjoint is a *bra*,

$$\langle\psi|.$$

The inner product of two vectors is therefore written

$$\langle\phi|\psi\rangle,$$

and is a scalar. It satisfies

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*.$$

In particular,

$$\langle\psi|\psi\rangle = \|\psi\|^2,$$

so a normalized state satisfies $\langle\psi|\psi\rangle = 1$.

The product in the opposite order,

$$|\phi\rangle\langle\psi|,$$

is an *outer product*. It is an operator: acting on a ket $|\chi\rangle$ it gives

$$(|\phi\rangle\langle\psi|)|\chi\rangle = |\phi\rangle\langle\psi|\chi\rangle.$$

A linear operator $\hat{A}$ acts on kets from the left,

$$|\phi\rangle = \hat{A}|\psi\rangle.$$

Taking the Hermitian adjoint reverses the order,

$$\langle\phi| = \langle\psi|\hat{A}^\dagger.$$

For a Hermitian operator, $\hat{A}^\dagger = \hat{A}$. Observables in quantum mechanics are represented by Hermitian operators, and the expectation value of $\hat{A}$ in a normalized state $|\psi\rangle$ is

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle.$$

For an orthonormal basis $\{|i\rangle\}$,

$$\langle i | j \rangle = \delta_{ij},$$

and every vector can be expanded as

$$|\psi\rangle = \sum_i c_i |i\rangle, \quad c_i = \langle i | \psi \rangle.$$

Equivalently, the basis satisfies the completeness relation

$$\hat{I} = \sum_i |i\rangle \langle i|.$$

In quantum chemistry we also frequently use *nonorthogonal* basis functions, such as atomic orbitals. If the basis is $\{|b_i\rangle\}$, its overlap matrix is

$$S_{ij} = \langle b_i | b_j \rangle.$$

In this case the expansion coefficients cannot in general be obtained simply by projecting with $\langle b_i |$; the overlap matrix must also be taken into account.

Exercises

Exercise 13.1 (Bra-ket notation and column vectors). Consider the orthonormal basis $\{|1\rangle, |2\rangle, |3\rangle\}$. In this basis, the ket

$$|\psi\rangle = 2|1\rangle - i|2\rangle + 3|3\rangle$$

can be represented by a column vector.

- Write $|1\rangle$, $|2\rangle$, and $|3\rangle$ as column vectors.
- Write $|\psi\rangle$ as a column vector.
- Write the bra $\langle\psi|$ in bra-ket notation.
- Write $\langle\psi|$ as a row vector. Remember that taking the adjoint involves both transposing and complex conjugating.

- e. Calculate $\langle 2|\psi\rangle$, first using bra-ket notation and then by multiplying a row vector by a column vector.
- f. Calculate $\langle\psi|\psi\rangle$ using the row and column vectors. What is the norm $\|\psi\|$?

Exercise 13.2 (Reading simple bra-ket expressions). Consider two orthonormal states $|1\rangle$ and $|2\rangle$ and the state

$$|\psi\rangle = c_1|1\rangle + c_2|2\rangle.$$

- a. Write the corresponding bra $\langle\psi|$.
- b. Compute $\langle 1|\psi\rangle$.
- c. Compute $\langle 2|\psi\rangle$.
- d. Compute $\langle\psi|\psi\rangle$.
- e. What condition must c_1 and c_2 satisfy if $|\psi\rangle$ is normalized?

Exercise 13.3 (From coordinate vectors to Hilbert-space vectors). Let $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ be a basis for an n -dimensional Hilbert space V , and let $\{|i\rangle\}$ be the standard orthonormal basis of $\mathbb{C}^n$. Define

$$\hat{B} = \sum_{i=1}^n |b_i\rangle\langle i|.$$

- a. What are the domain and range of $\hat{B}$?
- b. Let

$$|x\rangle = \sum_i x_i|i\rangle \in \mathbb{C}^n.$$

Show that

$$\hat{B}|x\rangle = \sum_i x_i|b_i\rangle.$$

Interpret the numbers x_i .

- c. Find $\hat{B}^\dagger$.
- d. Show that

$$\hat{B}^\dagger \hat{B} = \sum_{ij} |i\rangle\langle b_i|b_j\rangle\langle j|.$$

- e. Explain why $\hat{B}^\dagger \hat{B}$ is the overlap matrix of the basis B , expressed as an operator on $\mathbb{C}^n$.

Exercise 13.4 (Molecular orbitals in a nonorthogonal atomic-orbital basis). Suppose a molecular orbital is expanded in two normalized atomic orbitals,

$$|\psi\rangle = c_1|\chi_1\rangle + c_2|\chi_2\rangle,$$

where

$$\langle\chi_1|\chi_1\rangle = \langle\chi_2|\chi_2\rangle = 1, \quad \langle\chi_1|\chi_2\rangle = \langle\chi_2|\chi_1\rangle = s,$$

with s real.

- Compute $\langle\psi|\psi\rangle$.
- Write the overlap matrix S in the basis $\{|\chi_1\rangle, |\chi_2\rangle\}$.
- Show that the normalization condition can be written

$$\mathbf{c}^\dagger \mathbf{S} \mathbf{c} = 1,$$

where $\mathbf{c} = (c_1, c_2)^T$.

- Consider the symmetric combination

$$|\psi_+\rangle = N_+(|\chi_1\rangle + |\chi_2\rangle).$$

Find the normalization constant N_+ .

- Similarly, find the normalization constant N_- for

$$|\psi_-\rangle = N_- (|\chi_1\rangle - |\chi_2\rangle).$$

- Show that $\langle\psi_+|\psi_-\rangle = 0$. Thus symmetric and antisymmetric combinations can be orthogonal even though the atomic orbitals from which they are constructed are not.

Exercise 13.5 (Nonorthogonal bases, dual vectors, and completeness). Let $\{|b_i\rangle\}_{i=1}^n$ be a linearly independent, but not necessarily orthogonal, basis of an n -dimensional Hilbert space. Define the overlap matrix

$$S_{ij} = \langle b_i | b_j \rangle.$$

Assume that S is invertible.

- a. Define vectors $|b^i\rangle$ by

$$|b^i\rangle = \sum_j |b_j\rangle (S^{-1})_{ji}.$$

Show that

$$\langle b_k | b^i \rangle = \delta_k^i.$$

The set $\{|b^i\rangle\}$ is called the *dual basis*.

- b. Show that

$$\hat{I} = \sum_i |b^i\rangle \langle b_i|$$

and hence also

$$\hat{I} = \sum_{ij} |b_i\rangle (S^{-1})_{ij} \langle b_j|.$$

- c. Let

$$|\psi\rangle = \sum_i c_i |b_i\rangle.$$

Show that the expansion coefficients are

$$c_i = \langle b^i | \psi \rangle.$$

- d. Define the projections

$$p_i = \langle b_i | \psi \rangle.$$

Show that

$$p_i = \sum_j S_{ij} c_j$$

and therefore, in matrix notation,

$$\mathbf{c} = S^{-1} \mathbf{p}.$$

- e. Explain why the familiar orthonormal-basis formula $c_i = \langle b_i | \psi \rangle$ is recovered when $S = I$.
- f. In quantum chemistry, atomic-orbital basis functions are generally nonorthogonal. Explain briefly why confusing the projections $\langle b_i | \psi \rangle$ with the expansion coefficients c_i can therefore lead to incorrect molecular-orbital coefficients.

14 Gram–Schmidt orthogonalization

Orthogonal projection leads to the Gram–Schmidt construction and the QR factorization of a full-rank matrix.

Let V be a vector space over $\mathbb{F}$ with $\dim(V) = n < +\infty$. Recall that every finite-dimensional vector space has a basis. In the lecture notes, it is claimed that every finite dimensional Hilbert space has a basis of orthonormal vectors! Recall, that in an orthonormal basis, computations are usually much simpler than in nonorthogonal bases.

In this exercise, we show that any set of linearly independent vectors can be orthonormalized. We work in the space $\mathbb{F}^n$ for simplicity. That is, let $B = [\mathbf{b}_1, \dots, \mathbf{b}_m] \subset \mathbb{F}^n$ be a set of $m \leq n$ linearly independent vectors, spanning a subspace $V = \text{span}(B)$. We denote the vector set by its basis matrix B . Equivalently, B is an arbitrary matrix in $M(n, m, \mathbb{F})$ of full rank m , and the vectors $\mathbf{b}_i$ are the columns of B .

We will find a set of orthonormal vectors $U = [\mathbf{u}_1, \dots, \mathbf{u}_m]$, again denoted by a matrix in $M(n, m, \mathbb{F})$, such that $\text{span}(B) = \text{span}(U)$. That is, the two sets of vectors are bases for the *same* subspace, but one of them is an orthonormal basis.

We first define the notion of orthogonal projection onto the span of a single vector: Let $\mathbf{y} \in \mathbb{F}^n$ be nonzero, $Y = \text{span}(\mathbf{y})$, and let $\mathbf{x} \in \mathbb{F}^n$.

Exercises

Exercise 14.1.

1. Show that the vector $\mathbf{x}_Y$ given by

$$\mathbf{x}_Y = \mathbf{y} \frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\langle \mathbf{y}, \mathbf{y} \rangle}$$

is the *unique* vector in Y , such that

$$\forall \mathbf{y}' \in Y, \quad \|\mathbf{x} - \mathbf{x}_Y\| \leq \|\mathbf{x} - \mathbf{y}'\|.$$

Hint: Find a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which you can study, by using the definition of $Y = \text{span}(\mathbf{y})$.

2. Show that

$$\mathbf{x} = \mathbf{x}_Y + \mathbf{x}_{Y^\perp},$$

where

$$\langle \mathbf{x}_{Y^\perp} | \mathbf{x}_Y \rangle = 0.$$

Illustrate with a picture of the situation in $\mathbb{R}^2$.

The vector $\mathbf{x}_Y$ is called the *orthogonal projection of $\mathbf{x}$ onto Y or $\mathbf{y}$* .

1. Explain that the orthogonal projection is given by acting with an operator,

$$\mathbf{x}_Y = P_Y \mathbf{x}, \quad P_Y = \mathbf{y} \langle \mathbf{y}, \mathbf{y} \rangle^{-1} \mathbf{y}^H.$$

The operator P_Y is called the *orthogonal projection operator onto $Y = \text{span}(\mathbf{y})$* .

1. Show that $P^2 = P$ and that $P^\dagger = P$.

The Gram–Schmidt orthogonalization proceeds recursively:

1. $\mathbf{u}_1 = \mathbf{b}_1 \langle \mathbf{b}_1, \mathbf{b}_1 \rangle^{-1/2}$ (normalization)
2. $\mathbf{u}'_2 = \mathbf{b}_2 - P_{\mathbf{u}_1} \mathbf{b}_2$ (removal of projection), and $\mathbf{u}_2 = \mathbf{u}'_2 \langle \mathbf{u}'_2, \mathbf{u}'_2 \rangle^{-1/2}$ (normalization)
3. $\mathbf{u}'_3 = \mathbf{b}_3 - P_{\mathbf{u}_1} \mathbf{b}_3 - P_{\mathbf{u}_2} \mathbf{b}_3$ (removal of projection), and $\mathbf{u}_3 = \mathbf{u}'_3 \langle \mathbf{u}'_3, \mathbf{u}'_3 \rangle^{-1/2}$ (normalization)
4. Continue recursively.
5. $\mathbf{u}'_m = \mathbf{b}_m - \sum_{i=1}^{m-1} P_{\mathbf{u}_i} \mathbf{b}_m$ (removal of projection), and $\mathbf{u}_m = \mathbf{u}'_m \langle \mathbf{u}'_m, \mathbf{u}'_m \rangle^{-1/2}$ (normalization)

We see that each $\mathbf{u}_i$ is given by projecting away from $\mathbf{b}_i$ the components of all the *previously generated* $\mathbf{u}_j$, $j < i$.

1. Show that all the generated vectors $\mathbf{u}_i$ are orthonormal.
2. Show that a set of orthonormal vectors are linearly independent.

3. Show that the span of the first k original vectors is the same as the span of the first k generated vectors. Show that

$$B = UR,$$

where $R \in M(m, m, \mathbb{F})$ is *upper triangular*, i.e., $R_{ij} = 0$ whenever $i > j$. This is called the QR decomposition of the operator B . (The “Q” is just our U .)

4. We have shown the existence of the QR decomposition when B had full rank. Can you show the existence of the decomposition when B does *not* have full rank? It always exists. Hint: What happens when B does not have full rank?

We now turn to a computational exercise:

1. Write a Python function to compute the QR decomposition of a matrix with full rank. In SciPy, the function `scipy.linalg.qr` is an advanced implementation with pivoting. To compare with your implementation, use the call

```
Q,R = qr(A,mode='economic',pivoting=False)
```

With `pivoting=False`, the function computes essentially the same as our presented algorithm. (Turning on pivoting stabilizes the algorithm, rearranging the vectors before orthogonalizing.)

2. Let $n \geq m > 0$ be integers, and define the matrix $A \in \mathbb{R}^{n \times m}$ by

$$A_{ij} = \left(\frac{i-1}{n-1} \right)^{j-1}.$$

This is the matrix of the m first monomials x^{j-1} evaluated at an equidistant grid of n points in the interval $[0, 1]$. Test your QR implementation on A for various n, m , say $m = 10$ and $m = 200$. In particular, compute the difference between your factorization and A , and compare also with the NumPy implementation.

15 The singular-value decomposition

The singular-value decomposition gives orthogonal matrix expansions and low-rank approximations, including image compression.

The SVD is a powerful linear algebra tool. In quantum chemistry it is used to compress electron-repulsion integrals, for example.

In this small exercise, we demonstrate its power applied to *image compression*. For this exercise, a good grayscale image is handy. You can find a nice picture of a parrot in Figure 15.1.

Recall first the *Hilbert–Schmidt* inner product on the set $M(n, m, \mathbb{F})$ of matrices:

$$\langle X, Y \rangle_{\text{HS}} = \text{Tr}(X^H Y)$$

The *trace* $\text{Tr}(A)$ is the sum of the diagonal elements of A .

Recall also the SVD: For $A \in M(n, m, \mathbb{F})$ and $k = \min(n, m)$, there are unitary matrices $U \in M(n)$ and $V \in M(m)$ and singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k \geq 0$ such that

$$A = U \Sigma V^H = \sum_{i=1}^k \mathbf{u}_i \sigma_i \mathbf{v}_i^H.$$

Here $\Sigma \in M(n, m, \mathbb{R})$ has the singular values in its upper $k \times k$ diagonal block and zeros elsewhere. Removing unused columns gives the economical SVD

$$A = U_{:, :k} \Sigma_{:, :k} V_{:, :k}^H.$$



Figure 15.1: A parrot

Exercises

Exercise 15.1.

1. How is the Hilbert–Schmidt inner product related to the Euclidean inner product and the standard Euclidean vector space?
2. Show that $\text{Tr}(ABC) = \text{Tr}(CAB)$ (cyclic invariance).
3. Let $\mathbf{u}_i$ and $\mathbf{v}_i$ be the left and right singular vectors of a matrix A . Show that the matrices $X^{ij} = \mathbf{u}_i \mathbf{v}_j^H$ form an orthonormal set in the Hilbert–Schmidt inner product.
4. Conclude that the SVD is an expansion of a matrix in an orthonormal basis.

The *truncated* SVD

$$A^{(m)} = \sum_{i=1}^m \mathbf{u}_i \sigma_i \mathbf{v}_i^H$$

is the best approximation to A of rank at most m in the Hilbert–Schmidt norm:

$$A^{(m)} = \arg \min_{\text{rank}(B) \leq m} \|B - A\|_{\text{HS}}.$$

1. Write a Python program/notebook that reads an image file and converts it to a matrix $A \in M(n, m, \mathbb{R})$. The module `imageio` can be useful.
2. Continue writing the program, so it computes the truncated and full SVD of A . You can use, say, `scipy.linalg.svd` to find the full SVD.
3. Plot the singular values. Discuss.
4. Show that the error in the Hilbert–Schmidt norm is

$$\|A^{(m)} - A\|_{\text{HS}} = \sqrt{\sum_{i=m+1}^k \sigma_i^2}.$$

5. Make visualizations of the truncated SVD of the image A , with various m . Compute the relative error $\|A^{(m)} - A\|_{\text{HS}} / \|A\|_{\text{HS}}$ for each m you consider. Discuss the image quality.

16 Polynomial vector spaces

Polynomials form finite-dimensional vector spaces on which differentiation, multiplication, and changes of basis act as linear maps.

In this exercise, we study a vector space that is not $\mathbb{F}^n$. We start out without an inner product, but supply this later.

Recall that a polynomial of degree n over $\mathbb{F}$ has the form

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

with $a_i \in \mathbb{F}$ and $a_n \neq 0$.

Exercises

Exercise 16.1. Let P_n be the set of all polynomials of degree less than or equal to n . Show that P_n is a vector space of dimension $n + 1$ over $\mathbb{C}$.

Exercise 16.2. Let $D : P_n \rightarrow P_n$ be the differentiation operator, i.e.,

$$(Dp)(x) = p'(x).$$

Show that D is indeed a linear operator.

Exercise 16.3. Does there exist an inverse of D ?

Exercise 16.4. Let $B = \{1, x, x^2, \dots, x^n\}$ be the basis of *monomials*. Using bra-ket notation, we write the basis operator as

$$\hat{B} = \sum_{i=1}^{n+1} |x^{i-1}\rangle \langle i|,$$

where $|i\rangle$ is a standard basis vector for $\mathbb{F}^{n+1}$. Compute the matrix of D relative to this basis. (Note carefully that we do not have an inner product defined.) That is, find a matrix T such that

$$D\hat{B} = \hat{B}T.$$

We now restrict our attention to the interval $x \in [-1, 1]$. That is,

$$V_n = \{p : [-1, 1] \rightarrow \mathbb{C} \mid p \text{ a polynomial of deg} \leq n\}.$$

The above subexercises did not depend on the domain of the polynomial.

We define an inner product on V ,

$$\langle p|q \rangle := \int_{-1}^1 \overline{p(x)}q(x) dx.$$

We will now define the *Legendre polynomials*.

Exercise 16.5. Show that $\langle \cdot | \cdot \rangle$ is indeed an inner product by checking the axioms.

Exercise 16.6. Compute the overlap matrix S of the monomial basis. Is the basis orthonormal?

Exercise 16.7. Formulate the Gram–Schmidt procedure using the bra-ket notation. Explain how the procedure generates an upper triangular matrix R such that

$$\hat{B} = \hat{U}R,$$

where $\hat{U}$ is the basis matrix of the orthonormal set of vectors.

Exercise 16.8. Use the Gram–Schmidt decomposition by hand to compute the first three *normalized Legendre polynomials* $|\hat{P}_i\rangle$, $i = 0, 1, 2$, defined using the Gram–Schmidt orthogonalization of $\{1, x, x^2\}$.

The Legendre polynomials are orthogonal polynomials $P_i : [-1, 1] \rightarrow \mathbb{R}$ obtained from the monomials by Gram–Schmidt orthogonalization and normalized by $P_i(1) = 1$. They are orthogonal but not normalized. The endpoint normalization gives

$$\int_{-1}^1 P_i(x)P_j(x) dx = \frac{2}{2i+1}\delta_{ij}.$$

With $P_0(x) = 1$ and $P_{-1} \equiv 0$, they satisfy the recurrence relation

$$(i+1)P_{i+1}(x) = (2i+1)xP_i(x) - iP_{i-1}(x).$$

From this, one can show:

$$P'_{i+1}(x) = \frac{2P_i(x)}{\|P_i\|^2} + \frac{2P_{i-2}(x)}{\|P_{i-2}\|^2} + \frac{2P_{i-4}(x)}{\|P_{i-4}\|^2} + \dots$$

Exercise 16.9. Compute the matrix of the differential operator D in the *normalized* Legendre polynomial basis, both by using matrix multiplication involving X , R , and R^{-1} , and also by using the derivative relation above. Compare the two explicitly using pen-and-paper calculations.

Exercise 16.10. Repeat the previous exercise with the operator D^2 .

Orthogonal polynomials define efficient *quadrature rules*. Consider approximating an integral by

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n f(x_i) w_i,$$

where the w_i are *weights* and the x_i are *nodes*. A unique choice of nodes and weights makes the rule exact for every polynomial of degree at most $2n - 1$. The nodes are the zeros of P_n , and the weights are

$$w_i = \frac{2}{(1 - x_i^2)[P_n'(x_i)]^2}.$$

The [Wikipedia page on Gauss–Legendre quadrature](#) contains more details.

The so-called Golub–Welsch algorithm is a simple algorithm based on diagonalization of a tridiagonal Hermitian matrix for computing the nodes and weights of Gaussian quadratures, see [the Wikipedia page on Gaussian quadrature](#). The interested student should check it out!

Part IV

Polynomials

17 Polynomials

Polynomials are among the simplest functions we encounter in mathematics, but they appear almost everywhere. A polynomial in one variable x has the form

$$P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

where the numbers $a_0, a_1, \dots, a_n$ are called the **coefficients**. If $a_n \neq 0$, we say that the polynomial has **degree** n . For example,

$$P(x) = 2x^3 - x^2 + 4x - 7$$

is a polynomial of degree 3. Polynomials of degree 1, 2, and 3 are commonly called linear, quadratic, and cubic polynomials, respectively.

Polynomials are important in their own right, but they also occur naturally in linear algebra and quantum mechanics. For example, the eigenvalues of a matrix A are found by solving the **characteristic equation**

$$\det(A - \lambda I) = 0.$$

See Chapter 9. For a finite-dimensional matrix, $\det(A - \lambda I)$ is a polynomial in λ . Finding eigenvalues therefore leads directly to the problem of finding the roots of a polynomial.

17.1 Roots and factors

A number r is called a **root** or **zero** of the polynomial P if

$$P(r) = 0.$$

For example, consider

$$P(x) = x^2 - 3x + 2.$$

Since

$$P(1) = 1 - 3 + 2 = 0$$

and

$$P(2) = 4 - 6 + 2 = 0,$$

both 1 and 2 are roots. Indeed, the polynomial can be factorized as

$$P(x) = (x - 1)(x - 2).$$

This illustrates an important general result, the **factor theorem**:

$$P(r) = 0 \iff (x - r) \text{ is a factor of } P(x).$$

Thus there is a close connection between **finding roots** and **factorizing polynomials**. If we know that r is a root of P , we know that

$$P(x) = (x - r)Q(x)$$

for some polynomial Q whose degree is one less than that of P .

For a quadratic polynomial

$$ax^2 + bx + c,$$

the roots can always be found using the quadratic “ABC-formula”,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

For polynomials of higher degree, finding the roots can be more difficult. A useful strategy is therefore to find one root first, remove its corresponding linear factor, and reduce the problem to a polynomial of lower degree.

17.2 Division of polynomials

Polynomials can be divided in much the same way as integers. For integers, division with remainder gives, for example,

$$17 = 3 \cdot 5 + 2.$$

Similarly, if we divide a polynomial $P(x)$ by another polynomial $D(x)$, we can write

$$P(x) = Q(x)D(x) + R(x),$$

where $Q(x)$ is the **quotient** and $R(x)$ is the **remainder**. The degree of the remainder must be smaller than the degree of the divisor D .

For example, let us divide

$$x^3 - 2x^2 - 5x + 6$$

by $x - 1$. Using polynomial long division, we write

For example, let us divide

$$P(x) = x^3 - 2x^2 - 5x + 6$$

by

$$D(x) = x - 1.$$

Polynomial long division can be arranged as follows:

$$\begin{array}{r}
 x^3 \quad -2x^2 \quad -5x \quad +6 : x - 1 = x^2 - x - 6 \\
 \underline{-x^3 \quad +x^2} \\
 -x^2 \quad -5x \quad +6 \\
 \underline{+x^2 \quad -x} \\
 -6x \quad +6 \\
 \underline{+6x \quad -6} \\
 0
 \end{array}$$

At each stage, we choose the next term in the quotient so that its product with the leading term of the divisor cancels the leading term on the left. Thus the quotient is

$$Q(x) = x^2 - x - 6,$$

and the remainder is zero. Therefore,

$$P(x) = (x - 1)(x^2 - x - 6).$$

Thus

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6).$$

The quadratic factor can now be factorized further:

$$x^2 - x - 6 = (x - 3)(x + 2),$$

so

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(x - 3)(x + 2).$$

Hence the three roots are

$$x = 1, \quad x = 3, \quad x = -2.$$

17.3 Finding roots

For simple polynomials with integer coefficients, it is often possible to guess a root by trying a few small integers. A more systematic version of this idea is provided by the **rational root theorem**. If

$$P(x) = a_n x^n + \cdots + a_1 x + a_0$$

has integer coefficients and a rational root

$$r = \frac{p}{q},$$

written in lowest terms, then p must divide the constant coefficient a_0 , and q must divide the leading coefficient a_n .

In the particularly common case of a **monic polynomial**, where $a_n = 1$, every rational root must therefore be an integer that divides the constant term. For

$$P(x) = x^3 - 2x^2 - 5x + 6,$$

the only possible integer roots are consequently

$$\pm 1, \pm 2, \pm 3, \pm 6.$$

Trying these candidates quickly reveals that $P(1) = 0$. We can then divide by $x - 1$, as above, and solve the remaining quadratic equation.

The general strategy for simple polynomial equations is therefore:

1. Look for a root, for example by testing the candidates suggested by the rational root theorem.
2. If r is a root, use the factor theorem to conclude that $(x - r)$ is a factor.
3. Divide the polynomial by $(x - r)$ using polynomial long division.
4. Find the roots of the polynomial of lower degree.
5. Continue until the polynomial has been completely factorized.

A polynomial of degree n has exactly n complex roots when roots are counted with their **multiplicities**. This is the **fundamental theorem of algebra**. Some or all of these roots may be real. For example,

$$x^2 + 1 = 0$$

has no real roots, but it has the two complex roots i and $-i$, and

$$x^2 + 1 = (x - i)(x + i).$$

This is one reason complex numbers arise naturally in mathematics: over the complex numbers, every polynomial can ultimately be factorized into linear factors.

Exercises

Exercise 17.1 (Polynomial long division: roots of a quartic polynomial). Consider the polynomials

$$P(x) = x^4 - 6x^3 + 3x^2 + 26x - 24$$

and

$$d(x) = x^2 + x - 2.$$

- Perform the polynomial long division of $P(x)$ by $d(x)$ and verify that the remainder vanishes.
- Use the result of part (a) and the quadratic formula to find all four roots of $P(x)$.
- Write $P(x)$ as a product of four linear factors.

Exercise 17.2 (Roots of a cubic polynomial by guessing and polynomial long division). Every cubic equation

$$ax^3 + bx^2 + cx + d = 0, \quad a \neq 0,$$

can in principle be solved by a general formula, but that formula is rather cumbersome. For simple polynomial equations there is often an easier approach.

The **rational root theorem** says that if a polynomial with integer coefficients has a rational root

$$x = \frac{p}{q},$$

written in lowest terms, then p must divide the constant term d , while q must divide the leading coefficient a .

Once one root x_0 has been found, $(x - x_0)$ is a factor of the polynomial. We can divide out this factor using polynomial long division, leaving a quadratic polynomial.

Consider

$$x^3 - 2x^2 - 5x + 6 = 0.$$

- Use the rational root theorem to list all possible rational roots. Test these candidates until you find one root.
- Divide $x^3 - 2x^2 - 5x + 6$ by the corresponding linear factor using polynomial long division.
- Determine the remaining two roots from the resulting quadratic polynomial.
- Write the original polynomial as a product of three linear factors.

Solution for Exercise 17.2

- The leading coefficient is 1 and the constant term is 6. Therefore, any rational root must be an integer that divides 6. The possible roots are

$$x = \pm 1, \pm 2, \pm 3, \pm 6.$$

Testing $x = 1$ gives

$$1^3 - 2(1)^2 - 5(1) + 6 = 1 - 2 - 5 + 6 = 0.$$

Thus $x = 1$ is a root, and therefore $(x - 1)$ is a factor.

- We divide

$$x^3 - 2x^2 - 5x + 6$$

by $x - 1$.

The first term of the quotient is

$$\frac{x^3}{x} = x^2.$$

Multiplying,

$$x^2(x - 1) = x^3 - x^2.$$

Subtracting gives

$$-x^2 - 5x + 6.$$

The next quotient term is

$$\frac{-x^2}{x} = -x.$$

Multiplying,

$$-x(x-1) = -x^2 + x.$$

Subtracting gives

$$-6x + 6.$$

The final quotient term is

$$\frac{-6x}{x} = -6.$$

Multiplying,

$$-6(x-1) = -6x + 6.$$

The remainder is zero. Therefore,

$$x^3 - 2x^2 - 5x + 6 = (x-1)(x^2 - x - 6).$$

c. We solve

$$x^2 - x - 6 = 0.$$

Using the quadratic formula,

$$x = \frac{1 \pm \sqrt{1+24}}{2} = \frac{1 \pm 5}{2}.$$

Hence

$$x = 3 \quad \text{or} \quad x = -2.$$

Together with the root from part (a), the three roots are therefore

$$x = 1, \quad x = 3, \quad x = -2.$$

d. The complete factorization is

$$x^3 - 2x^2 - 5x + 6 = (x-1)(x-3)(x+2).$$

Part V

Metric spaces and topology

18 Metric spaces

A metric measures distance through symmetry, positivity, and the triangle inequality.

A *metric* encodes the intuition behind measuring *distances* in a set M . It is a distance function $d(x, y)$, where $x, y \in M$.

The definition of a metric space is as follows:

Definition: Metric. Let M be a set. A function $d : M \times M \rightarrow \mathbb{R}$ is a *metric* if it satisfies the following axioms:

1. $d(x, y) = d(y, x)$
2. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$
3. $d(x, y) \leq d(x, z) + d(z, y)$

The pair (M, d) is a *metric space*. If M is a vector space, we say that (M, d) is a *metric vector space*.

In this exercise, we will try to understand why the axioms are as they are, i.e., what makes them encode the concept of a metric.

Exercises

Exercise 18.1.

1. Consider two persons X and Y located at x and y in a set M . They both measure the distance to the other person. What would happen if axiom a) is violated? Is this reasonable?
2. What would happen if axiom b) is violated? Is this reasonable?
3. A third person Z located at z enters the picture. Consider axiom c). What happens if this axiom is violated? Is this reasonable?

4. Let $M = \mathbb{R}^2$, and let d be Euclidean distance, “as the crow flies”. But there are other distance measures as well. Consider for example the “Manhattan distance”, given by

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|.$$

This has a standard interpretation in terms of cities with straight streets and taxis driving on these streets. Discuss.

5. Consider a king moving on a chessboard. The king can move one step at a time, either horizontally or vertically, or diagonally. Consider the *minimum number of steps* the king has to move in order to get from one point to another. See Figure 18.1 for an illustration. The metric is

$$d(\mathbf{x}, \mathbf{y}) = \max(|x_1 - y_1|, |x_2 - y_2|).$$

This is also called the “maximum metric”. Can you verify the metric axioms? How does the maximum metric compare to the Manhattan metric?

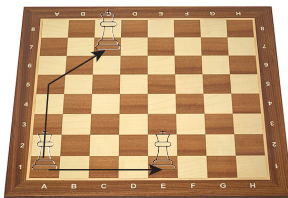


Figure 18.1: Illustration of “chessboard metric”: The king can move horizontally, vertically, and diagonally. The chessboard metric is the smallest number of steps the king must move in order to get from one point to another. Here, the king starts at A1, and two paths are shown: The optimal path to C7, and the optimal path to E1. Note that the king will move diagonally as much as he can, because he travels “faster” that way!

19 Open and closed sets in Euclidean space

Open balls define interior points, open sets, closed sets, boundaries, and closures in Euclidean space.

Exercises

Exercise 19.1. Show that any ϵ -ball is open.

Exercise 19.2. Show that $A \cup \partial A$ is closed.

Exercise 19.3. Show that if $A \subset \mathbb{R}^n$ is closed and a sequence $\mathbf{x}_i \in A$ converges to some $\mathbf{x} \in \mathbb{R}^n$, then $\mathbf{x} \in A$. This gives an equivalent characterization of closed subsets of $\mathbb{R}^n$.

Exercise 19.4. Show that the closure of A , the smallest closed set $\bar{A}$ that contains A , is equal to $A \cup \partial A$.

Exercise 19.5. Show that

$$A = \{(x, 0) \mid x \in [-1, 1]\} \subset \mathbb{R}^2$$

is closed.

Exercise 19.6. Show that $\mathbb{Q} \subset \mathbb{R}$ is neither open nor closed. Show that the boundary of $\mathbb{Q}$ is $\mathbb{R}$. Show that the interior of $\mathbb{Q}$ is empty. What is the closure of $\mathbb{Q}$?

Exercise 19.7. Let

$$A = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x < 1, \quad 0 \leq y < 1\}.$$

Compute the interior of A , the boundary of A , the closure of A .

Part VI

Calculus

Part VII

Multivariable calculus

20 Visualizing functions

In these exercises, you are given functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and asked to visualize them.

Exercises: paths and their curves

Exercise 20.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}^2$ be a path defined by

$$f(t) = [r(t) \cos(4t), r(t) \sin(4t)], \quad r(t) = \exp(-t).$$

Sketch the curve traced by the path for $0 \leq t \leq 2\pi$. You can check your sketch against, say, a Python plot.

Exercise 20.2. Let $f : (0, +\infty) \rightarrow \mathbb{R}^2$ be a path defined by

$$f(t) = [\ln(t), t].$$

Sketch the curve traced by the path. You can check your sketch against, say, a Python plot.

Exercise 20.3 (A heart-shaped curve). Consider the parametrically defined curve

$$x(t) = 16 \sin^3 t, \quad y(t) = 13 \cos t - 5 \cos(2t) - 2 \cos(3t) - \cos(4t),$$

where $0 \leq t \leq 2\pi$.

- Evaluate $(x(t), y(t))$ for $t = 0, \pi/2, \pi, 3\pi/2$, and 2π .
- Use these points, together with the symmetry of the curve, to make a rough sketch by hand.
- Plot the curve using Python. Use equal scaling on the two coordinate axes.

Exercises: level curves

Exercise 20.4. Let $f(x, y) = x^2 + y^2$. Sketch the level curves $\{(x, y) \mid f(x, y) = n\}$, for $n = 1$ and $n = 2$, $n = 3$, and $n = 4$.

Exercise 20.5. Let $f(x, y) = x + y + 2$. Sketch the level curves where $f(x, y) = 0$, 2 and 4. Can you sketch the graph of f ?

Exercise 20.6. Let $f(x, y) = x^2 - y^2$. The graph is called the *hyperbolic paraboloid*, or a saddle. Sketch the graph. Sketch the level curves $\{(x, y) \mid f(x, y) = n\}$, for $n = 0$ and $n = -1$, $n = 1$. Make sure that you get all “pieces”.

Exercise 20.7. Write a Python program for sketching the level curves in the hyperbolic-paraboloid exercise above, but use more closely spaced constants. There are tools in Matplotlib that are handy.

Exercise 20.8. Adapt your program to draw level curves of $(x, y) \mapsto (x^2 + 3y^2)e^{1-x^2-y^2}$.

Exercises: sections and level surfaces

Consider the hydrogen-atom eigenfunctions in atomic units. They are indexed by $n \geq 1$, $0 \leq \ell < n$, and $-\ell \leq m \leq \ell$. Their separated form is

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi),$$

with

$$R_{n\ell}(r) = N_{n\ell} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) e^{-\rho/2}, \quad \rho = \frac{2r}{n}.$$

Here $N_{n\ell}$ is a normalization constant. Coordinate conversion uses

$$r = \sqrt{x^2 + y^2 + z^2}, \quad x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

In particular, the $1s$ and $2p_z$ functions are given by

$$\psi_{1s}(x, y, z) = \frac{1}{\sqrt{\pi}} e^{-r(x, y, z)},$$

$$\psi_{2p_z}(x, y, z) = \frac{1}{4\sqrt{2\pi}} z e^{-r(x, y, z)/2}.$$

Exercise 20.9. Write a Python program to draw the level *surfaces* of hydrogen orbitals, $\{(x, y, z) \mid \psi(x, y, z) = c\}$. Choose interesting values of c . It can be interesting to color code surfaces of opposite positive and negative c .

Exercise 20.10. Write a program to visualize *sections* of the orbital graphs. For example, fix $y = y_{\text{const}}$ and plot $(x, z) \mapsto \psi(x, y_{\text{const}}, z)$.

21 Continuity and differentiability

Continuity and differentiability are fundamental concepts in calculus. For functions of several variables, the basic ideas are the same as for functions of one variable, but there are some important new features. In particular, a point in $\mathbb{R}^n$ can be approached from infinitely many directions, and a function can have partial derivatives without being differentiable.

Continuity

Consider a function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

We say that f is **continuous at $\mathbf{a} \in \mathbb{R}^n$** if

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

More precisely, f is continuous at $\mathbf{a}$ if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\|\mathbf{x} - \mathbf{a}\| < \delta \implies |f(\mathbf{x}) - f(\mathbf{a})| < \epsilon.$$

Here $\|\mathbf{x} - \mathbf{a}\|$ denotes the Euclidean distance between $\mathbf{x}$ and $\mathbf{a}$. For a function of two variables, $f(x, y)$, continuity at (a, b) therefore means that $f(x, y)$ approaches $f(a, b)$ whenever (x, y) approaches (a, b) .

An important difference from one-dimensional calculus is that (x, y) can approach (a, b) along infinitely many different paths. If different paths give different limiting values, the limit does not exist.

Partial derivatives

For a function of a single variable, the derivative is the slope, or rate of change, when that variable is increased by a small amount. For a function of several variables, however, we can study how the function changes when one variable is varied while all the other variables are held constant. This defines **partial derivatives**.

For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the partial derivative with respect to x at (a, b) is

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h},$$

provided that the limit exists. Similarly,

$$\frac{\partial f}{\partial y}(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

Thus, $\partial f / \partial x$ measures the rate of change of f in the x direction, while $\partial f / \partial y$ measures the rate of change in the y direction.

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the partial derivatives can be collected into the **gradient**

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}.$$

The gradient will play an important role throughout multivariable calculus. In quantum chemistry, for example, the gradient of an energy with respect to nuclear coordinates gives information about how the energy changes when the molecular geometry is displaced.

For functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the matrix of partial derivatives is called **the Jacobian**,

$$J_f = \frac{\partial(f_1, \dots, f_m)}{\partial(x_1, \dots, x_n)} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}.$$

The function f is differentiable at $\mathbf{x}_0$ if the Jacobian matrix exists and provides a good linear approximation to the function near that point:

$$f(\mathbf{x}_0 + \mathbf{h}) \approx f(\mathbf{x}_0) + J_f(\mathbf{x}_0)\mathbf{h},$$

Differentiability

In single-variable calculus, differentiability means that sufficiently close to a point, a function is well approximated by a straight line. The same idea applies in several dimensions: a differentiable function is locally well approximated by a **linear function**.

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we say that f is **differentiable at a** if

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + r(\mathbf{h}),$$

where the remainder $r(\mathbf{h})$ becomes negligible compared with $\|\mathbf{h}\|$ as $\mathbf{h} \rightarrow \mathbf{0}$:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|r(\mathbf{h})|}{\|\mathbf{h}\|} = 0.$$

Equivalently,

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + o(\|\mathbf{h}\|).$$

For a function $f(x, y)$, writing $\mathbf{h} = (h, k)$ gives the perhaps more familiar expression

$$f(a + h, b + k) \approx f(a, b) + \frac{\partial f}{\partial x}(a, b)h + \frac{\partial f}{\partial y}(a, b)k.$$

Geometrically, the graph $z = f(x, y)$ is therefore approximated near $(a, b, f(a, b))$ by its tangent plane.

It is important to distinguish **partial differentiability** from **differentiability**. The existence of all partial derivatives at a point does *not* by itself imply that the function is differentiable there. Partial derivatives examine the function only along the coordinate directions, whereas differentiability requires a single linear approximation that works for small displacements in *all* directions.

There is, however, a very useful sufficient condition: if all the first partial derivatives exist and are continuous in a neighbourhood of a point, then the function is differentiable at that point. Thus, for the smooth functions encountered in most applications, differentiability can usually be established by computing the partial derivatives and checking that they are continuous.

Exercises: Continuity

Exercise 21.1 (A first discontinuity). Consider

$$f(x) = \begin{cases} 1, & x > 0, \\ -1, & x \leq 0. \end{cases}$$

- Compute $f(-1)$, $f(0)$, and $f(1)$.
- For $x = 1$, what value does $f(x + h)$ have when h is sufficiently small? What is $\lim_{h \rightarrow 0} f(1 + h)$?
- Approach $x = 0$ from the left and from the right. What values does $f(h)$ take?
- For which $x \in \mathbb{R}$ does $\lim_{h \rightarrow 0} f(x + h)$ not exist? Explain in words.

Exercise 21.2 (Getting close in two dimensions). Let $\mathbf{a} = (1, 2)$ and $\mathbf{x} = (x, y)$.

- Write an expression for the Euclidean distance $\|\mathbf{x} - \mathbf{a}\|$.
- Compute this distance for $\mathbf{x} = (1.1, 2)$, $(1, 1.9)$, and $(1.06, 2.08)$.
- Which of these points lie inside the disk $\|\mathbf{x} - \mathbf{a}\| < 0.1$?
- In your own words, explain what $\mathbf{x} \rightarrow \mathbf{a}$ means.

Exercise 21.3 (Continuity by direct substitution). For each function below, evaluate the indicated limit by direct substitution. State briefly why this is allowed.

- $\lim_{(x,y) \rightarrow (1,2)} (x + 2y)$.
- $\lim_{(x,y) \rightarrow (1,2)} x^2 y$.

- c. $\lim_{(x,y) \rightarrow (0,0)} e^{-(x^2+y^2)}.$
- d. $\lim_{(x,y) \rightarrow (1,1)} \frac{x+y}{1+x^2+y^2}.$

Exercise 21.4 (Testing two paths). Consider

$$f(x, y) = \frac{x^2}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

- Set $y = 0$ and simplify $f(x, 0)$ for $x \neq 0$.
- Set $x = 0$ and simplify $f(0, y)$ for $y \neq 0$.
- What value is approached along each axis as $(x, y) \rightarrow (0, 0)$?
- Can $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist? Why or why not?

Exercise 21.5 (Approaching a point with Python). Consider $f(x, y) = x^2 + y^2$.

- By hand, compute $f(0.1, 0.1)$, $f(0.01, 0.01)$, and $f(0.001, 0.001)$.
- Write a short Python loop that evaluates $f(t, t)$ for $t = 10^{-1}, 10^{-2}, \dots, 10^{-6}$.
- What value do the results appear to approach?
- Repeat for the path $(x, y) = (t, 2t)$. Does the limiting value change?

Exercise 21.6 (Why polynomials are continuous). Use standard properties of continuous functions to show that every polynomial $p : \mathbb{R} \rightarrow \mathbb{R}$ is continuous. Then explain briefly why a polynomial in two variables, such as $p(x, y) = 3x^2y - 2xy + 7$, is also continuous on $\mathbb{R}^2$.

Exercise 21.7 (A radial limit). Consider

$$f(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

- Introduce $r^2 = x^2 + y^2$ and rewrite the function in terms of r .
- Determine $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ using the one-variable limit $\lim_{t \rightarrow 0} \sin t/t = 1$.
- How could f be defined at $(0, 0)$ so that the resulting function is continuous there?

Exercise 21.8 (A path-dependent limit). Consider

$$f(x, y) = \frac{x^2}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

- Evaluate the limit along the line $y = 0$.
- Evaluate the limit along the line $x = 0$.
- Conclude whether $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists.
- As an optional visualization, make a surface plot or contour plot near the origin and relate what you see to parts a–c.

Exercise 21.9 (Bounding a function near the origin). Consider

$$f(x, y) = \frac{2x^2y}{x^2 + y^2}, \quad (x, y) \neq (0, 0).$$

- Show that $|f(x, y)| \leq 2|y|$.
- Use $|y| \leq \sqrt{x^2 + y^2}$ to obtain a bound in terms of $\|(x, y)\|$.
- Use the bound to prove with an ϵ – δ argument that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

Exercise 21.10 (Making a function continuous). For $(x, y) \neq (0, 0)$, define

$$f(x, y) = \frac{x^2y^2}{x^2 + y^2}.$$

- Show that $0 \leq f(x, y) \leq y^2$.
- Determine $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.
- Choose a value for $f(0, 0)$ that makes f continuous at the origin.
- Check your conclusion numerically along the paths $y = x$ and $y = x^2$.

Exercise 21.11 (Continuity of a model potential energy). A simple model potential energy is

$$E(x, y) = \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2 + cxy,$$

where k_x , k_y , and c are constants.

- Explain why E is continuous everywhere.
- Evaluate $\lim_{(x,y) \rightarrow (0,0)} E(x, y)$.
- Let $k_x = 2$, $k_y = 3$, and $c = 1$. Compute $E(0.1, 0)$, $E(0, 0.1)$, and $E(0.1, 0.1)$.
- What do these values suggest about the energy surface close to the origin?

Exercises: Differentiability

Exercise 21.12 (First partial derivatives). Let $f(x, y) = x^2 + 3xy + y^2$.

- To compute $\partial f / \partial x$, treat y as a constant. Find $\partial f / \partial x$.
- To compute $\partial f / \partial y$, treat x as a constant. Find $\partial f / \partial y$.
- Evaluate both partial derivatives at $(1, 2)$.
- Write the gradient $\nabla f(1, 2)$ as a column vector.

Exercise 21.13 (Partial derivatives as one-dimensional derivatives). Let $f(x, y) = x^2y + y^3$ and consider the point $(1, 2)$.

- Hold $y = 2$ fixed and define $g(x) = f(x, 2)$. Write down $g(x)$ and compute $g'(1)$.
- Hold $x = 1$ fixed and define $h(y) = f(1, y)$. Write down $h(y)$ and compute $h'(2)$.
- Compute $\partial f / \partial x$ and $\partial f / \partial y$ directly and check your answers from a and b.
- Explain in one sentence what is held fixed when taking a partial derivative.

Exercise 21.14 (Using a gradient for a small displacement). Suppose an energy $E(x, y)$ has

$$E(1, 1) = 5, \quad \nabla E(1, 1) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

- For a small displacement $\mathbf{h} = (0.01, 0)$, use the linear approximation to estimate the change in energy.
- Repeat for $\mathbf{h} = (0, 0.01)$.
- Repeat for $\mathbf{h} = (0.01, 0.01)$.
- Estimate $E(1.01, 1.01)$.

Exercise 21.15 (A first Jacobian). Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be

$$F(x, y) = \begin{pmatrix} x + y \\ xy \end{pmatrix}.$$

- Identify the two component functions F_1 and F_2 .
- Compute $\partial F_1 / \partial x$ and $\partial F_1 / \partial y$.
- Compute $\partial F_2 / \partial x$ and $\partial F_2 / \partial y$.
- Assemble the four derivatives into the Jacobian matrix $J_F(x, y)$ and evaluate it at $(1, 2)$.

Exercise 21.16 (Partial derivatives of a radial quotient). For $(x, y) \neq (0, 0)$, let

$$f(x, y) = \frac{xy}{\sqrt{x^2 + y^2}}.$$

- Compute $\partial f / \partial x$.
- Compute $\partial f / \partial y$.
- Evaluate the gradient at $(1, 1)$.
- Is the formula defined at the origin? What additional question would have to be answered before discussing differentiability there?

Exercise 21.17 (Linear approximation). Let $f(x, y) = e^x \cos y$.

- Compute $\nabla f(x, y)$.
- Find the linear approximation to f near $(0, 0)$.
- Use it to estimate $f(0.02, 0.03)$.
- Compare with a Python evaluation of $\exp(0.02) * \cos(0.03)$.

Exercise 21.18 (Tangent plane to an energy surface). Consider $E(x, y) = x^2 + xy + 2y^2$.

- Compute $\nabla E(x, y)$.
- Find $E(1, 1)$ and $\nabla E(1, 1)$.
- Write the tangent-plane approximation to E near $(1, 1)$.
- Use it to estimate $E(1.02, 0.99)$ and compare with the exact value.

Exercise 21.19 (Checking differentiability efficiently). Let

$$f(x, y) = x^3 \sin y + e^{xy}.$$

- Compute the two first partial derivatives.
- Explain why these partial derivatives are continuous everywhere.
- Use the sufficient condition from the text to conclude that f is differentiable everywhere.
- Compute $\nabla f(0, 0)$ and write the linear approximation near the origin.

Exercise 21.20 (A numerical gradient with finite differences). Let $f(x, y) = x^2 + xy + 3y^2$.

- Compute $\nabla f(1, 2)$ analytically.
- Approximate $\partial f / \partial x$ at $(1, 2)$ by

$$\frac{f(1+h, 2) - f(1, 2)}{h}$$

for $h = 10^{-1}, 10^{-2}, 10^{-3}$.

- Do the same for $\partial f / \partial y$.

- d. Write a tiny Python script for b and c. How do the approximations behave as h becomes smaller?

Exercise 21.21 (Checking multivariable limits along different paths). Define

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

- Along a straight line $y = mx$, show that $f(x, mx) \rightarrow 0$ as $x \rightarrow 0$.
- Along the curved path $y = x^2$, determine the limiting value.
- Is f continuous at the origin?
- What does this example teach you about checking multivariable limits using paths?

Exercise 21.22 (Partial derivatives without continuity). Define

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

- Show that f is continuous at the origin by finding a useful bound for $|f(x, y)|$.
- Compute $\partial f / \partial x(0, 0)$ and $\partial f / \partial y(0, 0)$ directly from the definitions.
- Test differentiability at the origin along the path $(x, y) = (t, t)$.
- Reconcile your result with the statement that continuous first partial derivatives are sufficient for differentiability.

Exercise 21.23 (Explore a two-dimensional energy landscape). Consider the model energy

$$E(x, y) = (x^2 - 1)^2 + y^2 + \frac{1}{2}xy.$$

- Compute the gradient $\nabla E(x, y)$.
- Use Python to make a contour plot of E on $-2 \leq x \leq 2$, $-2 \leq y \leq 2$.

- c. On the same grid, make a quiver plot of $-\nabla E$, or plot a separate vector field if that is clearer.
- d. Locate the stationary points numerically or analytically. Which appear to be minima, maxima, or saddle points? Describe how the gradient field reflects this.

Exercise 21.24 (A numerical experiment with the definition of differentiability). Let $f(x, y) = \sin(xy) + x^2$ and let $\mathbf{a} = (1, 1)$.

- a. Compute $f(\mathbf{a})$ and $\nabla f(\mathbf{a})$.
- b. For a displacement $\mathbf{h}$, define the linearization error

$$R(\mathbf{h}) = f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

- c. Write a short Python program that evaluates $|R(\mathbf{h})|/\|\mathbf{h}\|$ for $\mathbf{h} = t(1, 2)$ and $t = 10^{-1}, \dots, 10^{-7}$.
- d. Repeat for a few random directions. What numerical behavior do you expect if f is differentiable at $\mathbf{a}$? Do you observe it?

22 Functions of a vector

In quantum chemistry, one often views multivariable functions as **functions of one or more vectors**. The archetypal example is the Rayleigh quotient in a non-orthonormal basis. Let $\mathbf{H}$ and $\mathbf{S}$ be Hermitian $n \times n$ matrices, with $\mathbf{S}$ positive definite. Think of $\mathbf{H}$ as the matrix of a one-electron Hamiltonian in a non-orthonormal AO basis $\{\chi_\mu\}_{\mu=1}^n$, and $\mathbf{S}$ as the overlap matrix. Their elements are

$$H_{\mu\nu} = \langle \chi_\mu | \hat{H} | \chi_\nu \rangle, \quad S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle.$$

For

$$\psi = \sum_{\mu=1}^n c_\mu \chi_\mu, \quad c_\mu \in \mathbb{C},$$

we collect the coefficients in the vector $\mathbf{c} \in \mathbb{C}^n$. The expectation value of $\hat{H}$ is

$$\langle \hat{H} \rangle = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\mathbf{c}^\dagger \mathbf{H} \mathbf{c}}{\mathbf{c}^\dagger \mathbf{S} \mathbf{c}}.$$

Recall that $\mathbf{c}^\dagger$ is the Hermitian transpose (conjugate transpose) of $\mathbf{c}$. The last expression is the **Rayleigh quotient**

$$R(\mathbf{c}) = \frac{\mathbf{c}^\dagger \mathbf{H} \mathbf{c}}{\mathbf{c}^\dagger \mathbf{S} \mathbf{c}}.$$

Since $\mathbf{H}$ is Hermitian and $\mathbf{S}$ is positive definite, $R(\mathbf{c})$ is real for every nonzero $\mathbf{c}$. Notice also that multiplication of $\mathbf{c}$ by a nonzero scalar does not change the quotient:

$$R(a\mathbf{c}) = R(\mathbf{c}), \quad a \neq 0.$$

Thus the Rayleigh quotient depends on the *direction* represented by $\mathbf{c}$, but not on its overall normalization or phase.

The gradient of a vector-dependent function

A function of a real vector $\mathbf{x} = (x_1, \dots, x_n)^T$ is simply a multivariable function written using vector notation,

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(\mathbf{x}) = f(x_1, \dots, x_n).$$

Its gradient is the vector of partial derivatives

$$\nabla_{\mathbf{x}} f = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}.$$

For example, if

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} = \sum_i a_i x_i,$$

then

$$\nabla_{\mathbf{x}} f = \mathbf{a}.$$

A particularly important case is a quadratic form. For a real symmetric matrix $\mathbf{A}$,

$$f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

has gradient

$$\nabla_{\mathbf{x}} f = 2\mathbf{A}\mathbf{x}.$$

One way of seeing this is to write the quadratic form with indices,

$$f(\mathbf{x}) = \sum_{ij} x_i A_{ij} x_j,$$

and differentiate with respect to one component x_k . The variable occurs twice, giving two contributions. Symmetry of $\mathbf{A}$ makes the two contributions equal.

In quantum chemistry the vectors are often complex. A real-valued function such as $R(\mathbf{c})$ then depends on both the coefficients c_i and their complex conjugates c_i^* . A convenient notation is to temporarily treat $\mathbf{c}$ and $\mathbf{c}^*$ as independent variables. For example,

$$f(\mathbf{c}, \mathbf{c}^*) = \mathbf{c}^\dagger \mathbf{A} \mathbf{c} = \sum_{ij} c_i^* A_{ij} c_j.$$

Differentiating with respect to c_k^* gives

$$\frac{\partial f}{\partial c_k^*} = \sum_j A_{kj} c_j,$$

or, in vector notation,

$$\frac{\partial}{\partial \mathbf{c}^*} (\mathbf{c}^\dagger \mathbf{A} \mathbf{c}) = \mathbf{A} \mathbf{c}.$$

Similarly,

$$\frac{\partial}{\partial \mathbf{c}} (\mathbf{c}^\dagger \mathbf{A} \mathbf{c}) = \mathbf{A}^T \mathbf{c}^*.$$

This way of differentiating with respect to a complex variable and its conjugate is closely related to **Wirtinger differentiation**. For the elementary applications here, the main practical rule is simple: when differentiating with respect to $\mathbf{c}^*$, regard $\mathbf{c}$ as constant, and vice versa.

We can now differentiate the Rayleigh quotient. Using the quotient rule,

$$\frac{\partial R}{\partial \mathbf{c}^*} = \frac{(\mathbf{H} \mathbf{c})(\mathbf{c}^\dagger \mathbf{S} \mathbf{c}) - (\mathbf{c}^\dagger \mathbf{H} \mathbf{c})(\mathbf{S} \mathbf{c})}{(\mathbf{c}^\dagger \mathbf{S} \mathbf{c})^2}.$$

Since $R(\mathbf{c}) = (\mathbf{c}^\dagger \mathbf{H} \mathbf{c})/(\mathbf{c}^\dagger \mathbf{S} \mathbf{c})$, this can be written more transparently as

$$\frac{\partial R}{\partial \mathbf{c}^*} = \frac{(\mathbf{H} - R\mathbf{S})\mathbf{c}}{\mathbf{c}^\dagger \mathbf{S} \mathbf{c}}.$$

At a stationary point the derivative vanishes. Therefore

$$(\mathbf{H} - R\mathbf{S})\mathbf{c} = 0,$$

or

$$\mathbf{H}\mathbf{c} = R\mathbf{S}\mathbf{c}.$$

Thus stationary points of the Rayleigh quotient are generalized eigenvectors of $\mathbf{H}$ with respect to $\mathbf{S}$, and the stationary value of the quotient is the corresponding generalized eigenvalue. This is one of the fundamental variational ideas underlying quantum chemistry.

The Hessian of a vector-dependent function

The gradient describes the first-order change of a scalar function. The **Hessian** describes its second-order change. For a real-valued function $f(\mathbf{x})$, the Hessian is the matrix

$$\mathbf{H}_f(\mathbf{x}) = \nabla_{\mathbf{x}}^2 f, \quad (\mathbf{H}_f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

For example, for the quadratic function

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c,$$

where $\mathbf{A}$ is real and symmetric,

$$\nabla f = \mathbf{A}\mathbf{x} + \mathbf{b}, \quad \mathbf{H}_f = \mathbf{A}.$$

The Hessian enters the multivariable Taylor expansion. For a small displacement $\Delta \mathbf{x}$,

$$f(\mathbf{x} + \Delta\mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^T \mathbf{H}_f(\mathbf{x}) \Delta\mathbf{x} + O(\|\Delta\mathbf{x}\|^3).$$

At a stationary point the gradient is zero, so the leading change is usually quadratic:

$$f(\mathbf{x} + \Delta\mathbf{x}) - f(\mathbf{x}) \approx \frac{1}{2} \Delta\mathbf{x}^T \mathbf{H}_f \Delta\mathbf{x}.$$

The signs of the Hessian eigenvalues therefore tell us about the local shape of the function. Positive eigenvalues correspond to directions in which the function curves upward, negative eigenvalues to directions in which it curves downward, and zero eigenvalues indicate flat directions that require more care.

For functions of complex vectors, the complete second-derivative structure can be written using derivatives with respect to both $\mathbf{c}$ and $\mathbf{c}^*$. We will not develop the full formalism here. The important idea is the same as in the real case: near a stationary vector, the second-order change tells us how sensitive the function is to small changes in the vector. For the Rayleigh quotient, this connects energy curvature to gaps between eigenvalues.

Exercises

Exercise 22.1 (From components to a gradient). Let $\mathbf{x} = (x_1, x_2)^T$ and

$$f(\mathbf{x}) = x_1^2 + 3x_2^2.$$

- Write f without vector notation, as a function $f(x_1, x_2)$.
- Compute $\partial f / \partial x_1$.
- Compute $\partial f / \partial x_2$.
- Collect the two partial derivatives into the gradient $\nabla_{\mathbf{x}} f$.
- Evaluate the gradient at $\mathbf{x} = (1, 2)^T$.

Exercise 22.2 (Differentiating a quadratic form). Consider

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}.$$

- Multiply out $\mathbf{Ax}$.
- Multiply out $\mathbf{x}^T \mathbf{Ax}$ and write f in terms of x_1 and x_2 .
- Differentiate the expression from part b with respect to x_1 and x_2 .
- Verify explicitly that

$$\nabla_{\mathbf{x}} f = 2\mathbf{Ax}.$$

- Use a short Python script to evaluate f and its gradient at $\mathbf{x} = (1, -1)^T$.

Exercise 22.3 (A first Rayleigh quotient). Let

$$\mathbf{H} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{S} = \mathbf{I}, \quad \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where c_1 and c_2 are real.

- Show that

$$R(\mathbf{c}) = \frac{c_1^2 + 3c_2^2}{c_1^2 + c_2^2}.$$

- Evaluate R for $\mathbf{c} = (1, 0)^T$, $(0, 1)^T$, and $(1, 1)^T$.
- Verify directly that multiplying $\mathbf{c}$ by 2 does not change R .
- The stationary vectors satisfy $\mathbf{Hc} = R\mathbf{Sc}$. Insert $\mathbf{c} = (1, 0)^T$ and determine R .
- Repeat part d for $\mathbf{c} = (0, 1)^T$. What familiar matrix problem have you recovered?

Exercise 22.4 (Treating c and c^* as independent variables). When differentiating expressions involving a complex number c and its complex conjugate c^* , it is often useful to **formally treat c and c^* as independent variables**.

This means that when differentiating with respect to c , we regard c^* as a constant, and when differentiating with respect to c^* , we regard c as a constant.

- a. Using this rule, compute

$$\frac{\partial c}{\partial c}, \quad \frac{\partial c^*}{\partial c}.$$

- b. Similarly, compute

$$\frac{\partial c}{\partial c^*}, \quad \frac{\partial c^*}{\partial c^*}.$$

- c. Consider the function

$$f(c, c^*) = c^*c.$$

Compute

$$\frac{\partial f}{\partial c}$$

by treating c^* as a constant.

- d. Compute

$$\frac{\partial f}{\partial c^*}$$

by treating c as a constant.

- e. Now consider

$$g(c, c^*) = c^*Ac,$$

where A is a constant number. Compute $\partial g / \partial c$ and $\partial g / \partial c^*$.

- f. Let $c = x + iy$. Check that $c^*c = x^2 + y^2$. Thus, although we treated c and c^* as independent when differentiating, are they actually independent complex numbers?

Exercise 22.5 (From the Rayleigh quotient to an eigenvalue problem). Let

$$R(\mathbf{c}) = \frac{\mathbf{c}^\dagger \mathbf{H} \mathbf{c}}{\mathbf{c}^\dagger \mathbf{S} \mathbf{c}},$$

where $\mathbf{H}$ and $\mathbf{S}$ are Hermitian and $\mathbf{S}$ is positive definite.

- a. Define

$$N(\mathbf{c}) = \mathbf{c}^\dagger \mathbf{H} \mathbf{c}, \quad D(\mathbf{c}) = \mathbf{c}^\dagger \mathbf{S} \mathbf{c}.$$

Show that

$$\frac{\partial N}{\partial \mathbf{c}^*} = \mathbf{H}\mathbf{c}, \quad \frac{\partial D}{\partial \mathbf{c}^*} = \mathbf{S}\mathbf{c}.$$

- b. Use the quotient rule to show that

$$\frac{\partial R}{\partial \mathbf{c}^*} = \frac{(\mathbf{H} - R\mathbf{S})\mathbf{c}}{\mathbf{c}^\dagger \mathbf{S} \mathbf{c}}.$$

- c. Show that a stationary point satisfies

$$\mathbf{H}\mathbf{c} = R\mathbf{S}\mathbf{c}.$$

- d. Suppose $\mathbf{c}$ satisfies $\mathbf{H}\mathbf{c} = \varepsilon\mathbf{S}\mathbf{c}$. Show directly that $R(\mathbf{c}) = \varepsilon$.
 e. Explain in words why minimizing the Rayleigh quotient is closely connected to finding the lowest-energy eigenvector.

Exercise 22.6 (Bonding and antibonding orbitals of H_2^+). Consider a minimal basis for H_2^+ consisting of two normalized hydrogen 1s orbitals, χ_A and χ_B . In this basis, suppose

$$\mathbf{H} = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix},$$

where all quantities are real and $0 < s < 1$.

- a. Consider the two coefficient vectors

$$\mathbf{c}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Write the corresponding molecular orbitals in terms of χ_A and χ_B . Which one has the same sign on the two atoms, and which one changes sign?

- b. Show that $\mathbf{c}_+$ and $\mathbf{c}_-$ satisfy the generalized eigenvalue equation $\mathbf{H}\mathbf{c} = \varepsilon\mathbf{S}\mathbf{c}$.
 c. Show that their energies are

$$\varepsilon_+ = \frac{\alpha + \beta}{1 + s}, \quad \varepsilon_- = \frac{\alpha - \beta}{1 - s}.$$

- d. Take $\alpha = -0.50$, $\beta = -0.20$, and $s = 0.30$ in atomic units. Compute the two energies. Which orbital is lower in energy?
- e. Normalize $\mathbf{c}_+$ and $\mathbf{c}_-$ with respect to the overlap matrix, so that $\mathbf{c}^T \mathbf{S} \mathbf{c} = 1$.
- f. Use Python and `scipy.linalg.eigh(H, S)` to calculate the generalized eigenvalues and eigenvectors numerically, and compare with your results.

Exercise 22.7 (Exploring an energy surface). Let

$$\mathbf{H} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad \mathbf{S} = \mathbf{I}.$$

Because the Rayleigh quotient is independent of the length of $\mathbf{c}$, restrict the coefficient vector to the unit circle,

$$\mathbf{c}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}.$$

- a. Insert $\mathbf{c}(\theta)$ into the Rayleigh quotient and show that

$$R(\theta) = \cos^2 \theta + 4 \sin^2 \theta.$$

- b. Differentiate $R(\theta)$ and find its stationary points for $0 \leq \theta < 2\pi$.
- c. Identify which stationary points are minima and which are maxima. Relate the corresponding vectors to the eigenvectors of $\mathbf{H}$.
- d. Write a short Python script that plots $R(\theta)$ for $0 \leq \theta \leq 2\pi$. Mark or print the minimum and maximum values.
- e. What does this example suggest about the relationship between the smallest/largest eigenvalues and the minimum/maximum of the Rayleigh quotient?

Exercise 22.8 (Second-order variation of the Rayleigh quotient). Let $\mathbf{H}$ be a real symmetric matrix with orthonormal eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ and eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

For simplicity take $\mathbf{S} = \mathbf{I}$. Consider a normalized eigenvector $\mathbf{u}_k$ and a small displacement in a perpendicular direction,

$$\mathbf{c} = \mathbf{u}_k + \delta \mathbf{v}, \quad \mathbf{u}_k^T \mathbf{v} = 0,$$

where $|\delta| \ll 1$ and $\|\mathbf{v}\| = 1$.

a. Show that

$$\mathbf{c}^T \mathbf{c} = 1 + \delta^2.$$

b. Use $\mathbf{H}\mathbf{u}_k = \lambda_k \mathbf{u}_k$ and $\mathbf{u}_k^T \mathbf{v} = 0$ to show that

$$\mathbf{c}^T \mathbf{H} \mathbf{c} = \lambda_k + \delta^2 \mathbf{v}^T \mathbf{H} \mathbf{v}.$$

c. Hence show that

$$R(\mathbf{c}) = \frac{\lambda_k + \delta^2 \mathbf{v}^T \mathbf{H} \mathbf{v}}{1 + \delta^2}.$$

d. Use $(1 + \delta^2)^{-1} = 1 - \delta^2 + O(\delta^4)$ to obtain the expansion

$$R(\mathbf{c}) = \lambda_k + \delta^2 (\mathbf{v}^T \mathbf{H} \mathbf{v} - \lambda_k) + O(\delta^4).$$

Why is there no term proportional to δ ?

e. Now choose $\mathbf{v} = \mathbf{u}_j$ with $j \neq k$. Show that

$$R(\mathbf{u}_k + \delta \mathbf{u}_j) = \lambda_k + \delta^2 (\lambda_j - \lambda_k) + O(\delta^4).$$

- f. Use this result to explain why the lowest eigenvector is a minimum of the Rayleigh quotient, the highest eigenvector is a maximum, and an eigenvector in the middle generally has both uphill and downhill directions.
- g. Choose a diagonal 3×3 matrix with three different eigenvalues and verify the second-order formula numerically with a short Python script. Try several decreasing values of δ and examine how rapidly the approximation improves.

Part VIII

Complex numbers and complex analysis

23 Complex numbers

Complex numbers extend the real numbers by introducing a number i satisfying

$$i^2 = -1.$$

They are indispensable in quantum mechanics: wave functions, state vectors, matrix elements, Fourier transforms, and time evolution are naturally complex-valued. Before using complex numbers in these settings, it is important to be comfortable with their elementary arithmetic and geometry.

23.1 Basic facts

A **complex number** has the form

$$z = a + ib,$$

where $a, b \in \mathbb{R}$. The number a is the **real part** and b is the **imaginary part**:

$$\operatorname{Re} z = a, \quad \operatorname{Im} z = b.$$

Notice that the imaginary part is the real number b , not ib .

Two complex numbers are equal precisely when their real and imaginary parts are equal:

$$a + ib = c + id \iff a = c \text{ and } b = d.$$

Real numbers are special complex numbers: $a = a + i0$. A **purely imaginary** number has zero real part, for example $3i$ or $-i/2$.

23.1.1 Arithmetic

Addition and subtraction are performed by collecting real and imaginary parts:

$$(a + ib) + (c + id) = (a + c) + i(b + d),$$

$$(a + ib) - (c + id) = (a - c) + i(b - d).$$

For multiplication, use the ordinary distributive law together with $i^2 = -1$:

$$(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc).$$

The powers of i repeat every four steps:

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

23.1.2 Complex conjugation and absolute value

The **complex conjugate** of

$$z = a + ib$$

is

$$\bar{z} = a - ib.$$

Geometrically, conjugation reflects a complex number across the real axis.

Multiplying a number by its conjugate gives a nonnegative real number:

$$z\bar{z} = (a + ib)(a - ib) = a^2 + b^2.$$

The **absolute value**, or **modulus**, is therefore

$$|z| = \sqrt{a^2 + b^2},$$

so that

$$|z|^2 = z\bar{z}.$$

This identity is useful for division. If $w \neq 0$, multiply numerator and denominator by $\bar{w}$:

$$\frac{z}{w} = \frac{z\bar{w}}{w\bar{w}} = \frac{z\bar{w}}{|w|^2}.$$

The denominator is then real.

Some useful conjugation rules are

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w}, \quad \overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}.$$

23.1.3 The complex plane

We can represent $z = a + ib$ as the point (a, b) in the **complex plane**. The horizontal axis is the real axis and the vertical axis is the imaginary axis. The modulus $|z|$ is the distance from the origin to z .

A nonzero complex number can also be described by its modulus $r = |z|$ and an angle θ , called an **argument**:

$$z = r(\cos \theta + i \sin \theta).$$

The angle is not unique: adding $2\pi k$, with $k \in \mathbb{Z}$, gives the same point.

Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

allows the compact **polar form**

$$z = re^{i\theta}.$$

This makes multiplication especially simple:

$$r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

Thus, multiplying complex numbers multiplies their lengths and adds their angles. Similarly,

$$\frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

For an integer n , De Moivre's formula follows:

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

Complex numbers also have multiple roots. For example, the equation $z^n = 1$ has the n solutions

$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

These are equally spaced around the unit circle. This simple fact is an early example of how algebra and geometry fit together beautifully in the complex plane.

Exercises

Exercise 23.1 (Real and imaginary parts). For each complex number, state its real and imaginary parts.

a. $z = 3 + 2i$

b. $z = -4 + 5i$

c. $z = 7$

d. $z = -3i$

e. $z = \frac{1}{2} - \frac{3}{4}i$

Exercise 23.2 (First arithmetic with complex numbers). Compute and write each answer in the form $a + ib$.

a. $(2 + 3i) + (4 - i)$

b. $(5 + 2i) - (1 + 7i)$

c. $3(2 - i)$

d. $i(2 + 3i)$

e. $(1 + i)(1 - i)$

f. $(2 + i)^2$

Exercise 23.3 (Powers of i). Use the repeating pattern of powers of i to simplify the following.

a. i^2

b. i^3

c. i^4

d. i^5

e. i^{10}

f. i^{2026}

Exercise 23.4 (Conjugates and absolute values). For each number z , find $\bar{z}$, $|z|$, and $z\bar{z}$.

- a. $z = 3 + 4i$
- b. $z = 1 - i$
- c. $z = -2i$
- d. $z = -5$

Exercise 23.5 (Finding your way in the complex plane). Sketch a complex plane and mark the following numbers. For each number, also state its distance from the origin.

- a. $1 + i$
- b. $-1 + i$
- c. $-1 - i$
- d. $1 - i$
- e. 2
- f. $-2i$

What geometric operation changes z into $\bar{z}$?

Exercise 23.6 (Division of complex numbers). Write each quotient in the form $a + ib$ by multiplying numerator and denominator by the conjugate of the denominator.

- a. $\frac{1}{1 + i}$
- b. $\frac{2 + i}{1 - i}$
- c. $\frac{3 - 2i}{2 + i}$

- d. Show directly that $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$ for every $z \neq 0$.

Exercise 23.7 (Working with conjugation). Let $z, w \in \mathbb{C}$.

- a. Verify, by writing $z = a + ib$, that $z + \bar{z} = 2 \operatorname{Re} z$.
- b. Verify that $z - \bar{z} = 2i \operatorname{Im} z$.
- c. Use these results to express $\operatorname{Re} z$ and $\operatorname{Im} z$ using only z and $\bar{z}$.

- d. Show that $\overline{zw} = \overline{z}\overline{w}$.
- e. Show that $|zw| = |z||w|$.

Exercise 23.8 (Cartesian and polar form). For each number, find a modulus r and an argument θ , and write the number as $re^{i\theta}$.

- a. $1 + i$
- b. $-1 + i$
- c. $-i$
- d. $-\sqrt{3} + i$
- e. Convert $2e^{i\pi/3}$ back to the form $a + ib$.

Exercise 23.9 (Multiplication as rotation and scaling). Let

$$z = 2e^{i\pi/6}, \quad w = 3e^{i\pi/4}.$$

- a. Find zw in polar form.
- b. Find z/w in polar form.
- c. What happens to the modulus when we multiply by w ?
- d. What happens to the argument when we multiply by w ?
- e. Explain geometrically why multiplication by a complex number can be viewed as a combination of scaling and rotation.

Exercise 23.10 (Roots of unity: algebra becomes geometry). Solve

$$z^4 = 1.$$

- a. Write 1 as $e^{2\pi ik}$, where $k \in \mathbb{Z}$.
- b. Find the four distinct fourth roots of 1 in exponential form.
- c. Write them in the form $a + ib$.
- d. Plot the four roots in the complex plane.
- e. What geometric figure do they form?

Exercise 23.11 (A first glimpse of quantum-mechanical phase). In quantum mechanics, multiplying a state by a complex number of modulus one,

$$e^{i\theta},$$

is called multiplication by a **phase factor**.

Let

$$z = 3 + 4i.$$

- Show that $|e^{i\theta}| = 1$.
- Show that $|e^{i\theta}z| = |z|$.
- Compute iz and locate both z and iz in the complex plane.
- What rotation takes z into iz ?
- Compute $e^{i\pi}z$. What geometric operation has occurred?

This preservation of magnitude under multiplication by a phase is one reason complex numbers appear so naturally in quantum mechanics.

Exercise 23.12 (An excursion: the Mandelbrot iteration). Complex arithmetic can produce unexpectedly intricate geometry. Starting from $z_0 = 0$, choose a complex number c and repeatedly calculate

$$z_{n+1} = z_n^2 + c.$$

The Mandelbrot set consists of those values of c for which this sequence does not escape to infinity.

- For $c = 0$, calculate $z_0, z_1, \dots, z_5$.
- For $c = -1$, calculate $z_0, z_1, \dots, z_6$. What pattern do you see?
- For $c = 1$, calculate terms until $|z_n| > 2$.
- Write a short Python function that takes c and a maximum number of iterations and performs this iteration. Stop if $|z_n| > 2$.
- Optional: sample many values $c = x + iy$ in a rectangle of the complex plane and make an image in which each point is represented by the number of iterations required before $|z_n| > 2$. Explore the boundary of the Mandelbrot set.

Solutions

Solution for Exercise 23.1

- a. $\operatorname{Re} z = 3, \operatorname{Im} z = 2.$
- b. $\operatorname{Re} z = -4, \operatorname{Im} z = 5.$
- c. $\operatorname{Re} z = 7, \operatorname{Im} z = 0.$
- d. $\operatorname{Re} z = 0, \operatorname{Im} z = -3.$
- e. $\operatorname{Re} z = \frac{1}{2}, \operatorname{Im} z = -\frac{3}{4}.$

Solution for Exercise 23.2

- a. $6 + 2i.$
- b. $4 - 5i.$
- c. $6 - 3i.$
- d. $2i + 3i^2 = -3 + 2i.$
- e. $(1 + i)(1 - i) = 1 - i^2 = 2.$
- f. $(2 + i)^2 = 4 + 4i + i^2 = 3 + 4i.$

Solution for Exercise 23.3

- a. $-1.$
- b. $-i.$
- c. $1.$
- d. $i.$
- e. Since $10 = 4 \cdot 2 + 2, i^{10} = i^2 = -1.$
- f. Since $2026 = 4 \cdot 506 + 2, i^{2026} = i^2 = -1.$

Solution for Exercise 23.4

- a. $\bar{z} = 3 - 4i$, $|z| = 5$, and $z\bar{z} = 25$.
- b. $\bar{z} = 1 + i$, $|z| = \sqrt{2}$, and $z\bar{z} = 2$.
- c. $\bar{z} = 2i$, $|z| = 2$, and $z\bar{z} = 4$.
- d. $\bar{z} = -5$, $|z| = 5$, and $z\bar{z} = 25$.

Solution for Exercise 23.5

The points are $(1, 1)$, $(-1, 1)$, $(-1, -1)$, $(1, -1)$, $(2, 0)$, and $(0, -2)$, respectively. Their distances from the origin are $\sqrt{2}, \sqrt{2}, \sqrt{2}, \sqrt{2}, 2, 2$.

Complex conjugation reflects a point across the real axis: $(a, b) \mapsto (a, -b)$.

Solution for Exercise 23.6

- a.
$$\frac{1}{1+i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2} = \frac{1}{2} - \frac{1}{2}i.$$
- b.
$$\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{2} = \frac{1+3i}{2}.$$
- c.
$$\frac{3-2i}{2+i} = \frac{(3-2i)(2-i)}{5} = \frac{4-7i}{5}.$$
- d. Since $z\bar{z} = |z|^2$,
$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2}.$$

Solution for Exercise 23.7

Write $z = a + ib$.

a.

$$z + \bar{z} = (a + ib) + (a - ib) = 2a = 2 \operatorname{Re} z.$$

b.

$$z - \bar{z} = (a + ib) - (a - ib) = 2ib = 2i \operatorname{Im} z.$$

c.

$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}.$$

d. If $z = a + ib$ and $w = c + id$, then

$$zw = (ac - bd) + i(ad + bc),$$

so

$$\overline{zw} = (ac - bd) - i(ad + bc) = (a - ib)(c - id) = \bar{z}\bar{w}.$$

e.

$$|zw|^2 = (zw)\overline{zw} = z\bar{z}w\bar{w} = |z|^2|w|^2.$$

Both sides are nonnegative, so $|zw| = |z||w|$.

Solution for Exercise 23.8

a. $r = \sqrt{2}$, $\theta = \pi/4$, so $z = \sqrt{2}e^{i\pi/4}$.

b. $r = \sqrt{2}$, $\theta = 3\pi/4$, so $z = \sqrt{2}e^{3\pi i/4}$.

c. $r = 1$, $\theta = -\pi/2$, so $z = e^{-i\pi/2}$.

d. $r = 2$, $\theta = 5\pi/6$, so $z = 2e^{5\pi i/6}$.

e.

$$2e^{i\pi/3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}.$$

Solution for Exercise 23.9

a.

$$zw = 6e^{i(\pi/6+\pi/4)} = 6e^{5\pi i/12}.$$

b.

$$\frac{z}{w} = \frac{2}{3}e^{i(\pi/6-\pi/4)} = \frac{2}{3}e^{-i\pi/12}.$$

c. Multiplication by w multiplies the modulus by $|w| = 3$.

d. It increases the argument by $\pi/4$.

e. For $w = re^{i\theta}$, multiplication by w changes $\rho e^{i\phi}$ into $r\rho e^{i(\phi+\theta)}$. Thus lengths are scaled by r and angles are rotated by θ .

Solution for Exercise 23.10

a.

$$1 = e^{2\pi i k}, \quad k \in \mathbb{Z}.$$

b. Taking fourth roots gives

$$z = e^{2\pi i k/4} = e^{i\pi k/2}.$$

The four distinct solutions are obtained for $k = 0, 1, 2, 3$:

$$1, \quad e^{i\pi/2}, \quad e^{i\pi}, \quad e^{3i\pi/2}.$$

c. These are

$$1, \quad i, \quad -1, \quad -i.$$

d. They lie at $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.

e. They form the vertices of a square centered at the origin.

Solution for Exercise 23.11

- a. Using Euler's formula,

$$|e^{i\theta}| = |\cos \theta + i \sin \theta| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1.$$

- b.

$$|e^{i\theta}z| = |e^{i\theta}||z| = |z|.$$

- c.

$$iz = i(3 + 4i) = -4 + 3i.$$

Thus z is the point $(3, 4)$ and iz is $(-4, 3)$.

- d. Multiplication by $i = e^{i\pi/2}$ is a counterclockwise rotation by $\pi/2$.

- e. Since $e^{i\pi} = -1$,

$$e^{i\pi}z = -z = -3 - 4i.$$

This is a rotation by π about the origin.

Solution for Exercise 23.12

- a. For $c = 0$, every term is zero:

$$z_0 = z_1 = \cdots = z_5 = 0.$$

- b. For $c = -1$,

$$z_0 = 0, \quad z_1 = -1, \quad z_2 = 0, \quad z_3 = -1, \quad z_4 = 0, \quad z_5 = -1, \quad z_6 = 0.$$

The sequence alternates between 0 and -1 .

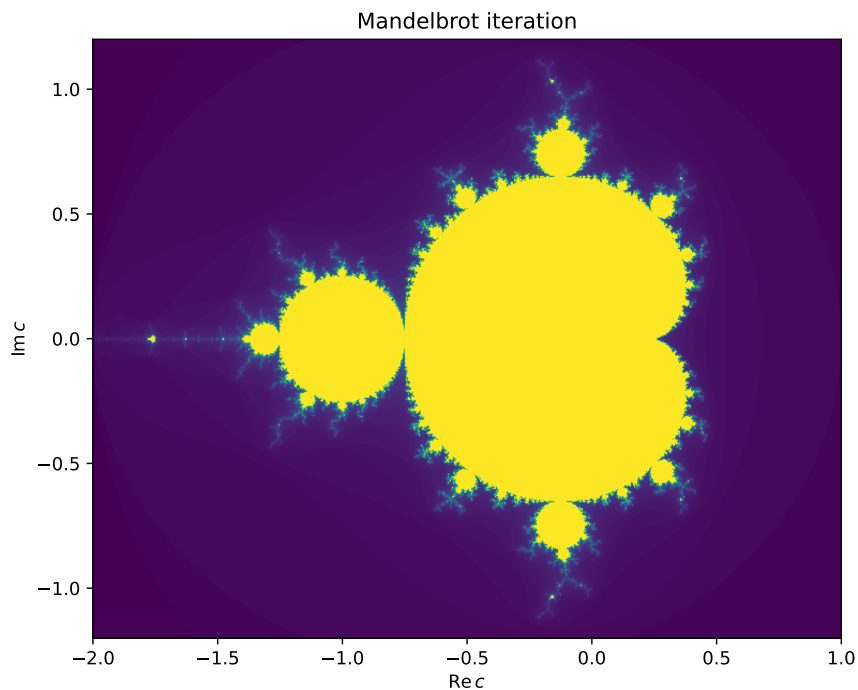
- c. For $c = 1$,

$$z_0 = 0, \quad z_1 = 1, \quad z_2 = 2, \quad z_3 = 5.$$

Thus $|z_3| > 2$.

- d. One possible implementation is

- e. A compact visualization is



The most intricate structure appears near the boundary between points that escape rapidly and points that remain bounded for many iterations.

24 Complex arithmetic and geometry

Cartesian and polar forms connect complex arithmetic with plane geometry, rotations, dot products, and cross products.

Exercises

Complex numbers

Exercise 24.1. Let $z = x + iy \in \mathbb{C}$, and compute the real and imaginary parts of z^2 , z^3 , and z^4 , as functions of x and y .

Exercise 24.2. Compute a closed-form expression for the real and imaginary parts of z^n .

Exercise 24.3. Compute closed-form expression for z^{-1} , exhibiting the real and imaginary parts as functions of x and y . (Assuming $z \neq 0$). Try to compute closed-form expressions of z^{-2} and z^n as well.

Exercise 24.4. A complex number $z = x + iy$ can be written in polar form,

$$x = r \cos(\theta), \quad y = r \sin(\theta).$$

Write down the rule for $z^n = R \cos \phi + iR \sin \phi$ using angle θ and modulus r .

Exercise 24.5. Consider the polynomial equation $f(z) = z^3 + 1 = 0$. Find all the roots in the complex plane using polar coordinates. Sketch the roots in a coordinate system.

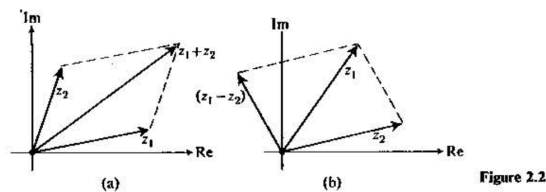


Figure 24.1: The parallelogram rule. (From Butkov.)

Basic calculations

Exercise 24.6. Let $z = x + iy \in \mathbb{C}$. Illustrate the addition $w = z + \bar{z}$ in a coordinate system using the parallelogram rule.

Exercise 24.7. Recall our earlier exercise, where $\mathbb{C}$ was regarded as the real vector space $\mathbb{R}^2$. Let $z_i = x_i + iy_i$, and denote the corresponding vectors in $\mathbb{R}^2$ by $\mathbf{v}_i$. Show that

$$\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = \Re(\bar{z}_1 z_2) = \Re(z_1 \bar{z}_2).$$

Exercise 24.8. Another product in $\mathbb{R}^2$ (and indeed $\mathbb{R}^3$) is the *cross product*:

$$\mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{vmatrix},$$

the matrix determinant. Show that

$$\mathbf{v}_1 \times \mathbf{v}_2 = \Im(\bar{z}_1 z_2) = -\Im(z_1 \bar{z}_2).$$

Exercise 24.9. Show that the function $z \mapsto iz$ is a counterclockwise rotation by $\pi/2$. Find the function that performs a clockwise rotation by the same angle.

25 Subsets of the complex plane

The modulus and real and imaginary parts describe curves and regions in the complex plane.

Visualize the following subsets of $\mathbb{C}$. Decide which sets are open, closed, or neither.

Exercises

Exercise 25.1. $\{z \mid |z - 1| = 1\}$

Exercise 25.2. $\{z \mid |iz - 1| < 1\}$

Exercise 25.3. $\{z \mid |z - i| + |z + i| < 4\}$

Exercise 25.4. $\{z^2 \mid |z - i| = 1\}$

Exercise 25.5. $\{z \mid \Re(e^{i\pi/2}z) > 0\}$

Exercise 25.6. $\{z \mid \Im(z^2) > 0\}$

26 The complex exponential

The functional equation for the exponential and the Cauchy–Riemann equations determine the complex exponential.

In this section, we give some relatively easy exercises to illustrate the important concepts of complex analytic functions.

In this exercise, we derive the complex exponential function. The starting point is the real exponential function, which is the unique function $f(x) = \exp(x)$ that satisfies

$$f(x_1 + x_2) = f(x_1)f(x_2), \quad f(0) = 1.$$

We desire to generalize $\exp(x)$ to the complex plane, so that $\exp(x + 0i) = \exp(x) \in \mathbb{R}$.

To avoid confusion, we call this unknown function $f : \mathbb{C} \rightarrow \mathbb{C}$.

Exercises

Exercise 26.1. Show that $f(x + iy) = \exp(x)f(iy)$. (Thus, we need to determine $f(iy)$.)

Exercise 26.2. We set $f(iy) = A(y) + iB(y)$. Show that

$$A(y) = B'(y), \quad B(y) = -A'(y),$$

and hence that $A''(y) = -A(y)$. Hint: Cauchy–Riemann.

Exercise 26.3. The general solution to the ODE $A'' = -A$ is

$$A(y) = \alpha \cos(y) + \beta \sin(y),$$

where α and β are constants to be determined. Show that $\alpha = 1$ and $\beta = 0$ are the only constants compatible with $f(z)$ being a generalization of the real exponential function.

Exercise 26.4. Conclude that

$$f(x + iy) = e^x(\cos y + i \sin y)$$

Exercise 26.5. Show that $(\exp(z))' = \exp(z)$, using Cauchy–Riemann, and that $\exp(z)$ is everywhere analytic.

Exercise 26.6. Show that the equation

$$\exp(z) = w$$

has infinitely many solutions for any $w \neq 0$.

Exercise 26.7. Show that $\exp(z) \neq 0$.

27 Complex power series

A **power series** is an infinite series of the form

$$\sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots,$$

where the variable z may be a complex number. We can think of a power series as a polynomial with infinitely many terms. As with any infinite series, an important question is whether the partial sums approach a finite limit. Typically, a power series converges when z lies inside some circle $|z| < R$ in the complex plane, where R is called the **radius of convergence**.

The simplest example is the **geometric series**

$$1 + z + z^2 + z^3 + \cdots = \frac{1}{1 - z}, \quad |z| < 1.$$

Thus, inside the unit circle, the function $1/(1 - z)$ can be represented exactly by an infinite sum of powers of z . Truncating the series after finitely many terms gives a polynomial approximation to the function (the MacLaurin series).

Many important functions can be represented by power series. In particular, the complex exponential is defined by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots.$$

Unlike the geometric series, this series converges for every complex number z ! Substituting imaginary arguments into the exponential series leads directly to the familiar power series for sine and cosine and provides another way of understanding Euler's formula $e^{ix} = \cos x + i \sin x$.

Power series are especially useful because they can often be manipulated much like ordinary polynomials. We can add and multiply them, substitute expressions for their variables, and use them to investigate functions that may initially appear problematic. For example, the expressions $(e^z - 1)/z$ and $\sin z/z$ appear to be undefined at $z = 0$, but their power-series representations reveal what happens there and show how the apparent singularity can be removed.

Exercises

Exercise 27.1 (Partial sums of the geometric series). Consider the geometric series

$$1 + z + z^2 + z^3 + \cdots.$$

For a finite integer $N \geq 0$, define the partial sum

$$S_N(z) = \sum_{n=0}^N z^n = 1 + z + z^2 + \cdots + z^N.$$

- Write down $S_0(z)$, $S_1(z)$, $S_2(z)$, and $S_3(z)$.
- Multiply $S_N(z)$ by z and write down the resulting expression.
- Subtract $zS_N(z)$ from $S_N(z)$ and show that

$$(1 - z)S_N(z) = 1 - z^{N+1}.$$

- Hence show, for $z \neq 1$, that

$$S_N(z) = \frac{1 - z^{N+1}}{1 - z}.$$

Exercise 27.2 (Convergence of the geometric series). From Exercise 27.1,

$$S_N(z) = \frac{1 - z^{N+1}}{1 - z}.$$

- Let $z = \frac{1}{2}$. What happens to z^{N+1} as $N \rightarrow \infty$? Find the limit of $S_N(z)$.

b. Let $z = \frac{i}{2}$. What happens to $|z^{N+1}|$ as $N \rightarrow \infty$?

c. Show that if $|z| < 1$, then

$$|z^{N+1}| = |z|^{N+1} \longrightarrow 0.$$

d. Conclude that

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \quad |z| < 1.$$

e. What happens when $z = 1$? What happens when $z = -1$? Do these series converge?

f. The condition $|z| < 1$ describes a region in the complex plane. What does this region look like?

Exercise 27.3 (The exponential, sine, and cosine series). The complex exponential is represented by the power series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots.$$

Euler's formula states that, for real x ,

$$e^{ix} = \cos x + i \sin x.$$

a. Substitute $z = ix$ into the power series for e^z and write out the first seven terms.

b. Use

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

to separate the real and imaginary parts of the series.

- c. By comparing with Euler's formula, show that

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

and

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

- d. Write out the first four nonzero terms of each series.
 e. Use the series to obtain polynomial approximations to $\cos x$ and $\sin x$ containing terms up to and including order x^5 .

Exercise 27.4 (Approximating sine and cosine by polynomials). Define the partial sums

$$C_N(x) = \sum_{n=0}^N (-1)^n \frac{x^{2n}}{(2n)!}, \quad S_N(x) = \sum_{n=0}^N (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

- a. Write down $C_0(x)$, $C_1(x)$, and $C_2(x)$ explicitly.
 b. Write down $S_0(x)$, $S_1(x)$, and $S_2(x)$ explicitly.
 c. Plot $\cos x$ together with $C_0(x)$, $C_1(x)$, $C_2(x)$, and $C_3(x)$ over the interval $-2\pi \leq x \leq 2\pi$.
 d. Make a corresponding plot for $\sin x$ and $S_0(x)$, $S_1(x)$, $S_2(x)$, and $S_3(x)$.
 e. Describe what happens to the quality of the approximation as more terms are included. Is the approximation equally good close to $x = 0$ and far away from $x = 0$?

Exercise 27.5 (A removable singularity). Consider

$$f(z) = \frac{e^z - 1}{z}, \quad z \neq 0.$$

At first sight, this expression is not defined at $z = 0$.

- a. Write down the power series for $e^z - 1$.

- b. Divide the series by z and show that, for $z \neq 0$,

$$f(z) = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \frac{z^3}{4!} + \cdots.$$

- c. Write the result using summation notation.
d. What value does the power series suggest that we should assign to $f(0)$?
e. Define $f(0)$ using your answer to part d. Explain why the apparent singularity at $z = 0$ is then removed.

Exercise 27.6 (The function $\sin z/z$). Consider the function

$$f(z) = \begin{cases} \frac{\sin z}{z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

- a. Start from the power series

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots$$

and divide by z .

- b. Show that

$$f(z) = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \frac{z^6}{7!} + \cdots.$$

- c. Write this series in summation notation.
d. Use the series to explain why the choice $f(0) = 1$ is natural.
e. What is the coefficient of z^2 in the power series? What is the coefficient of z^4 ?
f. Show directly from the power series that $f(-z) = f(z)$. Thus f is an even function.

Exercise 27.7 (The Cauchy product of power series). Consider two power series

$$A(z) = \sum_{n=0}^{\infty} a_n z^n, \quad B(z) = \sum_{n=0}^{\infty} b_n z^n.$$

Their product can be found in the same way as the product of two polynomials, by multiplying terms and collecting equal powers of z .

- a. Write out the terms of $A(z)$ and $B(z)$ up to order z^3 . Multiply the two expressions and show that the first four coefficients of $A(z)B(z)$ are

$$a_0 b_0, \quad a_0 b_1 + a_1 b_0, \quad a_0 b_2 + a_1 b_1 + a_2 b_0, \quad a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0.$$

- b. In general, show that the coefficient c_n of z^n is

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

This multiplication rule is called the **Cauchy product**.

- c. Apply the Cauchy product to two copies of the exponential series

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Show that the coefficient of z^n in $e^z e^z$ is

$$c_n = \sum_{k=0}^n \frac{1}{k!(n-k)!}.$$

- d. Use

$$\frac{1}{k!(n-k)!} = \frac{1}{n!} \binom{n}{k}$$

and the identity

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

to simplify c_n .

- e. Use your result to show directly from the power series that

$$e^z e^z = e^{2z}.$$

28 Singularities and contour integrals

Laurent series classify isolated singularities and determine contour integrals around poles.

Exercises

Singularities

Exercise 28.1. Consider the function $f(z) = \frac{1}{z(z-1)}$. How many poles does f have, and where are they? What are the order of the poles? Can you write down the Laurent series of f around $z = 0$?

Exercise 28.2. Does the function $f(z) = 1/\sin(z)$ have a pole? Where? What order?

Exercise 28.3. Find the singularities of $f(z) = e^{-1/(z-1)^2}$.

Line integrals

Let

$$f(z) = \frac{1}{z^3 - (2 + i)z^2 + (1 + 2i)z - i}.$$

Exercise 28.4. Let $\Gamma_\epsilon(w)$ be the simple closed curve defined by the counter-clockwise border of the ϵ -ball $B_\epsilon(w)$. Write down a parameterization (a path) for this curve.

Exercise 28.5. Identify the poles $P = \{w_1, w_2, \dots\}$ of $f(z)$ and their order. Hint: A zero of the denominator is $z = i$.

Exercise 28.6. What is the value of

$$\oint_{\Gamma_\epsilon(0)} f(z) \, dz$$

for $\epsilon = 1/2$?

Exercise 28.7. Find the value of the line integrals

$$\oint_{\Gamma_\epsilon(w_i)} f(z) \, dz$$

for $\epsilon = 1/2$. Hint: Do not attempt the integral directly; compute the relevant Laurent coefficient. It can be useful to use the Taylor expansion

$$\frac{1}{a-z} = \frac{1}{a} \frac{1}{1-z/a} = \frac{1}{a} \sum_{n=0}^{\infty} (z/a)^n.$$

Part IX

Series expansions and matrix functions

29 Taylor Series

Taylor series provide one of the most important ways of approximating a smooth function by a polynomial. The central idea is to find a polynomial that approximates the function locally: near a point a , we choose a polynomial whose value and derivatives agree with those of the function at a up to some order n .

For a sufficiently differentiable function f , the **Taylor polynomial** of degree n about a is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Thus,

$$T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

When $a = 0$, the expansion is called a **Maclaurin series**. Three particularly important Maclaurin series are

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots,$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$

and

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

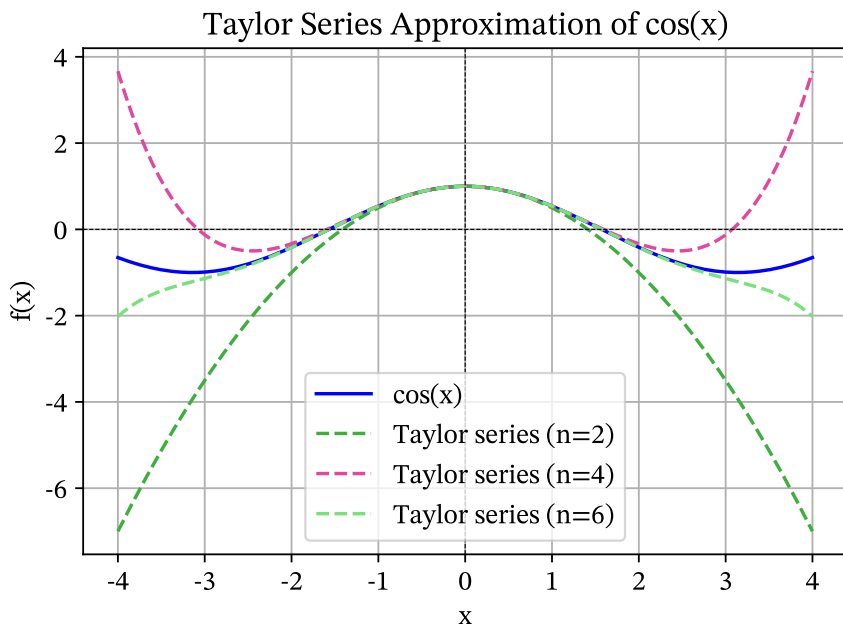


Figure 29.1: Taylor series approximations of $\cos(x)$ about $x=0$. The dashed lines show the Taylor polynomials of order 2, 4, and 6.

Keeping only a few terms gives a **low-order approximation**. For example, close to $x = 0$,

$$e^x \approx 1 + x, \quad \sin x \approx x, \quad \cos x \approx 1 - \frac{x^2}{2}.$$

These approximations are local. Thus, they usually become less accurate as we move farther from the expansion point. Increasing the order of the Taylor polynomial generally improves the approximation within a suitable region, but it does not guarantee accuracy far from the expansion point.

The difference between the function and its Taylor polynomial is the **remainder**,

$$R_n(x) = f(x) - T_n(x).$$

One useful form is the Lagrange remainder,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1},$$

where ξ lies somewhere between a and x . Although ξ is usually unknown, a bound on $|f^{(n+1)}|$ gives a bound on the error:

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1},$$

provided $|f^{(n+1)}(t)| \leq M$ between a and x .

Taylor series can be generalized to higher dimensions, i.e., functions of several variables.

In molecular physics and quantum chemistry, Taylor expansions appear quite often. An important example is the expansion of a potential-energy function around an equilibrium geometry. Keeping terms only through second order gives the harmonic approximation. Taylor series also extend naturally to functions of matrices and operators, such as the exponential operator.

Exercises

Exercise 29.1 (Approximating the exponential function).

- Write down the general Taylor polynomial $T_n(x)$ about $x = 0$.
- For $f(x) = e^x$, compute $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$.
- Construct the Taylor polynomials $T_1(x)$, $T_2(x)$, and $T_3(x)$ for e^x about $x = 0$.
- Use T_1 , T_2 , and T_3 to approximate $e^{0.1}$.
- Compare your approximations with $e^{0.1} \approx 1.105170$. Which polynomial gives the best approximation?
- Repeat the calculation for e^1 . Why are the low-order approximations less accurate at $x = 1$ than at $x = 0.1$?

Exercise 29.2 (Sine and cosine near the origin).

- Compute the first four derivatives of $f(x) = \sin x$ and evaluate them at $x = 0$.
- Use these derivatives to obtain the Taylor polynomial for $\sin x$ through the cubic term.
- Similarly, obtain the Taylor polynomial for $\cos x$ through the quadratic term.
- Use the lowest nonconstant approximations to estimate $\sin(0.1)$ and $\cos(0.1)$.
- Compare with $\sin(0.1) \approx 0.0998334$ and $\cos(0.1) \approx 0.995004$.
- Explain why $\sin x \approx x$ is called the small-angle approximation. Angles are measured in radians.

Exercise 29.3 (Controlling the error). Consider the Taylor polynomial $T_n(x)$ for e^x about $x = 0$.

- Write $T_n(x)$.
- Use the Lagrange remainder to show that, for $0 \leq x \leq 1$,

$$|R_n(x)| \leq \frac{e}{(n+1)!} x^{n+1}.$$

- Find the smallest n for which this bound guarantees an error below 10^{-3} when approximating $e^{1/2}$.
- Compute the corresponding approximation to $e^{1/2}$ and compare it with $e^{1/2} \approx 1.64872127$. Compare the actual error with the error bound.

Exercise 29.4 (A Gaussian near its maximum). The Gaussian function $f(x) = e^{-x^2}$ is closely related to the Gaussian functions used extensively in quantum chemistry.

- Starting from the Taylor series for e^u , substitute $u = -x^2$ and obtain the Taylor polynomial for e^{-x^2} through order x^6 .
- Write the quadratic, quartic, and sixth-order approximations explicitly.
- Evaluate these three approximations at $x = 0.5$ and compare them with $e^{-0.25} \approx 0.778801$.
- Repeat at $x = 1$, where $e^{-1} \approx 0.367879$. What do you observe?

- e. Explain why no finite Taylor polynomial can reproduce the behavior $e^{-x^2} \rightarrow 0$ as $|x| \rightarrow \infty$.

Exercise 29.5 (The harmonic approximation). Let $V(R)$ be the potential energy of a diatomic molecule as a function of the internuclear distance R . Suppose R_e is an equilibrium bond length and a local minimum of V .

- Taylor-expand $V(R)$ about R_e through fourth order in the displacement $q = R - R_e$.
- Use the equilibrium condition to simplify the expansion.
- Show that retaining terms only through second order gives

$$V(R) \approx V(R_e) + \frac{1}{2}kq^2,$$

and identify the force constant k .

- What physical information is contained in the cubic and quartic terms that is absent from the harmonic approximation?
- Consider the Morse potential

$$V(R) = D_e \left(1 - e^{-a(R-R_e)}\right)^2.$$

Expand it through fourth order in $q = R - R_e$ and identify its quadratic, cubic, and quartic coefficients.

- Explain qualitatively why the harmonic approximation is expected to work best for small-amplitude vibrations near R_e .

Exercise 29.6 (From the exponential to rotations). Consider the matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The exponential of a matrix is defined by the same power series as the scalar exponential,

$$e^{\theta A} = I + \theta A + \frac{\theta^2 A^2}{2!} + \frac{\theta^3 A^3}{3!} + \cdots.$$

- a. Compute A^2 , A^3 , and A^4 , and identify the repeating pattern.
- b. Separate the even and odd powers in the Taylor series for $e^{\theta A}$.
- c. Recognize the scalar Taylor series that occur and show that

$$e^{\theta A} = I \cos \theta + A \sin \theta.$$

- d. Hence show that

$$e^{\theta A} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

- e. Write approximations to $e^{\theta A}$ obtained by truncating the exponential series after terms of order θ^2 and θ^4 .
- f. Explain how this example illustrates the connection between the Taylor series of the exponential, sine, and cosine functions.

30 Matrix exponentials and operator exponentials

Matrix and operator exponentials occur throughout quantum mechanics and quantum chemistry. They appear in time evolution, unitary transformations, orbital rotations, coupled-cluster theory, and many other settings.

For a square matrix or linear operator A , the exponential is defined by its power series

$$e^A = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots = \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

This definition is the direct analogue of the exponential series for a number. The important difference is that matrices and operators generally do **not commute**. If A and B are operators, it is therefore generally true that

$$\text{In general: } e^{A+B} \neq e^A e^B.$$

If A and B commute, however, the familiar rule is recovered:

$$[A, B] = AB - BA = 0 \implies e^{A+B} = e^A e^B = e^B e^A.$$

The inverse of an exponential is particularly simple:

$$(e^A)^{-1} = e^{-A}.$$

Consequently,

$$e^A e^{-A} = e^{-A} e^A = I.$$

This result does not require any additional commutation assumptions, since A always commutes with itself.

A central operation in quantum mechanics is a **similarity transformation**

$$\tilde{B} = e^A B e^{-A}.$$

To understand such expressions, define the commutator

$$[A, B] = AB - BA.$$

Repeated or nested commutators are written, for example,

$$[A, [A, B]], \quad [A, [A, [A, B]]].$$

The exponential similarity transformation can then be expanded as

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!}[A, [A, B]] + \frac{1}{3!}[A, [A, [A, B]]] + \dots$$

This is sometimes called the **Hadamard lemma**. Physicists sometimes call it the **Baker–Campbell–Hausdorff formula**, although the two are not exactly the same.

The above expansion is especially useful when the nested commutators eventually vanish or follow a simple pattern, such as in coupled-cluster theory.

A closely related problem is to combine two exponentials into one. In general,

$$e^A e^B = e^C,$$

where C is not simply $A + B$. The **Baker–Campbell–Hausdorff (BCH) expansion** begins

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + \dots$$

Thus, the corrections to the familiar scalar rule are expressed entirely in terms of commutators. When $[A, B] = 0$, all these corrections disappear and $C = A + B$.

These formulas are useful not only for formal manipulations. They provide a systematic way of understanding how noncommuting transformations combine, and why operator ordering matters in quantum mechanics.

Exercises

Exercise 30.1 (Exponential of a diagonal matrix). Consider the matrix

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

- Calculate A^2 .
- Calculate A^3 .
- Use the power-series definition $e^A = I + A + A^2/2! + \dots$ to show that

$$e^A = \begin{pmatrix} e^a & 0 \\ 0 & e^b \end{pmatrix}.$$

- Find e^{-A} .
- Verify explicitly that $e^A e^{-A} = I$.

Exercise 30.2 (An exponential whose series terminates). In coupled-cluster theory, the cluster operator T is nilpotent. This means that some power of T vanishes. In this exercise, we explore a simple example of a nilpotent matrix and its exponential. Consider

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

- Calculate N^2 .
- Calculate N^3 .
- Use the power-series definition of the exponential to calculate e^N .
- Calculate e^{-N} .
- Multiply your two matrices and verify that $e^N e^{-N} = I$.

- f. Explain why the exponential series in this example contains only finitely many nonzero terms.

Exercise 30.3 (Exponentials of commuting matrices). Consider

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad B = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}.$$

- Calculate AB and BA .
- Calculate the commutator $[A, B]$.
- Calculate e^A , e^B , and e^{A+B} .
- Verify directly that

$$e^A e^B = e^{A+B}.$$

- Which property of A and B makes the familiar exponential rule work in this example?

Exercise 30.4 (A matrix exponential and rotations). Consider

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

- Calculate J^2 , J^3 , and J^4 .
- Use the power series of $e^{\theta J}$ and separate terms containing even and odd powers of J .
- Recognize the Taylor series of sine and cosine and show that

$$e^{\theta J} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

- Show that $e^{\theta J} e^{\phi J} = e^{(\theta+\phi)J}$.
- Explain why the result in part d does not contradict the fact that $e^A e^B \neq e^{A+B}$ in general.

Exercise 30.5 (Anti-Hermitian matrices and unitary matrices). A matrix A is **anti-Hermitian** if

$$A^\dagger = -A,$$

and a matrix U is **unitary** if

$$U^\dagger U = UU^\dagger = I.$$

- a. Show from the power-series definition of the matrix exponential that

$$(e^A)^\dagger = e^{A^\dagger}.$$

- b. Let A be anti-Hermitian and define $U = e^A$. Show that

$$U^\dagger = e^{-A}.$$

- c. Use the result from part b to prove that U is unitary.
d. Show that if H is Hermitian and α is real, then $A = i\alpha H$ is anti-Hermitian. Conclude that

$$U = e^{i\alpha H}$$

is unitary.

- e. Consider the Hermitian matrix

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Calculate $U = e^{i\theta H}$ explicitly and verify directly that $U^\dagger U = I$.

- f. The converse is also true: every unitary matrix can be written as the exponential of an anti-Hermitian matrix. Suppose that U has an orthonormal basis of eigenvectors with eigenvalues $e^{i\theta_j}$. Explain how to construct an anti-Hermitian matrix A such that $U = e^A$.

Exercise 30.6 (A terminating similarity transformation). Let

$$A = \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} b & 0 \\ 0 & c \end{pmatrix}.$$

- Calculate the commutator $[A, B]$.
- Calculate the nested commutator $[A, [A, B]]$.
- Use the expansion

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2}[A, [A, B]] + \dots$$

to calculate $e^A B e^{-A}$.

- Calculate e^A and e^{-A} directly and verify your result by matrix multiplication.
- Why did the commutator expansion terminate after only a few terms?

Exercise 30.7 (A simple BCH expansion). Suppose that two operators A and B satisfy

$$[A, B] = C,$$

and that C commutes with both A and B :

$$[A, C] = [B, C] = 0.$$

- Write the BCH expansion for $\log(e^A e^B)$ through the first nested commutators.
- Show that all nested commutators appearing after $[A, B]$ vanish.
- Conclude that

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]}.$$

- Similarly, find an expression for $e^B e^A$ as a single exponential.

e. Use your answers to show that

$$e^A e^B = e^B e^A e^{[A,B]}.$$

f. What does this result reduce to when $[A, B] = 0$?

Exercise 30.8 (Similarity transformation of an eigenoperator). Suppose A and B satisfy

$$[A, B] = \lambda B,$$

where λ is a scalar.

a. Show that

$$[A, [A, B]] = \lambda^2 B.$$

b. Show more generally that the n -fold nested commutator of B with A is

$$\underbrace{[A, [A, \dots, [A, B] \dots]]}_{n \text{ times}} = \lambda^n B.$$

c. Insert this result into the commutator expansion of $e^A B e^{-A}$.

d. Sum the resulting series and show that

$$e^A B e^{-A} = e^{\lambda} B.$$

e. In the harmonic oscillator, the number operator $\hat{N}$ and creation operator $a^\dagger$ satisfy

$$[\hat{N}, a^\dagger] = a^\dagger.$$

Use the result above to evaluate

$$e^{\alpha \hat{N}} a^\dagger e^{-\alpha \hat{N}}.$$

f. The annihilation operator satisfies $[\hat{N}, a] = -a$. Find $e^{\alpha \hat{N}} a e^{-\alpha \hat{N}}$.

Exercise 30.9 (BCH expansion for noncommuting quantum operators). Let

$$A = i\alpha\sigma_x, \quad B = i\beta\sigma_y,$$

where α and β are real and the Pauli matrices satisfy

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k.$$

- a. Calculate $[A, B]$.
- b. Calculate $[A, [A, B]]$.
- c. Calculate $[B, [B, A]]$.
- d. Use the BCH expansion

$$\log(e^A e^B) = A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + \dots$$

to obtain the exponent through terms of total degree three in α and β .

- e. Explain why the BCH series does not terminate in this case.
- f. Interpret the calculation in terms of successive rotations: why should two rotations about different axes not simply combine by adding their generators?

Part X

Numerical methods

31 Dual numbers and automatic differentiation

Dual numbers adjoin a nonzero element ε with $\varepsilon^2 = 0$, which encodes first derivatives algebraically.

We now consider the number system called *the dual numbers*, often written $\mathbb{R}(\varepsilon)$. The dual numbers form an algebraic extension of $\mathbb{R}$ with a nonzero element ε such that $\varepsilon^2 = 0$:

$$\mathbb{R}(\varepsilon) = \{x + \varepsilon y \mid x, y \in \mathbb{R}\}.$$

Their construction resembles $\mathbb{C} = \mathbb{R}(i)$, but $i^2 = -1$ while $\varepsilon^2 = 0$. Dual numbers support automatic differentiation.

Exercises

Exercise 31.1.

1. Compute the multiplication law in terms of real and “dual” parts, i.e., $z_1 z_2 = z = x + \varepsilon y$, and find x and y . Compute the addition law for $z_1 + z_2$.
2. Show that $\mathbb{R}(\varepsilon)$ is not a field.
3. Compute the law for all positive powers of $z = x + \varepsilon y$, and for negative powers when $x \neq 0$.
4. Consider $z = x + \varepsilon$ and evaluate its positive powers, and its negative powers when $x \neq 0$. How is each dual part related to the corresponding real part?
5. Let $p(x)$ be a real polynomial and extend it to dual arguments. Show that

$$p(x + \varepsilon) = p(x) + \varepsilon p'(x).$$

6. Let $q(x)$ be a polynomial in negative powers of x . Show, for $x \neq 0$, that $q(x + \varepsilon) = q(x) + \varepsilon q'(x)$.
7. Let $p(x)$ and $q(x)$ be polynomials, and consider the *rational function* $f(x) = p(x)/q(x)$. Where $q(x) \neq 0$, extend the rational function to $\mathbb{R}(\varepsilon)$ and prove that $f(x + \varepsilon) = f(x) + \varepsilon f'(x)$.

32 Newton's method

Newton's method replaces a nonlinear system by its first-order Taylor approximation at each iteration.

In the next exercise, we derive the Newton–Raphson method for root finding.

Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a function of class C^2 . We seek $\mathbf{x} \in \Omega$ such that $f(\mathbf{x}) = 0$.

Assume we have an initial guess $\mathbf{x}_0 \in \Omega$.

Exercises

Exercise 32.1. Explain why the following polynomial is a good approximation to $f(\mathbf{x})$ near $\mathbf{x}_0$:

$$p(\mathbf{x}) = f(\mathbf{x}_0) + J(\mathbf{x} - \mathbf{x}_0), \quad J = Df(\mathbf{x}_0).$$

Exercise 32.2. Newton's method now finds a new guess $\mathbf{x}_1$ by finding the root of the Taylor polynomial, i.e., $p(\mathbf{x}_1) = 0$. Write down a formula for $\mathbf{x}_1$, assuming that the matrix J has an inverse.

Exercise 32.3. Suppose the true solution is $\mathbf{x}_* \in \Omega$, and assume that the error in $\mathbf{x}_0$ is sufficiently small. Find an estimate for the error of $\mathbf{x}_1$. Assuming that the error can be converted to significant digits n , what is the number of significant digits in $\mathbf{x}_1$? Hint: Look at the remainder term.

Exercise 32.4. Suppose successive iterations are performed. How many digits do you have after k iterations?

Exercise 32.5. Make a Python implementation of Newton's method and test it on the following function:

$$f(x, y) = (e^x y^3 - 1, y^2 - \sin(x) - 1).$$

It can be an idea to try and visualize the function. Two roots are:

$$\{(4.38277027, 0.23202192), (0, 1)\}$$

Try different initial guesses $\mathbf{x}_0$, close to the roots and further away. How many iterations do you need to achieve machine precision for your guesses? Can you find more roots?

33 Complex-step differentiation

Complex-step differentiation estimates derivatives without the subtractive cancellation of a centered finite difference.

Let $f(x)$ be a real-valued function, assumed to be analytic near some x , i.e., it agrees with its Taylor series in the vicinity of x .

Exercises

Exercise 33.1.

1. Explain why there exists a complex analytic function near $x + 0i$ that agrees with f on the real axis.

A classical method for computing a numerical derivative of a C^1 function is the finite difference approximation:

$$f'(x) \approx \delta_h f(x) := \frac{f(x+h) - f(x-h)}{2h}.$$

The error is $\mathcal{O}(h^2)$. In the rest of the exercise, let f be given by

$$f(x) = \exp(x),$$

and we wish to differentiate around $x = 1$.

1. Write a small program that uses standard double precision floating point arithmetic to compute the finite difference derivative for step lengths from the interval $h \in [10^{-9}, 10^{-1}]$.
2. Make a plot of the absolute value of the error in the approximation as function of h . Use log scales.

The *complex step method* utilizes complex analyticity to avoid the cancellation errors seen in the finite difference scheme. It relies on the function in question being implemented/implementable using complex arithmetic.

1. Show that

$$f(x + ih) = f(x) + ihf'(x) - \frac{1}{2}h^2f''(x) - \frac{ih^3}{6}f'''(x) + \mathcal{O}(h^4).$$

2. Solve for $f'(x)$, and show that

$$f'(x) = \Im \frac{f(x + ih)}{h} + \mathcal{O}(h^2).$$

This is the complex step method.

3. In the plot from above, add a plot of the error of the complex step method. Compare.

Part XI

Functional analysis

34 Canonical commutation relations

35 Canonical commutation relations

In quantum mechanics, physical quantities such as position, momentum, and energy are represented by **linear operators** acting on quantum states. Unlike ordinary numbers, operators do not in general commute: the order in which two operators are applied can matter. Given two operators $\hat{A}$ and $\hat{B}$, their **commutator** is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

Thus $[\hat{A}, \hat{B}] = 0$ precisely when the two operators commute. Commutators play a fundamental role throughout quantum mechanics, both in expressing the mathematical structure of the theory and in deriving properties of quantum systems.

The most important example in elementary quantum mechanics is the **canonical commutation relation** between position and momentum. In one spatial dimension,

$$[\hat{x}, \hat{p}] = i\hbar\hat{1},$$

where $\hat{1}$ denotes the identity operator. In the position representation, the two operators act on a wavefunction ψ according to

$$(\hat{x}\psi)(x) = x\psi(x), \quad (\hat{p}\psi)(x) = \frac{\hbar}{i} \frac{d\psi(x)}{dx}.$$

The commutation relation follows directly from these definitions and the product rule for differentiation. It is important to understand this as an **operator identity**: it states that applying $[\hat{x}, \hat{p}]$ to any suitable wavefunction has the same effect as multiplying that wavefunction by $i\hbar$.

Canonical commutation relations also appear naturally when quantum systems are described using **creation and annihilation operators**. For bosons, creation and annihilation operators $\hat{a}_p^\dagger$ and $\hat{a}_p$ satisfy the canonical commutation relations

$$[\hat{a}_p, \hat{a}_q^\dagger] = \delta_{pq}\hat{1}, \quad [\hat{a}_p, \hat{a}_q] = 0, \quad [\hat{a}_p^\dagger, \hat{a}_q^\dagger] = 0.$$

These relations encode the algebraic structure associated with particles obeying Bose–Einstein statistics.

For fermions, ordinary commutators are replaced by **anticommutators**. The anticommutator is defined by

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A},$$

and fermionic creation and annihilation operators satisfy the canonical anti-commutation relations

$$\{\hat{a}_p, \hat{a}_q^\dagger\} = \delta_{pq} \hat{1}, \quad \{\hat{a}_p, \hat{a}_q\} = 0, \quad \{\hat{a}_p^\dagger, \hat{a}_q^\dagger\} = 0.$$

In particular, setting $p = q$ gives

$$(\hat{a}_p^\dagger)^2 = (\hat{a}_p)^2 = 0.$$

Thus a fermionic one-particle state cannot be occupied more than once. This is the algebraic manifestation of the Pauli exclusion principle and is one reason anticommutation relations are central to electronic-structure theory.

Commutation and anticommutation relations allow us to manipulate products of operators without referring to a particular matrix representation. For example, if

$$[\hat{A}, \hat{B}] = \hat{C},$$

then we can interchange the order of $\hat{A}$ and $\hat{B}$ according to

$$\hat{A}\hat{B} = \hat{B}\hat{A} + \hat{C}.$$

Similarly, canonical (anti)commutation relations can be used repeatedly to rearrange strings of creation and annihilation operators. Such manipulations occur throughout quantum chemistry, in particular in second quantization and many-body theory. The exercises in this chapter provide practice with these algebraic rules, beginning with the position–momentum commutator and progressing to bosonic and fermionic creation and annihilation operators.

Exercises

Exercise 35.1 (The canonical commutation relation in one dimension). Consider wavefunctions of one spatial variable,

$$\psi \in L^2(\mathbb{R}) = \mathcal{H}.$$

A linear operator $\hat{A}$ maps wavefunctions to wavefunctions,

$$\hat{A} : \mathcal{H} \rightarrow \mathcal{H}, \quad \psi \mapsto \hat{A}\psi.$$

Strictly speaking, some operators are defined only on suitable subspaces of $\mathcal{H}$.

The position and momentum operators are defined by their action on a wavefunction:

$$(\hat{x}\psi)(x) = x\psi(x)$$

and

$$(\hat{p}\psi)(x) = \frac{\hbar}{i} \frac{d\psi(x)}{dx}.$$

Products of operators mean composition,

$$(\hat{A}\hat{B})\psi = \hat{A}(\hat{B}\psi),$$

and the commutator of two operators is

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

- a. In the expression

$$(\hat{x}\psi)(x) = x\psi(x),$$

the letter x occurs in several different roles. Explain the difference between the operator $\hat{x}$, the argument x on the left-hand side, and the factor x on the right-hand side. What are the input and output of the operator $\hat{x}$? Why is it mathematically misleading to write simply $\hat{x} = x$?

- b. Show that the position operator $\hat{x}$ is linear. That is, show that for wavefunctions ψ, φ and scalars $a, b \in \mathbb{C}$,

$$\hat{x}(a\psi + b\varphi) = a\hat{x}\psi + b\hat{x}\varphi.$$

- c. Let ψ be an arbitrary sufficiently differentiable wavefunction. Evaluate

$$([\hat{x}, \hat{p}]\psi)(x)$$

by first expanding the definition of the commutator and then applying the definitions of $\hat{x}$ and $\hat{p}$. Use the product rule when necessary.

- d. Hence show that

$$[\hat{x}, \hat{p}] = i\hbar\hat{1},$$

where $\hat{1}$ is the identity operator. Explain why the right-hand side is written as $i\hbar\hat{1}$ rather than simply as the number $i\hbar$.